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Bridging Two Gaps: A Roadmap for Collaboration Between Occupational Physicians and General Practitioners

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Summary

Occupational physicians (OPs) and general practitioners (GPs) manage overlapping aspects of the same patient's health yet rarely communicate with one another. By assigning OPs a role in promoting cancer screening, Law 198/2025 has created in Italy a regulatory link between occupational surveillance and public health prevention—a link that cannot function without a structured exchange of data between the two professional roles. This commentary argues that the new regulatory framework, together with Legislative Decree 135/2024 on carcinogens, mutagens, and substances toxic to reproduction, and Legislative Decree 29/2024 on active aging, offers an unprecedented—yet narrow—window of opportunity to build formal collaboration between OPs and GPs in Italy. Another argument concerns the growing presence of foreign-born workers, who may carry undiagnosed endemic conditions and face language barriers that make OP-GP coordination essential for safe fitness-for-work assessments. Drawing on documented challenges from similar initiatives in the Netherlands, Germany, France, and Finland, we identify the main structural obstacles in the Italian context and propose a three-tiered roadmap for implementation, progressing from immediate consensus-based models to the long-term integration of occupational health data into the Electronic Health Record 2.0.

1. A Legislative Shift

Until recently, the statutory obligations of Italian occupational health physicians (OPs) under Legislative Decree 81/08 were limited to assessing occupational risks and conducting health surveillance of exposed workers. The relationship with the general practitioners (GPs) was left entirely to individual initiative, with predictable results: Persechino et al. [1] found that cooperation was perceived as difficult by most of the Italian OPs interviewed, hindered by a lack of time, the absence of communication channels, and unclear role boundaries.

Law 198/2025 (converting Decree-Law 159/2025), which entered into force on December 31, 2025, changes this landscape. New Article 25, Paragraph 1, Letter *a-bis* of Legislative Decree 81/08 requires occupational physicians to inform workers about the importance of cancer prevention and to promote participation in National Health Service (*SSN*) screening programs included in the Essential Levels of Care (*LEA*) [2]. For the first time, the workplace is legally recognized as the ideal setting for fulfilling public health functions that extend beyond occupational risk. This expanded mandate is difficult to fulfill without a structural link between

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the OP and the GP, who typically coordinates the patient's preventive care and holds information on screening participation. Without access to this information, the OP either duplicates the GP's work or acts blindly.

Two further legislative developments reinforce the need for integration. Legislative Decree 135/2024, which transposes EU Directive 2022/431, expands regulations on carcinogenic, mutagenic, and reprotoxic substances and emphasizes the need for post-exposure follow-up, which inevitably involves the GP once workplace monitoring has ended. Legislative Decree 29/2024 frames the employer's existing obligations under Legislative Decree 81/08 within a broader Workplace Health Promotion (WHP) model, referencing the WHO's approach to workplace health promotion and the National Prevention Plan (PNP) [4]. Although this does not create new obligations beyond those already established by Article 28 of Legislative Decree 81/08, it signals a policy expectation that occupational health should address population aging and chronic diseases—an expectation that presupposes coordination with primary care.

The policy context is further defined by the 2020-2025 National Prevention Plan (PNP), adopted through the State-Regions Agreement of August 6, 2020 [28], which serves as the central planning instrument for prevention and health promotion in Italy. The PNP adopts a *life-course* and *setting-based* approach and explicitly identifies the workplace as a privileged setting for health promotion interventions, citing the Total Worker Health® (TWH®) model [3, 26] as a reference framework for corporate health initiatives. This vision aligns with the ICOH International Code of Ethics, which mandates that occupational health professionals safeguard and promote workers' health and working capacity [27]. Complementing the PNP is the National Chronic Disease Plan (PNC), updated by the State-Regions Agreement of October 23, 2025 [29], which identifies prevention as a key element of chronic disease management and refers to PNP interventions to reduce modifiable risk factors. Taken together, the PNP and PNC establish a policy architecture in which the workplace is expected to contribute to population health goals, a contribution that, in operational terms, requires the OP and the GP to share information on the same patient.

A decisive step forward has been taken with the adoption of the new National Prevention Plan (PNP) 2026-2031 [30] through the State-Regions Agreement of May 21, 2026. The new PNP identifies seven priority intervention areas, including "Health, safety and well-being of workers". Building on the previous Plan but with greater operational ambition, the PNP 2026-2031 promotes the Italian Total Worker Health (ITWH) model as an integrated set of policies, programs, and practices that combine the prevention of occupational health and safety risks with actions to prevent acute and chronic harm, in pursuit of broader worker well-being. The new Plan reinforces the workplace as a privileged setting for health promotion and calls for optimizing integration among prevention services, primary care, territorial assistance, hospitals, and rehabilitation. This explicit reference to an integrated care pathway across settings provides the policy foundation for the OP-GP collaboration model proposed in our roadmap. Furthermore, the PNP 2026-2031 adopts an Evidence-Based Prevention (EBP) approach and mandates the development of Regional Prevention Plans by November 30, 2026, creating a regulatory window within which pilot OP-GP collaboration programs could be embedded at the regional level.

On the same date, the State-Regions Conference also approved, through a formal agreement, the first National Strategy for Occupational Health and Safety 2026-2030 [31]. This Strategy, aligned with the EU Strategic Framework 2021-2027 [5] and explicitly integrated with the PNP 2026-2031, is built on five strategic axes: addressing the transformations of work (digitalization, AI, ecological transition, new contractual forms); strengthening the resilience of the institutional system; enhancing the effectiveness of protections; supporting micro, small and medium-sized enterprises; and disseminating a culture of prevention from schools onward. The Strategy adopts the Vision Zero approach, introduced in Italy for the first time, according to which every workplace incident is preventable. Notably, the Strategy calls for strengthened coordination among supervisory bodies (Local Health Agencies, National Labor Inspectorate) and more targeted inspections, and its integration with the PNP creates a dual policy framework in which corporate TWH programs are no longer optional add-ons but embedded components of the national prevention architecture.

2. Four Arguments in Favor of Structured Collaboration

2.1. Continuity of Post-Exposure Care

When a worker exposed to CMR agents (carcinogenic, mutagenic or toxic for reproduction) retires or changes jobs, occupational health surveillance typically ceases, even though the disease's latency period continues, often for decades. The GP taking over the patient's care should be aware of the nature, duration, and intensity of previous occupational exposures to ensure adequate secondary prevention. This requirement is in line with the EU Strategic Framework for Occupational Safety and Health 2021-2027 [5] and the European Cancer Plan, both of which focus on lifelong prevention. In practice, however, this transfer of information is left to the worker's discretion, who may or may not provide the GP with a copy of the individual health and risk record or even the assessment issued by the OP following the end-of-employment medical examination. In practice, this rarely occurs, but it would be more logical for information to be transferred directly between the two healthcare professionals.

2.2. Coordination of Cancer Screening

The new Article 25 *a-bis* creates a practical coordination challenge. If the OP promotes cancer screening without knowing whether the worker has already been screened through the National Health Service (SSN), the intervention is redundant or misdirected. Conversely, the workplace offers a unique opportunity for access: workers who do not visit the GP office are nonetheless examined periodically by the occupational physician as part of health surveillance. The OP can reinforce GP recommendations, but only if both share basic information on the patient's preventive health status. As argued by Buijs et al. [6], patients often trust their GP more than the OP, making combined efforts more effective than isolated interventions.

2.3. Fitness for Work and Drug Interference

This is likely the most compelling yet least discussed argument in favor of collaboration between OP and GP. The OP *fitness-for-duty* assessment is designed to protect the health and safety of the individual worker and third parties, including colleagues and the public, when the job involves safety risks. The accuracy of the assessment depends on a comprehensive understanding of the worker's health status, including current medication regimens. Commonly used medications—sedatives, first-generation antihistamines, opioid analgesics, psychotropic drugs, and certain hypoglycemic agents—can impair alertness, reaction times, and decision-making. A worker who operates heavy machinery, drives commercial vehicles, or works at heights while taking such medications poses a significant safety risk.

Workers may conceal this information from the OP for understandable reasons: fear of losing their job, stigma, or simply because they do not recognize the professional relevance of their prescriptions. The GP, who manages the patient's medication history over time, holds the missing information. Targeted sharing of clinically relevant data—strictly limited to what affects workplace safety—could enable the OP to make truly protective fitness-for-work assessments and prevent workplace accidents that current information asymmetries leave unnoticed.

2.4. Foreign-Born Workers: Endemic Diseases and Language Barriers

A further dimension of the OP-GP collaboration gap concerns Italy's growing population of foreign-born workers. As of January 2024, approximately 5.3 million foreign residents, or 10% of the total population, are disproportionately represented in physically demanding and hazardous occupations—construction, agriculture, logistics, metalworking, and domestic care [7]. INAIL (National Institute for Insurance against Accidents at Work) data for 2025 show that injuries to foreign-born workers now account for 24.3% of all workplace injury reports, up from 23.5% in 2024, with a year-on-year increase of 3.7%, compared with a decrease of 0.5% among Italian-born workers. Fatal injuries in the foreign-born group also rose, from 176 to 182 reports [7].

These workers often come from countries where tuberculosis, hepatitis B and C, strongyloidiasis, schistosomiasis, and other parasitic infections are endemic. The ECDC evidence-based guidance on infectious disease screening in newly arrived migrants recommends voluntary screening for active and latent TB, HIV, HBV, HCV, strongyloidiasis, and schistosomiasis [8], while the OMEGA-NET position paper highlights that preventive services for infectious diseases, mental health, and chronic conditions are the most neglected areas in occupational health provision for migrants [9]. When these workers enter the Italian labor market, neither the OP nor the GP may be aware of pre-existing conditions that are clinically silent yet relevant to fitness evaluation—latent TB in a worker assigned to immunosuppressive dust environments, chronic HBV carriage in a healthcare auxiliary, or untreated parasitic infections that may interact with occupational chemical exposures.

The problem is compounded by language barriers. Workers with limited proficiency in Italian may be unable to communicate symptoms accurately to the OP during health surveillance, to understand the meaning of a fitness judgment with prescriptions or limitations, or to relay the OP's recommendations to their GP. Conversely, the GP—when a foreign-born worker is registered with the SSN—may prescribe treatments without knowledge of workplace exposures. The ILO has identified language and cultural barriers, together with precarious employment conditions, as key determinants of occupational health disparities among migrant workers [10]. The WHO Refugee and Migrant Health Toolkit explicitly recommends integrated health assessments that bridge occupational and primary care for this population [11].

For these workers, OP-GP data sharing is not merely useful but often indispensable. The GP may hold results from initial health assessments conducted upon entry, vaccination records from the country of origin, and diagnoses of endemic conditions. The OP needs this information to issue an informed fitness judgment. Conversely, the OP's knowledge of specific workplace exposures—silica dust, welding fumes, pesticides, solvents—is critical for the GP managing a patient whose baseline health profile already includes vulnerabilities. The consent-based exchange model proposed in this roadmap (Section 5, Level 1) should include multilingual consent forms and, where feasible, cultural mediation support to ensure that the worker's informed consent is genuine.

3. International Experience: Persistent Difficulties

The Italian situation is not unique in Europe: collaboration between occupational physicians and general practitioners has consistently proven difficult, and an honest assessment of the situation at the international level is necessary.

1. In Germany, Stratil et al. [12, 13] conducted focus groups with GPs, OPs, and rehabilitation physicians, identifying organizational barriers (lack of contact information, conflicting schedules), interpersonal obstacles (low trust in occupational physicians, perceived conflicts of interest with employers), and structural constraints (data privacy regulations). The proposed solutions included formal involvement of occupational physicians in rehabilitation, financial incentives for cooperation, and joint training programs, most of which have not yet been implemented years later.
2. In France, Verger et al. [14] surveyed national samples of GPs and OPs and found that although most endorsed collaboration, a significant proportion of GPs did not trust OPs' independence, a perception that inhibited referrals and data sharing. Moßhammer et al. [15] reported similar findings in Baden-Württemberg: telephone contact was the predominant mode of communication (used by 91% of OPs but only 49% of GPs), and GPs consistently underestimated the need for cooperation.

3. In Finland, primary care is organized into three parallel sectors—public, private, and occupational—all accessible to the working population. Reho et al. [16] found that patients who use multiple healthcare sectors in parallel face a significantly higher risk of disability retirement (OR 4.53), underscoring the clinical cost of fragmented care. Savinainen et al. [17] interviewed healthcare professionals in small Finnish municipalities and identified a lack of familiarity between OPs and GPs, inadequate information exchange, and insufficient time as the main barriers to cooperation.
4. In the Netherlands, interprofessional collaboration between the OP and the GP has been a policy goal for decades, yet Vossen et al. [18] recently documented that it remains largely unfulfilled. They assessed whether delegating collaborative tasks to assistant practitioners and practice nurses could overcome some barriers, but found that even these intermediary professionals still face difficulties in cross-sectoral communication.

The lesson from these experiences is that the value of collaboration between OPs and GPs is universally recognized, yet it has not been systematically achieved through calls for collaboration alone. Structural and regulatory interventions are needed.

4. Obstacles in the Italian Context

The specific structural characteristics of Italy present additional challenges that the three-tiered strategy outlined below must address.

- *The workload and incentive structure of GPs.* Italian GPs operate under National Collective Agreements, typically managing up to 1,500 patients each, with per-capita remuneration. Their time for patient visits is severely limited. Adding structured communication with OPs for their active patients requires dedicated time, financial compensation, or both. Without such incentives, the burden on GPs risks becoming yet another unfunded obligation.
- *The professional fragmentation of OPs.* Many Italian OPs work as independent professionals serving multiple companies simultaneously, often across different provinces. This model makes it difficult to establish stable relationships with the GPs of workers at a single company. Unlike integrated models (e.g., Finnish occupational health centers), the Italian system lacks a physical or institutional setting where the OP and the GP can interact naturally.
- *The FSE 2.0: Promises and Reality.* The Electronic Health Record 2.0 (*Fascicolo Sanitario Elettronico* 2.0) appears, in principle, to be the most appropriate infrastructural solution. However, the results of its implementation call for caution. Monitoring by the GIMBE Foundation in early 2026 revealed that no region had made all 20 required document types available; only 44% of citizens had consented to healthcare providers accessing their data; and significant interregional disparities persist, with the southern regions lagging considerably behind [19]. The Ministerial monitoring portal confirms that the Phase 3 deadline of March 31, 2026, was reached with substantial gaps in operational readiness across regions [20]. The *Profilo Sanitario Sintetico* (Patient Summary), which GPs are required to complete, has been criticized by the SNAMI medical union for the burden of manual compilation and for unclear medico-legal implications [21]. A section dedicated to occupational medicine within the FSE—the roadmap's long-term objective—is not currently provided for in any regulatory instrument and would require specific legislative action.
- *Privacy and Liability.* The OP serves as an independent data controller under the GDPR for health data collected during occupational health surveillance. Any sharing of data with GPs must comply with the principle of data minimization: only information strictly necessary for continuity of care, secondary prevention, or an accurate assessment of fitness may be exchanged. The employer must not

have access to clinical diagnoses or pharmacological data shared between the OP and the GP; the employer is entitled only to the final assessment of fitness. An unresolved issue concerns professional liability: if a GP shares pharmacological information with the OP and the OP nevertheless issues an inaccurate fitness-for-work assessment, or conversely if the GP omits relevant information, the attribution of liability remains poorly defined in Italian case law.

5. A Roadmap for Three-Tiered Implementation

Based on international evidence and the specific characteristics of the Italian system, we propose a phased strategy ranging from immediate feasibility to structural integration (Figure 1) .

- *Level 1: Consent-based exchange (immediate).* During routine occupational health visits, the OP asks the worker to sign a GDPR-compliant consent form authorizing the two-way sharing of clinically relevant information with their GP. The shared data is limited to a standardized minimum set: current occupational exposures, fitness limitations and prescriptions, screening adherence status (from the general practitioner to the occupational physician), and relevant pharmacological therapies affecting workplace safety (from the general practitioner to the occupational physician). The worker serves as an intermediary and retains full control. For foreign-born workers, consent forms should be available in the most commonly spoken languages, and, where feasible, the exchange should be supported by cultural mediation services. This model requires no regulatory changes and can be implemented immediately.
- *Level 2: InterSsociety Protocols (medium term).* Scientific societies representing occupational medicine and general practice should jointly develop standardized communication protocols that define: the minimum data set content, secure transmission procedures, pseudonymization requirements, and clinical scenarios that trigger mandatory information exchange (e.g., cessation of exposure to CMR substances, a new prescription of

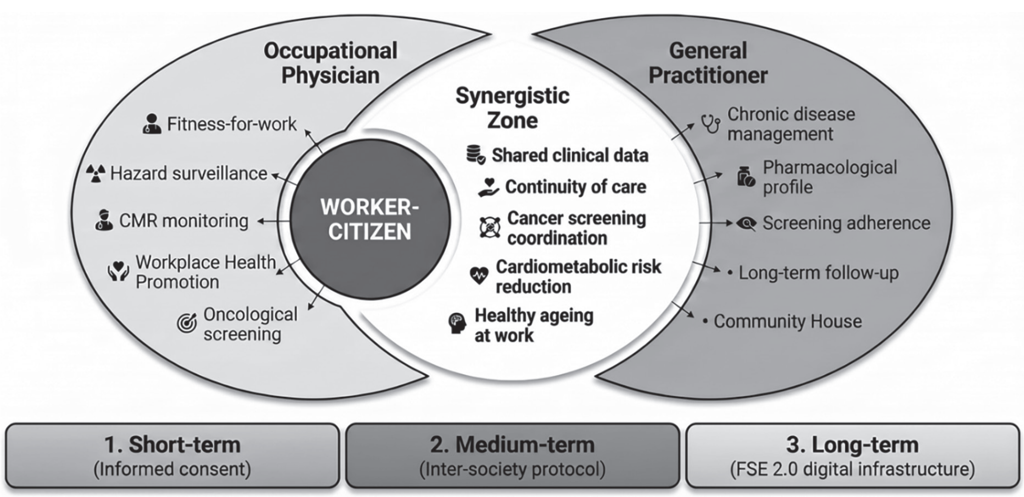

Figure 1. Conceptual framework for the synergistic collaboration between the Occupational Physician (OP) and the General Practitioner (GP) within a Total Worker Health® (TWH) approach, highlighting shared preventive domains (Synergistic Zone) and a three-tiered implementation strategy. CMR: carcinogen, mutagen, reprotoxic agents; WHP: Workplace Health Promotion; FSE 2.0: Fascicolo Sanitario Elettronico 2.0 (Electronic Health Record).

sedative medications for a worker performing safety-critical tasks). Pilot implementation could be hosted within the Community Centres (*Case della Comunità*) established under Ministerial Decree 77/2022 and funded by the PNRR, which provide physical locations for multidisciplinary healthcare teams. As documented by Leso et al. [22], Italian occupational physicians identify structured interdisciplinary collaboration and targeted training in WHP as their primary professional needs.

- *Level 3: Legislative and Digital Integration (long-term).* The goal is a systemic communication channel that does not depend on individual initiative. This requires two coordinated interventions: first, an amendment to Article 25 of Legislative Decree 81/08 mandating the sharing of clinically relevant health data between OP and GP for documented purposes (continuity of post-exposure care, accuracy of fitness assessments, coordination of screening); second, the creation of a role-based section dedicated to occupational health within the FSE 2.0. Given the current gaps in implementing the FSE [19-21], this level is realistic only once the basic infrastructure achieves reliable national coverage—a goal currently tentatively set for 2026 but likely to extend beyond that date.

6. Evidence from workplace health promotion

The clinical rationale for integrating occupational medicine and general practice is supported by recent evidence on the effectiveness of health promotion in occupational settings. The three-level meta-analysis by Godono et al. [23], which includes 44 studies and nearly 50,000 participants, demonstrated significant reductions in body mass index, blood pressure, and LDL cholesterol through workplace health promotion interventions. The effect was most pronounced when programs targeted high-risk populations, were led by qualified healthcare professionals, and were integrated into sustainable health promotion policies. Health promotion initiatives are precisely those that benefit from coordination between OPs and GPs: the OP identifies at-risk workers during surveillance, and the GP provides the clinical context and capacity for follow-up.

Magnavita et al. [24] and Talini and Baldasseroni [25] have emphasized that workplace health promotion activities in Italy require ongoing institutional commitment and must not replace mandatory health and safety measures. Health promotion programs require adequate follow-up to measure actual health benefits—a requirement that, once again, implies continuity between occupational medicine and primary care settings.

7. Conclusion

The convergence of Law 198/2025, Legislative Decree 135/2024, and Legislative Decree 29/2024 creates a legislative framework in which collaboration between occupational medicine and general practice is no longer optional but a prerequisite for legal compliance. However, legislation alone does not produce collaboration, as decades of experience in the Netherlands, Germany, France, and Finland demonstrate. The obstacles are structural: incompatible professional incentives, a lack of communication infrastructure, privacy constraints, and unresolved liability issues. The adoption of the PNP 2026-2031 and the National Strategy for Occupational Health and Safety 2026-2030 [30,31], both approved on May 21, 2026, reinforces this framework by explicitly embedding the TWH model within the national prevention architecture and mandating the integration of prevention services with primary care and territorial assistance. This policy direction is fully aligned with the roadmap proposed here.

The three-phase roadmap proposed here addresses these obstacles incrementally, from immediately feasible consensus-based exchanges to the long-term integration of occupational health data into the FSE 2.0. Its success depends on the commitment of scientific societies to develop shared protocols, of regional health authorities to create pilot programs within Community Health Centers, and of the legislature to extend Article 25 of Legislative Decree 81/08 to make bidirectional data sharing mandatory for defined clinical purposes.

A reform of the legal status of general practitioners is currently under consideration in Italy: whether they should remain self-employed practitioners under the *SSN* (with new rules) or, on a voluntary basis, switch to the status of civil servants within the *SSN*. In the proposals under discussion, references to 'prevention' and 'early diagnosis' of cancer are generic and geared toward proactive care for the chronically ill and frail, not toward post-exposure occupational health surveillance as per Articles 242 and 259 of Legislative Decree 81/08. The legislation should ideally be amended accordingly.

The worker and the citizen are one and the same person. Their health cannot be effectively managed by two professionals who do not communicate.

Conflicts of Interest: The author declares that there are no conflicts of interest.

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A Scoping Review of Workplace Violence Among Nurses in Indonesia: Implications for Educational and Reporting Models

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Keywords: Workplace Violence; Nurse; Hospitals; Indonesia; Models

Abstract

Background: *Workplace violence (WPV) is a critical issue in healthcare systems globally, disproportionately affecting nurses due to their frontline role in patient care. The International Council of Nurses recognizes WPV, including verbal abuse, threats, harassment, and physical assault, as a persistent threat to nurses' safety and well-being. In Indonesia, the magnitude of WPV against nurses remains underexplored, and there is limited evidence supporting the development of culturally responsive educational models tailored to local healthcare settings.* **Methods:** *A scoping review was conducted to analyze current research on workplace violence affecting nurses in Indonesian hospitals. Various electronic databases and gray literature were thoroughly examined using relevant keyword combinations: "Workplace Violence," "Nurse," "Hospital in Indonesia," and "Abuse." Studies were included if they (i) constituted original research, (ii) investigated WPV incidents within hospital environments in Indonesia, (iii) specifically focused on nurses, and (iv) utilized either quantitative or qualitative methodologies.* **Results:** *The review identified recurring themes related to the high prevalence of WPV among nurses in Indonesian hospitals. Factors contributing to WPV included a lack of institutional support, the absence of standardized reporting systems, and limited awareness and training. While a few studies proposed interventions, none presented a comprehensive or context-specific educational model for WPV prevention and reporting tailored to Indonesia's sociocultural setting.* **Conclusions:** *There is an urgent need for a structured, culturally adapted workplace violence educational model for nurses in Indonesia. Such a model should integrate reporting mechanisms and preventive strategies to ensure workplace safety and improve healthcare services delivered to the community.*

1. Introduction

Workplace violence (WPV) is a significant global issue in healthcare [1], notably impacting nurses who are on the front lines [2]. The International Council of Nurses (ICN) defines WPV as any incident where an employee is abused, threatened, or assaulted related to their work, including commutes. This violence directly threatens workers' physical and mental health [3]. Nurses face heightened risk due to their constant interactions with patients, families, and caregivers, often in emotionally charged or high-stress contexts [4].

Numerous empirical studies indicate that nurses report high rates of WPV. For instance, Gacki-Smith et al. and Esmaeilpour et al. found that over 50% of nurses had encountered some form of WPV, with verbal abuse being the most frequent [5-6].

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Likewise, Park et al. documented 12-month prevalence rates of verbal abuse (63.8%), threats (41.6%), physical assault (22.3%), sexual harassment (19.7%), and bullying (9.7%) across various clinical environments [7]. These statistics highlight a widespread global concern, further complicated by varying research methodologies that prevent straightforward comparisons [7]. A global integrative review indicated that nurses in the Anglo, Asian, European, and Middle Eastern regions experienced physical violence (31.8%), non-physical violence (62.8%), bullying (47.6%), and sexual harassment (17.9%) within the past year [8].

In Indonesia, WPV is also prevalent. National statistics reveal that 54.6% of nurses have faced non-physical violence, and 10% have experienced physical violence [9]. Nonetheless, many incidents remain unreported due to the normalization of violence, ignorance of reporting procedures, and the challenges of self-reporting [10–12]. A recent study in Aceh Province, affected by conflict, found that nurses regularly encounter verbal and physical aggression [10], primarily from patients and their families [13]. Despite the frequency of these violent occurrences, most nurses opt not to report, citing fears of retaliation, distrust in institutional responses, and a lack of awareness of reporting options [14]. This situation illustrates the "iceberg phenomenon" regarding WPV, where only a fraction of actual incidents get recorded and addressed [11].

The high rates and normalization of WPV, coupled with institutional inertia, necessitate an immediate and organized response [15]. To meet this challenge, researchers advocate for developing a Workplace Violence Education Model (WPVEM) tailored to the healthcare environment. This model educates healthcare professionals to identify early signs of violence, implement preventive measures, and respond effectively to incidents. According to OSHA, over 7.6 million WPV cases are reported in the United States annually, encompassing physical assaults, threats, intimidation, and verbal harassment, highlighting a critical need for comprehensive preventive measures [16]. Evidence indicates that structured WPV educational programs can substantially lower incident rates. For example, organizations that adopted NIOSH-endorsed WPV prevention frameworks experienced up to a 50% decline in WPV cases [17].

Various WPVEMs have been developed [16,18,19], including:

a. NOVA (Nurse Occupational Violence Awareness): This model adopts a prevention approach to lower WPV risk by focusing on four areas: physical environment, policies and procedures, employee recruitment and training, and conflict management;
b. AWAKE (Alert, Warn, and Knowledge for Employees): This model trains employees to recognize signs of violence, take preventive steps, and respond appropriately in violent situations. It emphasizes knowledge, skills, and behavior;
c. Prevention and Management of Violence in Healthcare: This model was designed to address violence in healthcare settings. It trains staff to identify early signs, understand risk factors, and develop prevention and response strategies.

Nevertheless, the availability of these models in Indonesia is limited. The sole formal legal structure is Law No. 12 of 2022 from the Ministry of Manpower, focusing on sexual violence but lacking comprehensive regulations for WPV in healthcare contexts [20]. Therefore, creating a localized and culturally appropriate WPVEM for Indonesian healthcare facilities is crucial for protecting workers, enhancing safety, and fostering a positive organizational culture.

This review examines WPV prevalence, reporting behaviors, psychological and organizational impacts, and strategies to mitigate WPV among nurses in Indonesia. It highlights the need for a structured educational model suited to the Indonesian healthcare sector. In addition, this review is important for uncovering shortcomings in current policies and reporting systems and for offering evidence-based recommendations to improve future practices, policy development, and research efforts. By reviewing recent literature and incorporating case studies from Aceh Province, this work contributes to growing knowledge on reducing WPV and promoting healthcare workers' well-being.

This scoping review aims to analyze empirical evidence on the incidence of WPV among nurses in Indonesian hospitals. Furthermore, guided by this Population-Concept-Context (PCC) framework, the scoping review addressed the following questions:

a. What types and prevalence of workplace violence are reported among nurses in Indonesian hospital settings?
b. What individual, organizational, and contextual factors are associated with workplace violence against nurses?
c. How do nurses report and respond to workplace violence, and what barriers to reporting are identified?
d. What institutional strategies, educational interventions, or reporting mechanisms have been described, and what gaps remain for developing a context-specific Workplace Violence Education Model?

2. Methods

2.1. Study Design

This study employed a scoping review design, conducted in accordance with established methodological guidance for scoping reviews and reported following the PRISMA-ScR reporting framework [21]. The review process was transparently documented using a PRISMA flow diagram (Figure 1) to ensure reproducibility and methodological rigor. To answer the research questions, we formulate this study using the PCC framework, which includes:

a. Population (P): Registered nurses working in hospital settings;
b. Concept (C): Workplace violence (WPV), encompassing physical violence, verbal abuse, threats, bullying, harassment, and sexual violence; associated factors; reporting behaviors; and institutional responses, including education and prevention strategies;
c. Context (C): Hospital settings in Indonesia, including emergency departments, intensive care units, inpatient wards, critical care units, operating rooms, and mental health facilities.

2.2. Search Strategy and Eligibility Criteria

Specific criteria ensured relevance and quality, including original, peer-reviewed articles from 2018 to 2025, in English or Bahasa, addressing WPV incidents among nurses in Indonesian hospitals. Qualitative and quantitative designs were suitable. Exclusions were applied to studies that were not fully available, were not in English or Bahasa, or did not focus on WPV among hospital nurses.

From December 5, 2024, to March 15, 2025, a thorough literature search was conducted across reputable databases (Elsevier, ResearchGate, and Google Scholar), as well as institutional libraries and national repositories [22]. A mix of keywords and Boolean operators helped locate relevant studies, including terms like "Workplace Violence", "Nurse", "Hospital in Indonesia" and "Abuse". The strategy was tailored to each database's features, as detailed in Supplementary Material S1. Reference lists of all selected articles were also reviewed for qualifying studies (Table 1).

2.3. Study Selection

All search results were imported into a structured spreadsheet for screening. The selection process involved two stages, which were completed by independent reviewers. In the first stage, titles and abstracts were used to eliminate irrelevant studies. In the second stage, full-text versions of eligible articles were reviewed. Reviewer discrepancies were resolved through discussion or a third reviewer. The process followed PRISMA guidelines, detailed in the flow diagram [23] (Figure 1).

2.4. Data Extraction, Quality Appraisal, and Data Synthesis

Both reviewers independently extracted data using a standardized form capturing study characteristics, including authors, year, location, design, sample size, types and frequencies of WPV, perpetrator and victim characteristics, contextual factors, and significant findings. Inconsistencies were discussed and resolved for accuracy.

Table 1. Matrix Review Literature.

Ref	Participant	Aims	Methods	Findings	Conclusions
[9]	The study involved 169 nurses employed in the emergency departments (EDs) of six Jakarta and Bekasi, Indonesia hospitals.	Examine the experiences of violent incidents committed by nurses in the ED in Indonesia.	Quantitative Research with data analyzed using descriptive and multivariate logistic regression.	As per the study findings, 10% of emergency nurses stated that they had experienced physical abuse, mainly perpetrated by patients. More than half of the nurses (54.6%) reported experiencing non-physical violence, with family members being the primary culprits. Most nurses (55.6%) were not encouraged to report WPV incidents, and only a tiny fraction (10.1%) had received any information or training on handling WPV situations.	The study's outcomes brought attention to the gravity of violence in the EDs of Indonesia. The findings suggest that management backing, support, and motivation to report violence, and the availability of WPV training could help reduce and control the abuse that nurses face while working in EDs.
[25]	The sample of this study comprised ten nurses who had been working in the ED for a minimum of six months and were not currently on maternity leave, as the nurse manager/head of the room.	To understand the psychological trauma experienced by ED nurses in the face of verbal abuse in the workplace.	Qualitative research using a phenomenological approach.	The themes found in this study are (i) knowledge, with sub-themes: types of verbal violence and forms of verbal violence; (ii) sources of verbal violence with sub-themes of perpetrators and causes of verbal violence; (iii) the influence of verbal violence with sub-themes of response when experiencing verbal violence, the effects of verbal violence, and coping in the face of verbal violence.	Verbal abuse in the workplace impacts the psychological performance of nurses. By understanding the psychological trauma of nurses, efforts are expected to arise to protect nurses from providing nursing care and improve the quality of service.

[26]	The 105 participating nurses came from the ED and ICU at four hospitals: Bitung, Budi Mulia Bitung, Hermana Lembean, and Maria W Maramis Airmadidi.	To analyze the relationship between violence and work stress in nurses in several EDs and ICUs of hospitals in Bitung City and North Minahasa.	Type of quantitative survey analytics research using a cross-sectional design. This study used the chi-square, Fisher's exact, and logistic regression tests.	As per the study findings, the most frequently encountered form of violence reported by the participants was verbal abuse (54.3%), followed by bullying (28.6%), physical violence and threats (19%), sexual harassment (11.4%), and harassment (8.6%). Perpetrators of violence are most often committed by the patient's family (63.5%). Furthermore, there is a meaningful association between all WPV, including physical, verbal, bullying, harassment, sexual harassment, threats, and work stress. Multivariate analysis showed that verbal abuse was the most dominant factor associated with job stress in nurses.	For hospital institutions, it is recommended to develop orientation and introduction programs for staff, design the environment, and work in a clear organizational structure, placement, training in WPV, clear job descriptions, and good conflict communication to prevent WPV from triggering work stress.
[27]	The respondents were 60 nurses who work in the public hospital in Luwuk.	Determine emergency response and wait times for outpatients with violence against nurses in the Luwuk Banggai General Hospital district.	Quantitative Analytical Surveys. The collected data were subjected to univariate and bivariate analysis, which included the Chi-Square Test.	The most dominant violence is committed by the patient's family against nurses in the form of verbal violence (85%), followed by physical violence (8.33%) and psychological violence (6.67%). Verbal violence is closely related to ridicule, cursing, innuendo, insults, and even being laughed at if the nurse is doing the wrong task. The root of the problem is triggered by the status or profession of a nurse who is considered low, and many indirectly get complaints about the issue of doctor services to nurses.	The relationship between emergency response time and aggressive behavior among nurses is to maximize the treatment of response time through ABCD procedures. Facilities, human resources infrastructure, and ED management improvements are also possible. The relationship between outpatient wait times and the quality of service includes specific dimensions. Namely, I will determine duty time, ease of meeting a patient's needs, convenience and hospitality, human resource aspects, paying attention, and others.

Table 1 (*Continued*)

Ref	Participant	Aims	Methods	Findings	Conclusions
[28]	The researcher used a total sampling technique involving 196 nurses from the Regional General Hospital of Meuraxa.	Determine factors associated with workplace intimidation among nurses in Banda Aceh City.	Cross-sectional quantitative research. Data were analyzed using univariate analysis, bivariate analysis (chi-square test), and multivariate analysis (multiple logistic regression).	Factors associated with the impact of bullying in the workplace include age = 0.001, year of service = 0.000, position = 0.001, and organizational climate = 0.000. Results of the multivariate analysis indicated that organizational climate was the most crucial factor for this issue, with an OR value of 3.483.	Bullying in the workplace is linked with age, seniority, position, and organizational climate. However, the most related factor is the organizational climate, where a poor environment versus a good one leads to workplace bullying. Leaders and decision-makers look forward to creating a supportive organizational climate and anti-bullying policies, creating a pleasant and reassuring workplace.
[29]	Respondents were nurses from the Aceh General Hospital; 391 out of 410 nurses participated (95%). Respondents were nurses from patient rooms, critical care units, EDs, and operating rooms.	To identify factors influencing verbal abuse among nurses in Aceh's government hospitals.	A quantitative design using a cross-cutting approach. The data gathered are then processed and analyzed using univariate, bivariate, and multivariate analysis.	This study revealed that the factors influencing nurse verbal abuse were (0.014), occupational status (0.021), work unit (0.188), shift (0.007), work stress (0.000), and organizational climate (0.012). The value of a Nagelkerke R-squared is 0.212. Indicates that each variable is most associated with verbal abuse has a 21.2% probability of influencing verbal abuse, and that 79.8% is another factor apart from these factors.	Addressing the issue of verbal abuse among nurses in the workplace requires assigning assignments based on a nurse's career path. Adopt a strong workplace attitude and improve the factors that can cause verbal violence in the workplace.

[30]	A study sample of 50 nurses from two hospitals (X & Y) was obtained.	To analyze WPV among nurses in an accredited hospital in Pekanbaru.	The type of research used for the present study is descriptive, quantitative, and analytical.	Finding a Hospital X: 100% of respondents stated "no evaluation of the effectiveness of the impact of violence at work policies" and "violence training in the workplace is not provided"; 92% of respondents said, "they did not know about policy changes based on evaluations of violence prevention programs in the workplace". Finding at the Hospital Y: 40% of respondents stated "the availability of verbal violence policies from patients or visitors"; 48% of respondents stated that "the policy of preventing violence in the workplace was needed related to incident reporting"; 36% of respondents said that "appropriate technical training is needed to reduce conflict"; 72% of respondents stated "the availability of follow-up support such as counseling for nurses who experience verbal/physical violence".	Hospital managers must demonstrate and document their commitment to the prevention of WPV. They should also prioritize preventing WPV, setting goals, targets, and effectiveness, providing adequate resources and support, and appointing leaders with the authority and knowledge to facilitate change and provide leadership. Ideally, employees with a different understanding of the workplace should be engaged in all aspects of the program. Employees are invited to communicate openly with management and to raise concerns without fear of reprisal.
[11]	Eight ED nurses were purposely sampled. The data gathered were analyzed by theme using the Miles and Huberman technique.	To explore WPV as a function of the internal situation of ED nurses.	A descriptive phenomenological approach was carried out utilizing in-depth interviews.	Three themes emerged from the study: (i) the nurse reporting dilemma, (ii) "ready or not," and (iii) the iceberg phenomenon. The conclusion demonstrates the internal factors that influenced nurses in reporting WPV.	The study highlighted that hospitals should remove the "normalization" of violence to reduce incidents and move from the "culture of blame" to the "culture of learning" to create a "culture of reporting." Furthermore, an educational and training program should be reinforced for all forms of unacceptable violence. Senior managers must also adequately protect and support staff who have experienced violence.

Table 1 (*Continued*)

Ref	Participant	Aims	Methods	Findings	Conclusions
[31]	Out of the total sample size of 161 individuals, 127 participants (78.9% response rate) from the Aceh Provincial Hospital participated in this study.	To explore the correlation between overtime and physical and emotional violence experienced by nurses working at the Aceh Provincial Hospital.	Quantitative approach with transversal study design. Data analysis used Univariate analysis (descriptive statistical tests) for time analysis. The bivariate analysis (chi-square test) revealed the relationship between the two variables.	The study found a significant association between overtime and the psychological abuse of nurses working at the Aceh Provincial Hospital ($p=0.000$). However, no such correlation was observed for physical abuse among the nurses at the hospital ($p=0.290$).	Overtime and psychological/ non-physical violence are linked to verbal harassment and bullying/ crowding. Nurses are tired of working overtime, thereby reducing their concentration at work. It will also affect the quality of service, which will affect the satisfaction of service recipients, which may lead to WPV.
[32]	Data were obtained from 120 RNs employed in a mental health hospital in West Java.	Examine the violence and related factors experienced by Indonesian nurses.	A descriptive transversal design was used in the present study.	46.7% of nurses reported verbal abuse, 29.2% physical assault, and 24.2% verbal and physical abuse. Of those who experienced both verbal and physical abuse, 27.6% were victims of acute care, and 32.1% were victims of routine treatment and unit rounds. Nurses with degrees were more likely to be physically assaulted and to experience physical and verbal abuse than those with an undergraduate degree. Nurses working in the ICU tended to experience more physical and verbal abuse than in other units.	Developing an electronic system is essential to assist nurses in requesting assistance from the safety team when patients are unruly, agitated, out of control, destructive, aggressive, life-threatening, and environmental. It is also essential to give nurses legal immunity from litigation and to create an ethics team that can protect the interests of nurses and patients.
[33]	The sample size in this study was 172 nurses using the snowball sampling approach in the Banda Aceh district.	Describe incidents of physical violence by nurses in Banda Aceh's district.	The present study uses a descriptive method and a transversal study approach.	Nurses had never been subjected to physical violence in the past 12 months. However, up to 7% of nurses witnessed physical abuse, and 2.9% reported incidents of physical abuse.	Implementing a reporting mechanism is vital to ensuring that nurses are adequately protected in a safe workplace so that quality health services can meet the community's interests.

[13]	The research applied the snowball sampling technique to gather information from 433 participants. The criteria for inclusion were nurses working at a single hospital with at least two years of experience.	This study aimed to collect information on both physical and non-physical violence against nurses in hospitals throughout Aceh Province while also pinpointing the individuals responsible for these acts.	This research employs a cross-sectional design and utilizes a quantitative, descriptive approach. Data was collected through web-based questionnaires distributed to selected respondents via Google Forms.	Nurses in hospitals across Aceh Province reported experiencing workplace violence, which included physical assault (15%) and sexual abuse (5.5%), significantly affecting their well-being. A significant number (64.4%) dealt with emotional abuse, 37.9% faced verbal threats, and 10.4% were subjected to verbal sexual harassment. In 60.3% of the incidents, relatives and patients' families were identified as the primary perpetrators.	Nursing associations are vital in providing guidance, regulatory frameworks, and educational materials regarding workplace violence. Their support is essential for identifying and addressing potentially violent behaviors swiftly, which significantly contributes to reducing violent incidents at work.
[14]	The study involved experienced nurses from public hospitals, including seven key informants. These participants were chosen through purposive sampling, where individuals were selected intentionally based on defined criteria for study inclusion.	This study seeks to comprehensively examine, analyze, and document workplace violence incidents and related experiences nurses face in hospitals in Aceh, Indonesia.	This qualitative research initiative aims to carefully identify the types of WPV nurses face, analyze the factors that contribute to these incidents, and thoroughly investigate current reporting procedures through detailed interviews.	Three key themes emerge from the analysis. The first theme addresses the nature and frequency of incidents, shedding light on their prevalence and defining characteristics. The second theme investigates the various forms of workplace violence faced by nurses, such as physical assault, emotional abuse, threats, verbal sexual harassment, and sexual abuse. Lastly, the third theme focuses on the difficulties of reporting incidents of workplace violence in the healthcare environment, highlighting the existing barriers and complexities.	Apparent and standardized regulations for reporting incidents of workplace violence, along with the essential support and infrastructure, are vital for managing these situations effectively. The ultimate goal is to encourage and assist nurses in reporting violent acts and managing them more effectively.

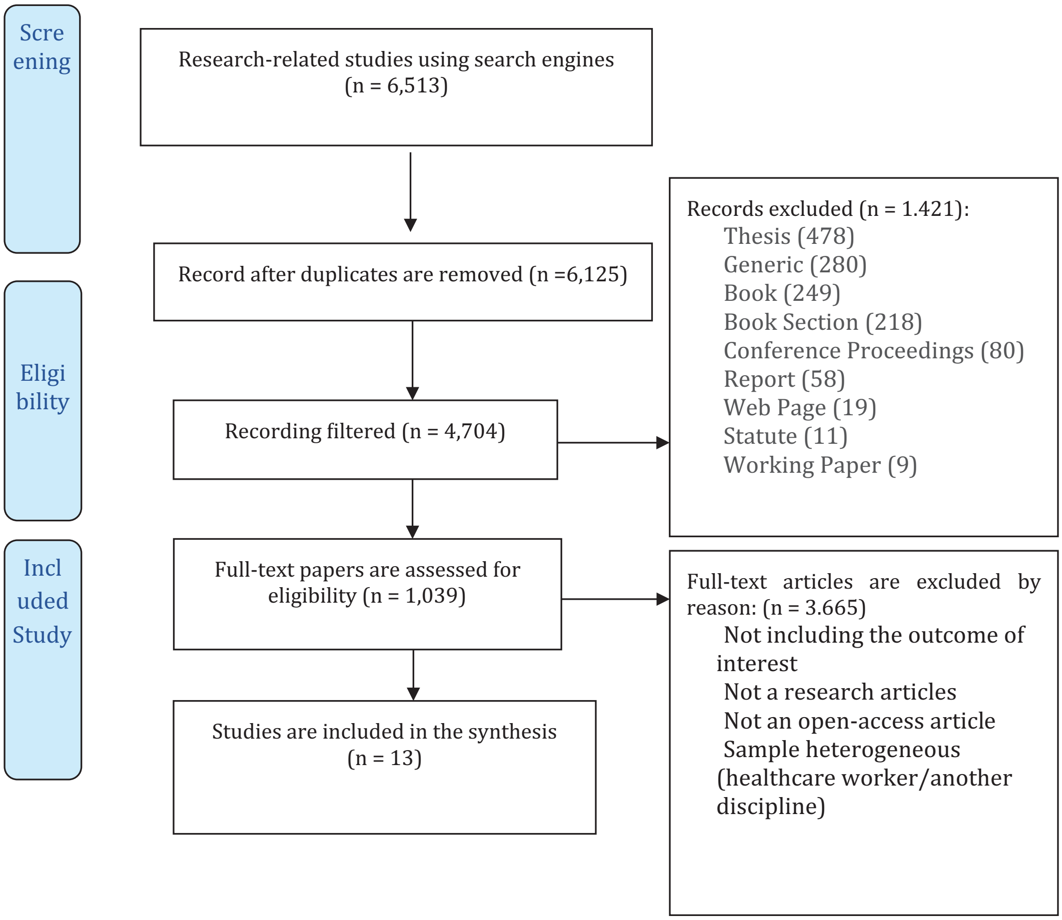

Figure 1. PRISMA flow diagram.

Critical appraisal tools evaluated methodological quality and risk of bias based on study design. The CASP checklist was used for qualitative studies, while the JBI checklist was used for quantitative analyses, categorizing risk as low, moderate, or high. Effect measures varied due to qualitative and quantitative designs. Quantitative studies reported prevalence rates, odds ratios (ORs), and confidence intervals (CIs), while qualitative studies employed thematic analysis to derive insights [24].

A narrative synthesis grouped and interpreted findings thematically to identify commonalities, differences, and contextual factors influencing WPV among nurses in Indonesian hospitals. Potential reporting biases were considered during quality appraisal. The certainty of the evidence from quantitative studies was assessed using the GRADE approach, evaluating study limitations, consistency, directness, precision, and publication bias, resulting in high, moderate, low, or very low overall ratings.

3. Results

A total of 13 studies were included in this scoping review, comprising quantitative cross-sectional surveys, qualitative phenomenological studies, and descriptive analyses. The studies were conducted across diverse hospital settings in Indonesia, including emergency departments, intensive care units, inpatient wards, operating rooms, and mental health hospitals. Most participants were registered nurses providing direct patient care, with several studies focusing specifically on high-risk units such as emergency and critical care departments. This diversity of settings allowed for a comprehensive mapping of workplace violence experiences across clinical contexts.

3.1. Incidence of WPV Among Nurses in Indonesia

Empirical studies reveal the widespread and complex nature of WPV against nurses in Indonesian hospitals [10]. Findings show that 10% of emergency nurses face physical violence, mainly from patients, while 54.6% experience non-physical violence, mostly from patients' families. Alarmingly, 55.6% are discouraged from reporting incidents, and only 10.1% have received WPV management training [9]. Verbal abuse is the most common form (54.3%), followed by bullying (28.6%), physical violence and threats (19%), sexual harassment (11.4%), and harassment (8.6%). Most acts are committed by patients' families (63.5%) and are linked to nurses' job stress, with verbal abuse being the most significant factor [26].

Verbal violence is the dominant type (85%), involving ridicule, cursing, and insults, often rooted in societal perceptions of nurses' lower professional status [27]. Another study revealed that 46.7% of nurses experienced verbal abuse, 29.2% physical assault, and 24.2% both forms. These incidents were more frequent in acute care (27.6%) and routine units (32.1%), particularly among nurses with advanced degrees and those in ICU settings [32]. More recent data indicated that while no nurses reported being physically abused in the prior year, 7% witnessed physical violence, and 2.9% formally reported it, reflecting ongoing underreporting and normalization [10, 33].

Regarding contributing factors, an earlier study identified age, years of service, job position, and organizational climate as significant, with the environment being the most influential (OR = 3.483) [28]. Zulfan et al. also reported that verbal abuse was significantly influenced by occupational status, shift work, work unit, job stress, and organizational climate, accounting for 21.2% of the variance [29]. Institutional evaluations further revealed systemic gaps. In one accredited hospital, 100% of respondents reported no policy evaluations or violence prevention training, and 92% were unaware of any policy revisions. Another hospital showed partial policy implementation, with 40% acknowledging verbal violence policies and 48% indicating a need for reporting mechanisms and conflict-resolution training [30]. Reinforcing this point, it is noted that institutional support remains limited, while cultural stigma hinders reporting [10]. They called for policy reforms, improved communication, and the development of a culture of accountability to reduce WPV in Indonesian healthcare environments [14].

3.2 Implication of WPVEM for Nurses in Indonesia

This review identifies critical challenges in addressing WPV among nurses in Indonesia, emphasizing the urgency of context-specific strategies that promote awareness, improve reporting, and facilitate effective educational interventions. A significant barrier is the persistent underreporting of WPV, often likened to an iceberg phenomenon in which many incidents go unreported due to internal and systemic constraints. These include fear of retaliation, normalization of violence in healthcare settings, limited managerial support, and a prevailing culture of silence within institutions [11].

WPV is especially prevalent in regions such as Aceh Province, where a history of armed conflict has intensified aggression toward healthcare workers. A previous study highlighted frequent occurrences of verbal and physical violence by patients and their families in Aceh [13]. A subsequent investigation revealed that many nurses avoid reporting such incidents due to a lack of trust in institutional processes, fear of repercussions, and limited knowledge of support mechanisms [14]. The seriousness of WPV in Indonesia is further underscored by data from the International Committee of the Red Cross, which recorded over 600 WPV incidents targeting healthcare workers during the COVID-19 pandemic, including events across Kalimantan, Java, and Sumatra involving both civilians and state actors [31, 34].

A qualitative study adds depth to these findings, which identified key themes from nurses' experiences, including their understanding of WPV, its verbal forms, the triggers of abuse, and coping strategies [25]. Similarly, recurrent concerns were the reluctance to report incidents, lack of preparedness, and continuation of the iceberg effect [11]. These results suggest the need for a comprehensive WPVEM tailored to address personal and institutional barriers.

Organizational responses must be proactive and inclusive. Meri and Mayenti advocated that hospital leaders set clear objectives, allocate appropriate resources, and appoint competent leaders, while ensuring broad employee engagement in anti-violence initiatives [24]. Cultivating a culture of open communication, shifting from blame to learning, is vital to fostering trust and reporting [11]. From technological and ethical perspectives, integrating electronic alert systems was proposed to enable rapid staff access to security during violent episodes [32]. Legal protections should also be established to shield staff who report violence in good faith, with ethics committees supporting both patient and healthcare provider rights [32, 33].

4. Discussion

Workplace violence against nurses remains a significant global concern, with international evidence indicating that up to 90% of nurses experience at least one form of violence during their careers [35]. Consistent with this global pattern, studies in Indonesia report a high prevalence of WPV in hospital settings. Zahra and Feng documented that 54.6% of nurses experienced non-physical violence and 10% reported physical violence, most often perpetrated by patients or their family members [9]. However, these figures likely underestimate the true magnitude of the problem because WPV is widely underreported. This phenomenon, often described as the "iceberg phenomenon", reflects systemic and institutional barriers that prevent incidents from being formally recorded or addressed [11].

Evidence from Aceh Province further illustrates the depth and complexity of this issue, showing that nurses in post-conflict settings are particularly vulnerable to verbal and physical violence, largely driven by interactions with patients and their families [13]. Follow-up findings showed consistently low reporting rates, caused by limited trust in institutional procedures, fear of retaliation, and a lack of awareness about available support systems [14]. These patterns are not exclusive to Indonesia and are consistent with wider national and global literature, suggesting that nurses often see violence as a normal part of their job and thus tend to avoid using formal reporting systems [11, 36].

The consequences of WPV extend beyond immediate physical harm to include substantial psychological and professional impacts. Repeated exposure to violence has been linked to reduced job satisfaction, emotional exhaustion, anxiety, depressive symptoms, and somatic complaints such as sleep disturbances and headaches [37-42]. At the organizational level, WPV leads to increased staff turnover, lower workforce morale, and reduced patient safety, impacting overall healthcare quality and system performance [39, 41].

Despite the severity of the issue, institutional efforts to address WPV in Indonesian hospitals are still insufficient. Data show that organizational support structures, such as formal reporting systems and post-incident counseling, are often inconsistently available or missing [14]. Studies show that many hospitals lack structured policies or educational interventions to prevent WPV [30]. Additionally, the absence of clear definitions of WPV, inconsistent enforcement of rules, and a culture of silence contribute to the normalization of violence [9, 43].

At the policy level, Indonesia has yet to establish a comprehensive framework to prevent and manage WPV in healthcare settings. Current legal instruments, including Law No. 12 of 2022, primarily address sexual violence and do not encompass the broader spectrum of physical, verbal, and psychological violence that nurses encounter [20]. This policy gap highlights the urgent need for a structured WPVEM. International frameworks like NOVA, AWAKE, and the Prevention and Management of Violence in Health Care model provide valuable insights by focusing on preventive education, early risk detection, de-escalation strategies, and organizational readiness [14, 15, 17, 18]. However, these models require contextual adaptation to align with Indonesia's sociocultural and institutional realities.

Successfully implementing a WPVEM in Indonesia relies heavily on strong institutional commitment. Hospital leadership is crucial for emphasizing violence prevention by setting clear goals, allocating sufficient resources, and appointing capable leaders to oversee policy enforcement [28, 30]. A positive organizational climate backed by strong anti-bullying policies is crucial for encouraging nurses to report incidents without fear of blame or retaliation. Additionally, technological tools such as electronic alert systems that enable quick access to security support during violent situations can further enhance protections for frontline staff [44]. Moreover, healthcare workers who report WPV in good faith should be protected by law, and ethics committees ought to be established to defend the rights of both staff and patients [32, 33].

Additionally, occupational medicine is a potentially useful yet underutilized component of WPV prevention in Indonesia [45]. It usually involves risk prevention, hazard detection, post-incident health assessments, and fostering safe work environments. However, its role in addressing WPV is limited because there are no specific rules, institutional mandates, or standardized clinical procedures concerning violence in healthcare [46]. Consequently, responses to WPV tend to be reactive and inconsistent, relying primarily on local managers' judgment rather than coordinated occupational health strategies [11]. Integrating occupational medicine into the WPVEM framework may enhance systematic risk assessments, health monitoring for nurses, documentation of violence outcomes, and cooperation with hospital management on prevention efforts [47]. This integration would help shift the focus from individual blame to a preventive, system-oriented approach that prioritizes worker safety and organizational accountability.

This study acknowledges certain limitations, such as the fact that most included studies are observational, which limits causal inference. Additionally, reliance on published research may have omitted relevant grey literature, and regional variations in reporting suggest that findings may not be universally applicable across Indonesia. Despite these issues, the practical, policy, and research implications are evident. Indonesian healthcare facilities should adopt structured, context-specific WPVEM systems; policymakers need to strengthen legal and reporting frameworks; and future research should evaluate the long-term effects of educational and occupational health initiatives aimed at reducing workplace violence. These measures will help build a safer, more adaptable healthcare system that protects both healthcare workers and patients.

5. Conclusion

WPV in hospitals is not new in healthcare, especially in nursing. The violence occurred regardless of age, gender, position, and other factors. WPV can affect psychological performance during hospital work. There needs to be facilities and institutions to channel and convey problems, especially violence against nurses, be it physical, psychological, sexual, or/racial violence, so that these incidents can be followed up and receive the appropriate punishment according to the level and severity of violence committed against nurses.

Policymakers and stakeholders should prioritize WPV prevention by creating education programs, encouraging incident reporting, and providing access to enhanced OSH training to help reduce violence against health workers. Additionally, facilities and infrastructure must be designed with policies that minimize violence against nurses and foster a safe, supportive work environment. Ultimately, adopting a culturally appropriate WPVEM supported by legal frameworks, digital technologies, and strong leadership is essential to reducing WPV and promoting safe, rights-based healthcare settings in Indonesia.

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Author Contribution Statement: AP conceptualized the study and drafted the manuscript. HK critically reviewed the research concept and contributed to manuscript editing. YY, MM, and AM were responsible for identifying relevant themes and conducting data extraction. All authors reviewed and approved the final version of the manuscript.

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APPENDIX

Supplementary Material: S1

Complete Search Strategy for Scoping Review

1. **Overview**
   This supplementary material includes the full and reproducible search strategy for this scoping review. It aims to thoroughly identify studies on workplace violence (WPV) among nurses in Indonesian hospital settings, following scoping review methodology and PRISMA-ScR guidelines.
2. **Databases and Sources**
   The literature search was conducted between December 5, 2024, and March 15, 2025, across the following sources:
   a. Elsevier (ScienceDirect)
   b. Google Scholar
   c. ResearchGate
   d. National & Institutional repositories (e.g., Garuda, Neliti)
   Additionally, manual searches of the reference lists of included studies were conducted to identify additional relevant articles.
3. **Search Keywords and Boolean Strategy**
   The search strategy was developed using a combination of keywords and Boolean operators based on the Population–Concept–Context (PCC) framework:
   a. Population: Nurse, Nurses, Nursing staff.
   b. Concept: Workplace violence, Occupational violence, Abuse, Aggression, Harassment, Bullying.
   c. Context: Hospital, Healthcare setting, Indonesia.
4. **Full Search Strings (Examples per Database)**
   4.1 Elsevier (ScienceDirect)
   (“workplace violence” OR “occupational violence” OR abuse OR aggression OR harassment OR bullying)
   AND (nurse OR nurses OR “nursing staff”)
   AND (hospital OR “healthcare setting”)
   AND (Indonesia)
   4.2 Google Scholar
   (“workplace violence” AND nurse AND hospital AND Indonesia)
   OR (“kekerasan di tempat kerja” AND perawat AND rumah sakit AND Indonesia)
   Additional filters applied:
   a. Year: 2018–2025
   b. Language: English OR Bahasa Indonesia
   4.3 ResearchGate
   (“workplace violence” OR abuse OR harassment)
   AND (nurse)
   AND (Indonesia hospital)
   4.1 National & Institutional Repositories
   (“kekerasan” OR “kekerasan di tempat kerja”)
   AND (perawat)
   AND (rumah sakit)
   AND (Indonesia)
5. **Search Limits and Filters**
   The following inclusion filters were applied:
   a. Publication years: 2018–2025.
   b. Language: English and Bahasa Indonesia.
   c. Study type: Original research (quantitative, qualitative, or mixed-methods).
   d. Setting: Hospital settings in Indonesia.
   e. Population: Registered nurses.
6. **Eligibility Criteria**
   6.1 Inclusion Criteria
   Studies were included if they met the following criteria:
   a. Original empirical research.
   b. Focused on workplace violence.
   c. Conducted in hospital settings in Indonesia.
   d. Included nurses as the primary population.
   e. Used quantitative, qualitative, or mixed-methods designs.
   f. Available in full-text.

6.2 Exclusion Criteria

Studies were excluded if:

a. The review does not focus specifically on nurses; for example, it considers mixed healthcare workers without separate analysis.
b. It was not conducted in hospital settings and excludes reviews, editorials, theses, conference abstracts, or reports.
c. Full-text versions are not available, and the publications are not written in English or Bahasa Indonesia.

7. **Study Selection Process**

All retrieved records were exported into a structured spreadsheet for screening. The selection process involved two stages: firstly, screening titles and abstracts to exclude irrelevant studies, and secondly, conducting a full-text review to determine eligibility.

This process was carried out independently by two reviewers, with any disagreements resolved through discussion or by involving a third reviewer. The selection process is illustrated in the PRISMA flow diagram included in the main manuscript.

8. **Additional Search Procedures**
   a. Backward citation tracking involved screening the reference lists of included studies.
   b. Manual search included reviewing key journals and relevant Indonesian publications.
   c. Grey literature was considered only in accessible institutional repositories.

9. **Notes on Search Adaptation**

The search strategy was adapted to each database's functionality, including:

a. Simplified keywords are used in Google Scholar, while Boolean operators and phrase searches are employed in ScienceDirect.
b. Repositories feature broader keyword combinations due to limited indexing.

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Combined Occupational Exposure to Lead and Other Metals and Risk of Lung Cancer

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Keywords: Lead; Arsenic; Cadmium; Co-Exposure; Lung Cancer; Interaction

Abstract

Background: *The evidence linking occupational exposure to lead and cancer in humans is suggestive. Existing studies examine individual exposure to lead, ignoring the potential joint effects of mixed metal exposure.* **Methods:** *We analyzed the data of a case-control study of lung cancer conducted in seven European countries and comprising 2,861 cases and 2,936 controls, with a detailed assessment of occupational exposure to arsenic, cadmium, chromium (VI), and lead, to estimate the odds ratio (OR) of lung cancer for combined exposure to lead and these other carcinogenic metals, after adjustment for potential confounders.* **Results:** *The results suggested a synergism between lead and arsenic or cadmium: ORs were around 1.0 for lead alone and above 2.0 for combined exposure. No interaction was suggested between lead and chromium (VI).* **Conclusion:** *Our results provide suggestive evidence of a possible interaction between occupational exposure to lead, arsenic, or cadmium and lung cancer risk, and highlight the importance of considering mixed metal exposures in occupational epidemiology.*

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1. Introduction

Lung cancer is one of the most serious global public health challenges. According to GLOBOCAN 2022 estimates, there were about 2.48 million new cases and 1.82 million deaths worldwide, making it the most commonly diagnosed cancer and the leading cause of cancer-related deaths. Globally, the age-standardized incidence rate (ASIR) was 23.6 per 100,000 people and the mortality rate (ASMR) was 16.8 per 100,000, with higher rates in males than females [1].

Tobacco smoking remains the predominant cause of lung cancer, responsible for approximately 85-90% of cases worldwide [2]. In addition to tobacco smoking, occupational exposures contribute to a measurable fraction of lung cancer incidence, especially in industrial and manufacturing settings [3]. The International Agency for Research on Cancer (IARC) has classified several metal agents as Group 1 carcinogens (carcinogenic to humans) with sufficient evidence for lung cancer, including arsenic, cadmium, and hexavalent chromium [4, 5]. These agents have been linked consistently to elevated lung cancer risk in multiple epidemiological studies of exposed workers [6, 7, 8]. In contrast, inorganic lead compounds are classified as Group 2A (probably carcinogenic to humans), based on limited epidemiological evidence for lung cancer, despite sufficient evidence from experimental and mechanistic studies [9].

In addition, previous epidemiological studies on occupational lead exposure and lung cancer have reported inconsistent findings, with some suggesting a modest increased risk and others showing no significant association after adjusting for smoking and other confounders [10, 11, 12, 13].

Although the association between occupational lead exposure and lung cancer remains uncertain [14], workers are often exposed to complex mixtures of metals and other hazardous substances, while occupational standards typically address single agents [15]. Also, mechanistic evidence indicates that metals may promote carcinogenesis through oxidative stress, DNA repair interference, and epigenetic dysregulation, suggesting that combined exposures could have additive or supra-additive effects [16]. Despite this biological plausibility, epidemiologic studies have typically addressed individual exposures, leaving important gaps in understanding how joint metal exposures influence lung cancer risk [17].

This study focuses on lead as the primary exposure and examines its interaction with arsenic, cadmium, and hexavalent chromium in relation to lung cancer risk, using additive-scale interaction measures such as the relative excess risk due to interaction (RERI) to assess whether co-exposure increases risk beyond the sum of individual effects. It also extends previous analyses [18] of the same dataset by specifically evaluating joint exposure effects particularly those involving lead which had not been assessed in earlier work.

2. Methods

2.1. Study Design and Population

The case-control study was conducted during 1998-2001 in 17 centres from 7 European countries (Czech Republic, Hungary, Poland, Romania, Russia, Slovakia, and UK). Incident, histologically verified cases of lung cancer, including trachea, were included, and controls were frequency-matched based on sex and age (± 3 years). Hospital-based controls were enrolled in all centers, except in Liverpool and Warsaw where population-based controls were selected. Hospital controls were selected from in-patients admitted for a pre-specified list of diseases which exclude other cancers and diseases associated with tobacco smoking and occupational factors. Face-to-face interviews were conducted by trained interviewers using a questionnaire that focused on lifestyle factors and occupations held for more than one year. A general questionnaire was administered for each job, covering tasks performed, machines used, the working environment, and the time spent on each task. A total of 18 specialized questionnaires were available to obtain more detailed information from subjects who had worked in occupations or industries associated with an increased risk of lung cancer. The total interview typically lasted 45-60 minutes.

2.2. Exposure Assessment

The exposure assessment in this study was conducted by industrial hygienists with expertise in the relevant industries, using a standardized protocol [19] derived from the Montréal study [20]. This approach involved evaluating exposure to 70 agents for each job, based on general and, when available, specialized questionnaires, combined with expert judgment informed by field experience. Exposure indices incorporated the experts' confidence in the presence of exposure, as well as its frequency and intensity, and ensured consistency in application through annual workshops and coding exercises. Using this framework, the distribution of occupational exposures observed in the study aligns with previously reported patterns from population-based case-control studies, including Siemiatycki et al. [20] in Montréal, where lead exposure was most frequently identified among mechanics, welders, machinists, painters, and metal-processing workers, while cadmium and chromium (VI) exposures were mainly associated with smelting, welding, and electroplating industries.

2.3. Statistical Analysis

Unconditional logistic regression modelling was used to study the relation between pair-wise combinations of ever-exposure to lead and arsenic, cadmium, or chromium (VI), and risk of lung cancer to calculate odds ratios (OR) and 95% confidence intervals (CI). Nickel exposure was excluded from the interaction analysis because no association was detected with lung cancer risk in this population. All models were adjusted for age (three categories), center, sex, cumulative cigarette smoking (six categories of pack-years), and exposure to the other metals. Smoking pack-years were calculated as the average number of cigarettes smoked per day multiplied by the number of years of smoking, divided by 20.

In each pair-wise analysis, subjects unexposed to lead and each of the other metals comprised the reference category. The interaction between exposure to lead and each of the other metals was tested according to the additive model, by calculating the relative excess risk due to interaction (RERI), and its 95% CI [21]. Different approaches were also explored for the categorization of cigarette smoking, which however did not alter the results on metal exposure. Given the relatively low number of subjects exposed to lead and the other metals, no stratified analyses were performed, e.g., by gender, age of diagnosis, histological type or stage of lung cancer or country. For the same reason, no analyses based on semi-quantitative indices of exposure were performed. We performed a sensitivity analysis, in which occupational exposure to asbestos, crystalline silica, nickel, wood dust and polycyclic aromatic hydrocarbons were also adjusted for. All analyses were performed using STATA package, Version 17.0. The parent study was approved by the relevant Ethics Committees.

3. Results

Table 1 provides details on the study population. Overall, 2,861 cases and 2,936 controls were included in the analysis, with a majority of men (77.1% of cases, 73.7% of controls). The median age at enrolment was 61 for both cases (interquartile range: 53-68) and controls (54-68). As expected, cigarette smoking, especially heavy smoking, was more prevalent among cases: the OR for the categories of pack-years shown in Table 1, compared to never smokers, were 2.70, 6.99, 10.57, 11.86, and 17.70, respectively.

The prevalence of ever-occupational lead exposure among controls was 2.4%, that of arsenic, cadmium, and chromium (VI) were 2.3%, 3.5%, and 8.7%, respectively. Lead exposure prevalence was higher among men (2.9% among controls) than women (0.3%); the main occupations of lead-exposed workers comprise mechanics, machinists, and painters.

The correlation coefficients among controls between exposure to lead, and that to arsenic, cadmium, and chromium (VI) were 0.07, 0.13, and 0.13, respectively (all $p < 0.05$).

The OR of lung cancer for lead exposure was 0.94 (95% CI 0.63-1.40), based on 66 exposed cases and 71 exposed controls. The results of the analysis of co-exposure to lead and the other metals are reported in Table 2. For both combined exposure to lead and arsenic and to lead and cadmium, there was a suggestion of a positive interaction, with RERI non-significantly greater than 0. ORs were

Table 1. Selected characteristics of the study population.

Characteristic		Cases		Controls	
		N	%	N	%
Total		2,861	100.0	2,936	100.0
Sex	Male	2,205	77.1	2,164	73.7
	Female	656	22.9	772	26.3
Country	Czech Republic	304	10.6	453	15.4
	Hungary	402	14.1	315	10.7
	Poland	800	28.0	841	28.6
	Romania	181	6.3	228	7.8
	Russia	600	21.0	600	19.3
	Slovakia	346	12.1	285	9.7
	United Kingdom	228	8.0	234	8.0
Age (years)	25-54	750	26.2	828	28.2
	55-64	1,042	36.4	973	33.1
	65+	1,069	37.4	1,135	38.7
Cumulative Cigarette Smoking (pack-years)	0 (Never Smoker)	274	9.6	1,041	35.5
	0.1-19.9	422	14.7	754	25.7
	20.0-29.9	526	18.4	394	13.4
	30.0-39.9	646	22.6	341	11.6
	40.0-49.9	480	16.8	230	7.8
	50.0+	513	17.9	176	6.0

in the order of 1.0 for exposure to lead alone, and above 2.0 for combined exposure. In the case of co-exposure to lead and chromium (VI), no interaction was suggested (RERI~0). The results of the sensitivity analysis with additional adjustment for exposure to other occupational agents provided results similar to those reported in Table 2. (not shown in detail).

4. Discussion

In this large multicenter case-control study, occupational exposure to lead alone was not associated with an increased risk of lung cancer. However, we observed suggestions of positive interaction on the additive scale between lead and arsenic, and between lead and cadmium. In both instances, the odds ratios for joint exposure exceeded 2.0, whereas the risk estimates for lead alone were close to unity. In contrast, no evidence of interaction was observed between lead and chromium (VI).

Previous studies reported limited or inconsistent associations between occupational lead exposure and lung cancer. Some studies suggested slightly elevated risks, for example, a meta-analysis of occupational studies reported a modest increase in lung cancer risk (RR = 1.29) among workers with high

Table 2. Odds ratio of lung cancer for exposure to lead and another metal.

Combination of metal exposure	Cases		Controls				RERI	
	N	%	N	%	OR	95% CI	Value	95% CI
Arsenic And Lead								
Neither	2,706	94.58	2,802	95.44	1.0	Ref.		
Only Arsenic	89	3.11	63	2.15	1.28	0.87-1.89		
Only Lead	55	1.92	67	2.28	0.87	0.57-1.33		
Both	11	0.38	4	0.14	2.46	0.68-8.84	1.30	-1.85,4.45
Cadmium And Lead								
Neither	2,663	93.08	2,770	94.35	1.0	Ref.		
Only Cadmium	132	4.61	95	3.24	1.24	0.89-1.73		
Only Lead	45	1.57	62	2.11	0.81	0.52-1.27		
Both	21	0.73	9	0.31	2.18	0.88-5.41	1.13	-0.87, 3.13
Chromium (VI) And Lead								
Neither	2,487	86.93	2,631	89.61	1.0	Ref.		
Only Chromium (VI)	308	10.77	234	7.97	1.20	0.97-1.49		
Only Lead	40	1.40	51	1.74	0.94	0.58-1.52		
Both	26	0.91	20	0.68	1.11	0.56-2.20	-0.03	-0.91, 0.85

Reference category: for each analysis, subjects unexposed to both metals.
OR, odds ratio, adjusted for center, sex, age, cumulative tobacco smoking.
CI, confidence interval.
Ref, reference category.
RERI, relative excess risk due to interaction.

lead exposure, although confounding by smoking and other factors was possible [10]. Cohort studies among lead-smelter workers also found elevated lung cancer incidence (SIR = 3.1) in highly exposed subgroups [11], and follow-up of workers biologically monitored for blood lead levels showed increasing risk with higher exposure [12]. In contrast, large case-control studies, such as those conducted in Montreal, did not find positive associations after adjustment for confounders [13].

In contrast to lead, arsenic, cadmium, and chromium (VI) are well-established lung carcinogens [4, 5]. Previous analyses from this multicenter study have confirmed that exposure to these metals increases lung cancer risk [19]. Although research on metal interactions is limited, an analysis of the SYNERGY pooling project, which comprises this dataset, reported that combinations of established lung carcinogens may further elevate risk, however lead was not considered [17].

Experimental and mechanistic evidence supports potential synergistic effects of co-exposure to metals such as lead, arsenic, cadmium, and chromium (VI). Arsenic and cadmium are known to induce oxidative stress, genomic instability, and epigenetic alterations, mechanisms broadly implicated in metal-induced carcinogenesis [22, 23]. Lead, although less clearly established as a human lung carcinogen, has been shown in experimental systems to interfere with DNA repair and promote oxidative damage [24, 25]. Combined exposure can amplify these effects, enhancing oxidative stress, disrupting signaling pathways such as MAPK/ERK, and inducing epigenetic dysregulation more strongly than exposure to individual agents [26]. Metals may also influence each other's toxicokinetics, including absorption, distribution, and intracellular binding, which can modify the effective dose at target tissues [27]. In our data, the low correlation coefficients between lead and other metals suggest that observed joint effects are unlikely to be solely due to co-occurrence of exposures, supporting the possibility of a true biological interaction. Mechanistic reviews further highlight that metals can converge on common molecular targets, including DNA repair systems and epitranscriptomic regulation, plausibly amplifying carcinogenic processes when co-exposure occurs [28, 29]. Our observation of a positive relative excess risk due to interaction (RERI) for lead with arsenic and cadmium, though not statistically uncertain, aligns with this framework. The absence of interaction between lead and chromium (VI) may reflect differences in predominant mechanisms, exposure patterns, or limited statistical power.

The strengths of this study include its large sample size, multicenter design across seven European countries, and detailed expert-based occupational exposure assessment using a standardized protocol [20]. All models were adjusted for detailed categories of cumulative cigarette smoking, the main risk factor for lung cancer. However, residual confounding by smoking cannot be fully ruled out, given its strong association with lung cancer in this population and the small numbers in some co-exposure groups. Although smoking was modeled in six pack-year categories, alternative categorizations produced similar results, indicating that findings were not driven by how smoking exposure was defined. Pack-years were calculated as packs per day multiplied by years smoked, with the highest category (≥50 pack-years) reflecting heavy long-term exposure. Additional sensitivity analyses adjusting for other occupational carcinogens gave similar results.

Several limitations should be acknowledged. First, the prevalence of joint exposure to lead and the other metals was low, resulting in small numbers in the co-exposed categories and consequently wide confidence intervals around both the odds ratios and the RERI estimates. This imprecision limits the strength of the conclusions regarding interaction and warrants cautious interpretation of the observed patterns. Despite this, the use of additive-scale measures such as the relative excess risk due to interaction (RERI) remains informative from a public health perspective, as it captures the excess risk attributable to joint exposure beyond the sum of individual effects and helps to identify combinations of exposures that may contribute disproportionately to disease burden. Second, exposure assessment was based on expert evaluation rather than quantitative measurements, and non-differential misclassification may have attenuated risk estimates. Third, although we adjusted for major occupational carcinogens, residual confounding by unmeasured workplace exposures cannot be excluded. Finally, the hospital-based design in most centers may have introduced selection bias, although the exclusion of patients admitted for tobacco-related diseases among controls mitigates this concern, as cases and controls may still have had smoking-related comorbidities (e.g., COPD), and it is unclear how selection bias would have distorted the observed interaction between lead and arsenic/cadmium, rather than the main effects of each metal.

5. Conclusion

In conclusion, in this large multicenter study, occupational lead exposure alone was not associated with an increased risk of lung cancer. We observed suggestive evidence of an interaction between lead and arsenic or cadmium, with joint exposure patterns consistent with an excess risk beyond that expected from individual effects, although estimates were imprecise due to the limited number of co-exposed subjects. No evidence of interaction was observed between lead and chromium (VI), suggesting that potential combined effects may differ across metal pairs. These findings should be interpreted as exploratory and hypothesis-generating, but are consistent with experimental evidence proposing synergistic mechanisms among certain metals. While the low prevalence of co-exposure limited statistical power, the observed patterns warrant further investigation. Larger studies with quantitative exposure assessment are needed to better clarify potential interactions and support occupational risk assessment for mixed-metal exposures.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Declaration of Interest: PB acted a private expert consultant in litigation involving exposure to metals and risk of lung cancer, unrelated to the present research. Other authors declare no conflicts of interest.

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A Portable Heat-Exchange-Based System for Assessing Digital Vascular Response to Cold

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Keywords: Finger Blood Flow; Strain-Gauge Plethysmography; Heat Exchange; Local Cooling; Peripheral Perfusion; Hand-Arm Vibration Syndrome; Vibration-Induced White Finger; Raynaud's Phenomenon; Occupational Health Surveillance

Abstract

Background: *Digital vascular assessment is relevant in occupational medicine, but current measurement methods may be complex and poorly portable.* **Objective:** *To validate a portable heat-exchange-based system for assessing cold-induced digital vascular response, using strain-gauge plethysmography as the reference method.* **Methods:** *Twenty healthy volunteers, 10 males and 10 females, underwent five subsequent sessions. Finger blood flow (FBF) was assessed by strain-gauge plethysmography, while simultaneously, fingertip thermal power (TP) was recorded with a portable thermoflowmeter. FBF was measured at baseline, while both FBF and TP were measured during application of 1-N contact force alone and during 1-N force combined with local cooling at 10 °C. The relationship between TP and FBF, after adjustment for covariates (age, sex, body mass index (BMI), and room temperature), was evaluated in the cooled condition using a linear mixed model including fixed and random effects.* **Results:** *Local cooling induced a significant vascular response, with FBF decreasing from 1.23 to 0.98 ml/100 ml/s ($p<0.0001$). In the cooled-condition model, TP was significantly associated with FBF, with each 1 ml/100ml/s increase in FBF corresponding to a 39.6 mW increase in TP ($p<0.0001$). Under cooling, TP showed lower within-subject variability than FBF (wsCV ≈0.19 vs ≈0.64). BMI was positively associated with TP (4.40 mW per kg/m^2, $p=0.007$), whereas age and sex were not significant predictors.* **Conclusions:** *These findings support the interpretation of TP as a perfusion-related thermal descriptor with potential applicability in occupational health surveillance. Further validation in subjects with digital vascular disorders is needed.*

1. Introduction

Exposure to hand-transmitted vibration represents a significant occupational hazard for the development of digital vascular disorders, such as vibration-induced white finger (VWF), which represents a secondary form of Raynaud's phenomenon [1]. Evidence of this phenomenon appeared as early as the 1900s, when Loriga (1911) and then Hamilton (1918) observed cases of digital pallor among workers exposed to pneumatic hammers, highlighting the role of mechanical and thermal

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stimuli in the genesis of vascular disorders [2]. In the following decades, several epidemiological and experimental studies demonstrated a strong association between exposure to hand-transmitted vibration and cold exposure and digital vasospasm [3]. VWF has a multifactorial pathogenesis, characterized by recurrent episodes of digital pallor, pain, and tingling, typically triggered by cold exposure. These manifestations result from vascular injury caused by repeated hand-transmitted mechanical stress, and are with increased autonomic nervous system activity and dysregulated vascular tone [1, 3-5]. Cold exposure acts as a powerful vasoconstrictor, which reduces blood flow and precipitates ischemic episodes [4, 6]. In occupational medicine, objective assessment of digital vascular function (e.g., measurement of finger systolic blood pressure with cold challenge by plethysmography, or digital thermal monitoring) is crucial as a complementary assessment in the early diagnosis of vibration-induced vascular disorders and to identify countermeasures. Although several validated instrumental techniques are available [7], many are difficult to use in routine practice. Strain-gauge plethysmography (SGP) is a well-established non-invasive method for measuring finger systolic blood pressure during cold provocation in vibration-exposed workers [8, 9] and for monitoring changes in finger blood flow induced by vibration and/or cold in experimental settings [10, 11]. Nevertheless, the use of SGP in clinical practice is limited by practical constraints, including the need for tightly controlled environmental conditions, technical complexity, long test duration, and limited portability. As a result, its application is mostly confined to specialized settings.

In this context, developing a new tool for occupational physicians to use during workplace health surveillance could be of great interest.

The present study aimed to provide a preliminary validation of a portable device that measures heat exchange between the fingertip and a cold surface to assess the digital vascular response to cold. The physiological rationale for this approach is based on the role of blood perfusion in heat transport from the core of the body to peripheral tissues. When a fingertip is placed in contact with a colder surface, the measured heat exchange depends not only on the imposed temperature gradient and the thermal properties of the skin-surface interface, but also on the amount of heat delivered to the fingertip by local circulation. Experimental evidence indicates that digital blood flow influences the skin cooling response during contact with cold surfaces, with different thermal responses observed under arterial occlusion and vasodilated conditions [12]. Therefore, thermal power measured by a thermoflowmeter is considered a potentially perfusion-related thermal descriptor and an indirect measure of digital blood flow.

2. Materials and Methods

2.1. Study Population

The study was conducted on a sample of 20 healthy volunteers (10 males and 10 females), aged 18 to 46 years, recruited among university students and hospital staff. Inclusion criteria required the absence of known cardiovascular or metabolic diseases. Individuals who were current or former smokers were excluded due to the well-established effects of smoking on peripheral vascular function. All participants provided written informed consent before inclusion in the study. The experimental procedures were conducted in accordance with the Declaration of Helsinki and approved by the ethical committee of Politecnico di Milano (auth. 41/2025).

2.2. Measurement Systems

Two measurement systems were used to assess digital vascular function: a strain-gauge plethysmograph (HVLab) for measuring finger blood flow (FBF), and a thermal power (TP) measurement system (thermoflowmeter, hereafter referred to as TFM) designed to quantify heat exchange between the fingertip and a temperature-controlled surface. The study aimed to investigate the relationship between TP and FBF under controlled experimental conditions and to assess the feasibility of using thermal measurements as an indirect indicator of peripheral perfusion.

2.2.1 Strain-Gauge Plethysmography (HVLab)

FBF was assessed by venous occlusion strain-gauge plethysmography, a reference method for the quantitative evaluation of peripheral circulation. An indium-in-silicone strain-gauge was placed circumferentially around the distal phalanx of the right index finger, while a pneumatic cuff positioned at the base of the finger was inflated to approximately 60 mmHg to occlude venous outflow while maintaining arterial inflow. Under these conditions, arterial inflow produces a progressive increase in finger volume, which is detected as a change in the strain-gauge's electrical resistance. FBF was estimated from the initial linear slope of the volume increase following cuff inflation, according to the venous occlusion principle described by Greenfield et al. [13]. Blood flow values are expressed as ml/100 ml/s.

2.2.2 Thermal Power Measurement System (TFM)

Digital perfusion was also evaluated using a system that measured TP exchange between the fingertip and a temperature-controlled surface. The sensing unit consists of a thermally conductive metal plate whose temperature is actively regulated by a thermoelectric module controlled via a proportional-integral feedback loop based on resistance temperature detectors. When the fingertip is placed in contact with the surface, heat is transferred from the tissue to the plate by conduction. The system measures the rate of heat transfer exchanged between the finger and the cooled interface, which is influenced by local blood perfusion [12]. A load cell integrated beneath the contact surface continuously measures the force applied by the fingertip. Monitoring contact force is essential because it affects thermal contact resistance and the effective contact area, making it a major source of variability independent of physiological changes [14]. The system continuously records TP (mW) and applied force (N) during the measurement period, allowing the assessment of both transient and steady-state thermal responses.

2.3 Experimental Protocol

All measurements were performed in a temperature-controlled laboratory environment. Upon arrival, participants underwent a 15-minute acclimatization period in a seated position. Anthropometric parameters (height and weight), brachial blood pressure, and baseline finger skin temperature (FST) were recorded before the experimental session. Brachial systolic and diastolic blood pressures were measured in the left upper arm using an auscultatory technique. FST was measured using K-type thermocouple at the beginning of the test: participants pinched one end of the thermocouple between their thumb and index finger. The room temperature (RT) was measured by the same thermocouple, which was placed on the table where the tests were performed. Participants were seated comfortably with their right hand positioned at heart level. FBF and TP measurements were performed on the right index finger. The positioning of the finger on the TFM surface for the acquisition of force and thermal signals, together with the plethysmographic setup, is shown in Figure 1 (left panel).

Each participant underwent a standardized experimental protocol consisting of three consecutive phases, performed within a single session and repeated on five separate days:

1. Baseline: No external force or thermal stimulus was applied. FBF was measured using strain-gauge plethysmography only.
2. Force (1-N): participants were asked to apply a target force of approximately 1-N on the thermoflowmeter surface at room temperature. During this phase, FBF and TP were recorded simultaneously.
3. Force + cold (1-N, 10 °C): participants were asked to apply a target force of approximately 1-N while the finger was in contact with the temperature-controlled surface set at 10 °C. Both systems recorded data simultaneously.

Each phase lasted 100 seconds. During each phase, FBF was measured by repeated venous occlusion cycles (10 measurements over 100 seconds), each including cuff inflation to 60 mmHg followed by recording of the fingertip volume increase. At the same time, the TFM continuously recorded TP (mW) and the applied force (N) throughout the entire phase. A schematic representation of the experimental protocol is provided in Figure 1 (right panel).

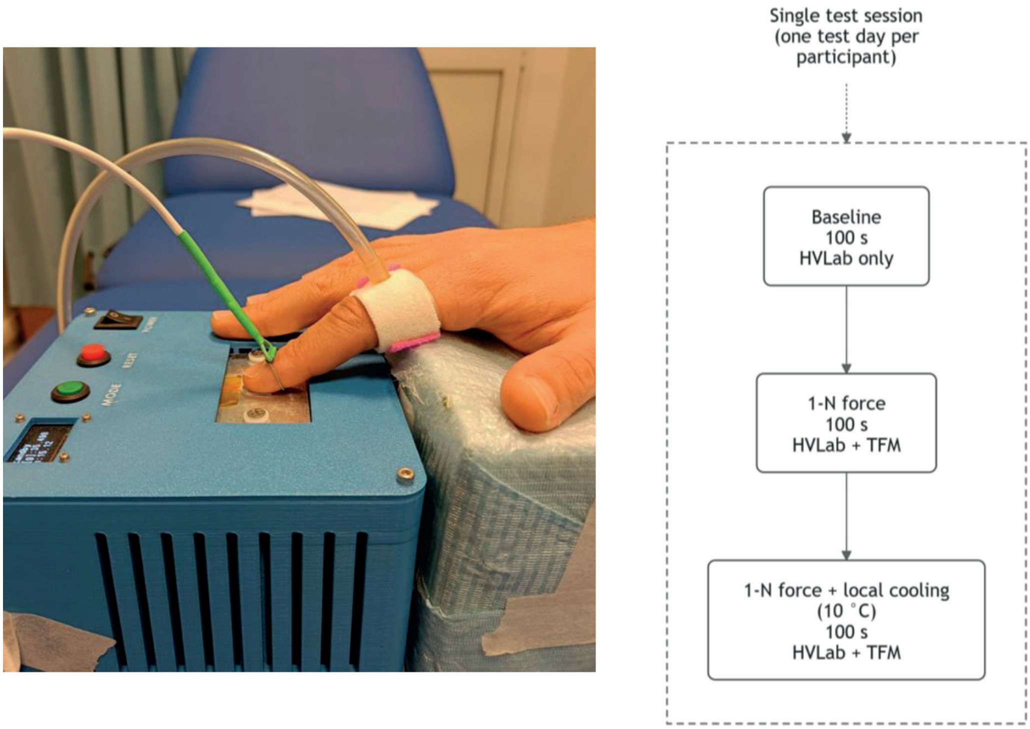

Figure 1. Left panel: positioning of the participant's right index finger on the TFM surface for simultaneous acquisition of contact force and TP; FBF was measured simultaneously using the strain-gauge plethysmograph (HVLab), with the strain-gauge positioned around the right index finger. Right panel: schematic representation of the experimental protocol during a single test session, consisting of three consecutive phases: baseline (HVLab only), application of a 1-N contact force (HVLab + TFM), and combined 1-N force with local cooling at 10 °C (HVLab + TFM). Each phase lasted 100 seconds.

2.4 Statistical Analysis

Statistical analyses were performed using the Stata software (version 19 SE; StataCorp, College Station, TX, USA). Continuous variables were summarized as means, standard deviations (SD), or 95% confidence intervals (95% CI). For FBF, the value used for each participant, session, and experimental condition was the mean of the 10 consecutive plethysmographic measurements acquired during each 100-second phase. For TP, the mean value was calculated from all samples recorded during the same period. Between-group comparisons (males *vs* females) for baseline characteristics were made using an unpaired Student's *t*-test. Within-subject comparisons between experimental conditions were assessed using a paired Student's *t*-test. A p-value <0.05 (two-sided) was set as the limit of statistical significance. The variability and repeatability of repeated measurements over the five consecutive experimental days were evaluated by calculating the repeatability coefficient (RC) and the within-subject coefficient of variation (wsCV), according to the Bland and Altman approach [15]. The relationship between TP and selected covariates, including age, sex, body mass index (BMI), RT, and FBF, was investigated using linear mixed-effects regression models with robust standard errors (rSE) to account for repeated measurements within subjects. The models included a random-intercept term at the subject level and a random slope on FBF. Model performance was assessed using R-squared values to estimate the proportion of variance explained by fixed and random effects.

3. Results

The individual characteristics of the study participants are reported in Table 1. Males and females were comparable in age and BMI, whereas males were significantly taller and heavier than females. Environmental and baseline physiological conditions were stable across the five experimental sessions. In the total sample, RT averaged 24.5°C (95% CI: 24.3-24.8). Similarly, baseline FST averaged 31.0°C (95% CI: 30.3-31.8), with limited variability (wsCV ≈0.097) and no significant sex-related differences (p=0.334).

FBF measurements under the different experimental conditions are summarized in Table 2. No significant differences were observed between baseline and the 1-N force condition in the total sample (mean 1.22 vs 1.23 ml/100 ml/s; p=0.90) or when stratified by sex (males: p=0.22; females: p=0.38). Within-subject variability was similar across experimental conditions. In the total sample, wsCV values were 0.59, 0.61, and 0.64 for baseline, 1-N force, and 1-N force plus cold conditions, respectively, with comparable repeatability coefficients. Exposure to local cooling was associated with a reduction in FBF. In the total sample, FBF decreased from 1.23 ml/100 ml/s (95% CI: 1.05-1.40) during the 1-N force condition to 0.98 ml/100 ml/s (95% CI: 0.83-1.13) during the combined force and cold condition (p<0.0001). FBF reduction was observed in both males and females.

TP results are presented in Table 3. In the total sample, TP increased markedly from 64.8 mW (95% CI: 54.0-75.6) under the 1-N force condition to 243.0 mW (95% CI: 230.6-255.4) during combined force and cold exposure (p<0.0001). TP increase was consistent across sexes and was associated with a reduction in within-subject variability under cold conditions (wsCV ≈0.19 vs ≈0.58 in the force-only condition), indicating improved repeatability when a thermal stimulus was applied.

The crude relationship between TP and FBF is illustrated in Figure 2. Linear regression analysis showed that TP increased with FBF according to the following fitted model: TP = 201.3 + 42.1(FBF). The association was statistically significant (χ^2(*df* 1) = 28.5, p < 0.0001) and explained a substantial proportion of the variance (R^2 = 0.56), indicating a moderate-to-strong relationship between the two variables. The multivariable relationships of TP with individual, physiological, and environmental factors, investigated using linear mixed-effects models, are shown in Table 4. In the full model, FBF remained a significant predictor of TP (39.6 mW per 1 ml/100 ml/s increase in FBF, rSE = 7.52, p < 0.0001). Sex was not significantly associated with TP (p = 0.84). In sex-stratified analyses, FBF was significantly associated with TP in both males and females, although with different estimated regression coefficients (36.6 mW per 1 ml/100 ml/s increase in FBF for males; 56.4 mW per 1 ml/100 ml/s increase in FBF for females; both p < 0.0001). Among the other covariates, BMI was positively associated with TP (4.40 mW per kg/m^2, p = 0.007), while RT showed a weaker positive effect (6.71 mW per °C, p = 0.064). Age was not significantly associated with TP. The full model explained a substantial proportion of the variance in TP (R^2 = 0.68), with contributions from the fixed effects (R^2 ≈ 0.51), the random intercept (≈ 0.10), and the random slope (≈ 0.06).

Table 1. Characteristics of the study participants. Data are given as means (SD).

	Females (n=10)	**Males (n=10)**	**Total sample (n=20)**
Age (yr)	34.6 (7.6)	35.6 (6.2)	35.1 (6.8)
Height (cm)	164 (5.4)	180 (9.7)^b	172 (11.1)
Weight (kg)	64.1 (12.5)	76.7 (12.1)^a	70.4 (13.6)
BMI (kg/m^2)	23.9 (4.6)	23.7 (2.9)	23.8 (3.7)

Males vs females: ^{a}*p=0.034;* ^{b}*p=0.0003.*

Table 2. Finger blood flow (FBF, ml/100 ml/s) measured over five consecutive days under baseline conditions, during application of a 1-N contact force, and during 1-N force combined with local cooling (10°C) in healthy subjects. The repeatability coefficients (RC) and the within-subject coefficients of variation (wsCV) for FBF are reported.

	FBF (ml/100 ml/s)		
Subjects	**Mean (95% CI)**	**RC (95% CI)**	**wsCV (95% CI)**
Males (n=10, obs=50)			
Baseline	1.13 (0.88-1.37)	2.10 (1.72-2.69)	0.67 (0.55-0.86)
1-N force	1.26 (1.02-1.50)	2.40 (1.97-3.08)	0.69 (0.50-0.88)
1-N force+cold 10°C	1.09 (0.87-1.32)[a]	2.15 (1.77-2.75)	0.71 (0.50-0.93)
Females (n=10, obs=50)			
Baseline	1.31 (1.04-1.59)	1.87 (1.54-2.40)	0.52 (0.42-0.66)
1-N force	1.20 (0.93-1.46)	1.65 (1.35-2.11)	0.50 (0.27-0.72)
1-N force+cold 10°C	0.88 (0.67-1.08)[b]	1.24 (1.01-1.58)	0.51 (0.27-0.75)
Total sample (n=20, obs=100)			
Baseline	1.22 (1.04-1.40)	1.99 (1.72-2.36)	0.59 (0.51-0.70)
1-N force	1.23 (1.05-1.41)	2.06 (1.79-2.44)	0.61 (0.45-0.76)
1-N force+cold 10°C	0.98 (0.83-1.14)[b]	1.75 (1.52-2.08)	0.64 (0.47-0.82)

95% CI: 95% confidence intervals.
Paired t-test (1-N force vs 1-N force+cold 10°C): [a]p=0.017; [b]p<0.0001.

Table 3. Thermal power (TP in mW) measured over five consecutive days in healthy males (n=10, obs=48) and females (n=10, obs=50) during exposure of the right index fingertip to 1-N push force alone or combined with local cooling to 10°C. The repeatability coefficients (RC) and the within-subject coefficients of variation (wsCV) for TP are reported.

	TP (mW)		
Subjects	**Mean (95% CI)**	**RC (95% CI)**	**wsCV (95% CI)**
Males (n=10, obs=48)			
1-N force	59.8 (42.8-76.8)	116.0 (93.2-153.3)	0.68 (0.28-1.08)
1-N force+cold 10°C	247.5 (229.6-265.5)[a]	142.0 (114.2-187.8)	0.20 (0.15-0.26)
Females (n=10, obs=50)			
1-N force	69.7 (55.7-83.6)	81.4 (66.2-105.7)	0.47 (0.27-0.67)
1-N force+cold 10°C	238.6 (220.9-256.3)[a]	119.0 (97.4-151.7)	0.18 (0.13-0.22)
Total sample (n=20, obs=98)			
1-N force	64.8 (54.0-75.6)	99.1 (84.9-119.1)	0.58 (0.37-0.78)
1-N force+cold 10°C	243.0 (230.6-255.4)[a]	130.0 (111.4-154.8)	0.19 (0.16-0.23)

95% CI: 95% confidence intervals.
Paired t-test (1-N force vs 1-N force+cold 10°C): [a]p<0.0001.

4. Discussion

The present study aimed to evaluate the feasibility and physiological relevance of a thermal power-based approach for assessing peripheral digital perfusion, using strain-gauge plethysmography as the reference method. The findings provide insight into both the robustness of the measurement protocol and the physiological interpretability of the thermal signal under controlled experimental conditions.

A preliminary aspect concerns the comparison between the baseline and the 1-N force condition. From a strictly physiological perspective, this comparison was not intended to reveal differences in vascular response, as no specific stimulus was

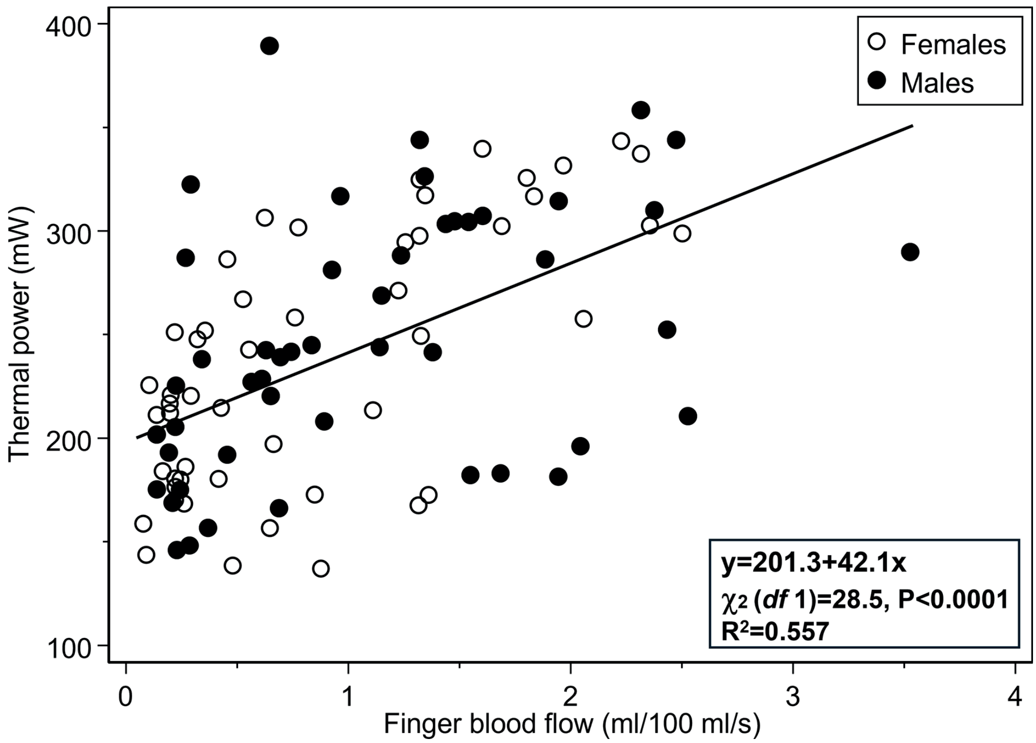

Figure 2. Association between thermal power (TP, mW) and finger blood flow (FBF, mL/100 mL/s) measured over five sessions in healthy males (n=10, observations=48) and females (n=10, observations=50) during exposure of the right index fingertip to a 1-N contact force combined with local cooling at 10 °C. The fitted line represents the crude regression of TP on FBF; the corresponding equation, test statistic, and R^2 are shown in the figure.

introduced. Rather, its primary purpose was methodological, namely, to validate the measurement protocol. Specifically, this test was designed to verify that the contact between the fingertip and the device surface, required for thermal measurements, did not interfere with the plethysmographic assessment of FBF, and that the selected force level was appropriate. The results support this assumption: FBF mean values were almost the same in the two conditions (1.22 vs 1.23 ml/100 ml/s in the total sample), with comparable within-subject variability (wsCV ≈ 0.59 vs 0.61). These findings indicate that a contact force of approximately 1-N does not produce measurable alterations in blood flow or in the strain-gauge signal under the adopted conditions. This is a desirable outcome, as moderate contact forces are required to ensure stable thermal contact while minimizing physiological perturbation. Therefore, the baseline *vs* 1-N comparison provided an internal validation of the experimental setup, suggesting that the measurement interface does not substantially bias measurements obtained under contact conditions.

In contrast, the combined exposure to a 1-N force and local cooling to 10°C produced a statistically significant reduction in FBF of about 20%. This effect was consistent across sexes and repeated sessions, indicating a robust vasoconstrictive response to cold exposure. These findings suggest that the cooling stimulus, even if limited to the fingertip, is sufficient to induce a measurable reduction in blood flow, indicating that digital circulation is highly sensitive to local temperature changes, despite its short duration. The magnitude and consistency of the observed effect support the adequacy of the experimental protocol for inducing physiologically meaningful changes in digital blood perfusion.

TFM measurements showed a marked, approximately fourfold, increase in TP under cooling

Table 4. Linear mixed-effects regression models of thermal power (TP in mW) on individual characteristics, room temperature (RT), and finger blood flow (FBF) in healthy males (n=10, obs=48) and females (n=10, obs=50) during exposure of the right index fingertip to 1-N push force and local cooling to 10°C. The model include a random-intercept term at the subject level and a random slope on FBF. Regression coefficients and robust standard errors (rSE) are adjusted for the effect of repeated measurements over time. The R-squared (R^2) measures for the models fit show the proportions of the total outcome (TP) variance attributed to fixed and random effects in the total sample and by sex.

	Males (n=10, obs=48)	**Females (n=10, obs=50)**	**Total sample (n=20, obs=98)**
Factors	**Coefficients (rSE)**	**Coefficients (rSE)**	**Coefficients (rSE)**
Age (yr)	3.47 (1.99)	1.17 (0.98)	1.29 (1.08)
Gender (M vs F)	-	-	-2.44 (12.28)
BMI (kg/m^2)	8.57 (2.05)^d	2.39 (1.01)^a	4.40 (1.64)^b
RT (°C)	15.3 (5.59)^c	3.08 (3.59)	6.71 (3.62)
FBF (ml/100ml/s)	36.6 (10.5)^d	56.4 (9.23)^d	39.6 (7.52)^d
Constant	180.1 (19.6)	153.4 (10.6)	175.9 (11.6)
R^2 measures			
fixed slopes	0.48	0.67	0.51
random intercept	0.13	0.01	0.10
random slope	0.10	0.09	0.06
whole model	0.71	0.77	0.68
residuals	0.29	0.23	0.32

$^a p=0.018$; $^b p=0.007$; $^c p=0.006$; $^d p<0.0001$.

conditions. This finding is expected based on the physics of heat transfer, as thermal power is directly driven by the temperature gradient between the tissue and the contact surface. Lower interface temperatures increase the gradient and, therefore, the conductive heat flux required to maintain thermal equilibrium. This is consistent with the theoretical framework of the measurement system, in which heat flux depends on both boundary temperature and the overall skin-contact thermal resistance. In addition to increasing signal magnitude, cooling substantially improved measurement repeatability. The wsCV for TP decreased from approximately 0.58 in the non-cooled condition to about 0.19 under cooling, indicating a marked reduction in variability. This improvement can be attributed to the active control of interface temperature, which stabilizes one of the main sources of experimental uncertainty. Moreover, the higher absolute TP values under cooling likely contribute to an improved signal-to-noise ratio. This is consistent with the system's design, which is specifically intended to operate under a controlled cooling regime. Moreover, the wsCV observed for TP in cooled conditions (≈0.19) was lower than that of FBF (≈0.64), suggesting that the thermal measurement may offer better repeatability under controlled boundary conditions.

In the cooled condition, univariate regression analysis showed a positive and statistically significant association between TP and FBF. The fitted model explained a substantial proportion of the variability in TP ($R^2 \approx 0.56$) and confirmed that higher FBF was associated with greater heat transfer during local cooling. When extending the analysis to multivariable linear mixed models with fixed and random effects, the association between TP and FBF remained significant in both the entire sample and after stratification by sex. Nevertheless, in the sex-stratified analysis, the association between TP and FBF was stronger in females than in males. A plausible explanation is that sex-related differences in fingertip skin structure influence heat transfer between the finger and the cooled surface. Previous studies have shown that the stratum corneum of the volar fingertips is thinner in women than in men [16]. The lower epidermal insulation may facilitate heat transfer from deeper, perfused tissues to the contact surface, thereby increasing the sensitivity of TP to changes in FBF. The positive association between BMI and TP is unlikely to reflect a direct effect of localized adiposity at the fingertip. If present, greater local adipose tissue would be expected to increase thermal insulation and potentially reduce, rather than increase, heat transfer to the cooled surface. Therefore, BMI may more plausibly serve as an indirect anthropometric marker of body size, finger morphology, tissue mass, or effective contact area, all of which influence thermal exchange during finger-surface contact [17]. The inclusion of random-intercept and random-slope terms allowed for accounting for both baseline differences between subjects and subject-specific variability in the TP-FBF relationship, leading to a more accurate and flexible estimation of the FBF effect. The persistence of a significant FBF effect after adjustment for covariates confirms its role as an independent predictor of thermal power. Moreover, from a methodological standpoint, these findings highlight the importance of controlling individual and environmental variables when interpreting thermal measurements. Although thermal power is strongly influenced by blood perfusion, it also depends on boundary conditions and tissue-specific properties. Therefore, standardized acquisition protocols and appropriate statistical modeling are essential to ensure robustness and comparability.

In this study, some limitations should be considered. The sample size was relatively small (n = 20), and only healthy subjects were included. In addition, although contact force and temperature were carefully controlled, residual variability related to finger positioning, effective contact area, and initial skin temperature cannot be ruled out.

5. Conclusion

The present study provides preliminary evidence supporting the feasibility and physiological relevance of a thermal power-based approach for assessing digital vascular response to cold. Under controlled experimental conditions, the proposed portable device detected a clear increase in thermal power during local cooling and showed improved repeatability when the cold stimulus was applied. At the same time, local cooling induced a measurable reduction in finger blood flow, confirming that the experimental protocol elicited a physiologically meaningful vascular response. The significant association between thermal power and finger blood flow measured by strain-gauge plethysmography provides preliminary support for the interpretation of TFM-derived thermal power as an indirect measure of digital blood perfusion.

These findings suggest that the TFM may represent a promising, non-invasive, and portable tool for the functional assessment of digital vascular response to cold. Its potential applicability in occupational health surveillance is particularly relevant, as current objective methods for evaluating vibration-induced vascular disorders are often limited by technical complexity, test duration, and poor portability. This study should be considered as an initial validation step. Further studies are needed in larger populations and, mainly, in subjects affected by primary or secondary Raynaud's phenomenon, VWF, or other digital vascular disorders, in order to (i) define diagnostic TFM thresholds, (ii) assess the sensitivity, specificity, and predictive value of the TFM method, (iii) evaluate its reproducibility in clinical and workplace settings, and (iv) ascertain the contribution of the TFM method to the health surveillance of workers exposed to hand-transmitted vibration.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Declaration of Interest: The authors declare no conflict of interest.

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Women's Work and Its Effects on Personal Well-being and Satisfaction With Life: A Cross-Sectional Comparative Study

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Keywords: Life Satisfaction; Subjective Wellbeing; Work-Life Balance; Working women

Abstract

Background: *Women's participation in the workforce has increased substantially, raising important questions about how employment is associated with their subjective well-being and satisfaction with life. This study aimed to assess differences in personal well-being and life satisfaction between working and non-working women and to identify their significant socio-demographic and occupational predictors.* **Methods:** *A cross-sectional comparative study was conducted among 402 women (192 working, 210 non-working) attending outpatient clinics at Mansoura University and primary healthcare centers. Data were collected using an interviewer-administered questionnaire covering sociodemographic factors, occupational characteristics, the Personal Well-being Index–Adult (PWI-A), and the Life Satisfaction Scale(LSS). Bivariate and binary logistic regression analyses identified the independent predictors.* **Results:** *Working women reported significantly higher well-being (PWI-A: 72.06 ± 13.03) and life satisfaction (LSS: 21.52 ± 5.76) than non-working women (PWI-A: 67.42 ± 10.14; LSS: 17.55 ± 5.91). University education (AOR = 2.81) and sufficient income (AOR = 2.94) were strong predictors of higher wellbeing. Subjective well-being (AOR = 21.67) was a significant predictor of life satisfaction. Among working women, professional (AOR = 4.76) and administrative/technical occupations (AOR = 7.40), and absence of shift work (AOR = 6.53) were associated with better wellbeing. Shorter working hours (≤40 h/week; AOR = 14.34) were linked to higher life satisfaction.* **Conclusion:** *In Mansoura, working women exhibit higher subjective well-being and life satisfaction than non-working women. These outcomes were associated with education, income, and occupational conditions. Improving work conditions may contribute to better well-being among working women.*

1. Introduction

Women constitute a substantial proportion of the population in all countries, and their social, economic, and professional participation is increasingly recognized as a key determinant of national development. In recent decades, the rising engagement of women in paid employment has emerged as

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an important demographic and socioeconomic shift. This trend warrants focused investigation, particularly in relation to women's subjective well-being, as their psychological and emotional states may be associated with productivity, organizational performance, and broader societal outcomes [1].

Well-being is a multidimensional concept that extends beyond the absence of illness and encompasses both emotional experiences and effective functioning in daily life. It includes the presence of positive affective states, such as happiness and contentment, alongside the ability to realize personal capacities, exercise autonomy, pursue meaningful goals, and maintain supportive interpersonal relationships [2].

Subjective well-being (SWB) refers to individuals' personal evaluations of their lives, incorporating both emotional reactions and cognitive judgments about overall life quality. As a comprehensive indicator of psychological adjustment, SWB has been consistently associated with favorable health outcomes, increased life expectancy, and stronger social connections [3]. Within this framework, life satisfaction represents the cognitive component of subjective well-being and reflects individuals' reflective appraisals of their lives as a whole. These appraisals are shaped by the perceived balance between personal aspirations and achieved outcomes and serve as an important marker of psychological well-being [4]. Furthermore, subjective well-being encompasses personal strengths and virtues that foster positive emotional states, including feelings of gratitude, fulfillment, and happiness [5].

A range of socio-demographic and occupational factors, including education, income, marital status, and employment status, has been associated with subjective well-being and life satisfaction among working women [5]. Empirical evidence suggests that women who participate in the workforce frequently report higher levels of life satisfaction compared with their non-working counterparts. Employment provides women with financial autonomy, enabling them to independently meet personal and family needs. This economic independence is often linked to enhanced feelings of security, stability, and control over living conditions. In contrast, women who are not engaged in paid work may experience reduced life satisfaction, potentially due to financial dependency and fewer opportunities for personal growth and self-realization [6].

Despite these potential benefits, employment may also introduce additional challenges for women, particularly when combined with traditional domestic responsibilities. The simultaneous management of occupational duties and household roles has been associated with increased stress, which may adversely affect subjective well-being and overall quality of life [7]. Working women often navigate competing expectations as employees, mothers, and spouses, making the reconciliation of work and family demands a persistent personal and social challenge across diverse cultural and organizational contexts [8]. The ability to effectively manage and balance work-related and non-work-related roles plays a critical role in fulfilling life responsibilities and contributes positively to overall subjective well-being [9]. So, it is crucial to explore the effect of women's work on personal well-being and satisfaction with life. To the best of the authors' knowledge, this is the first study to focus on women's work and its effects on personal well-being and life satisfaction among working and nonworking women in Egypt, in general, and in Mansoura, specifically. Thus, this study aimed to compare personal well-being (PWB) and satisfaction with life (SWL) of working vs. non-working women and their associated factors.

2. Methods

2.1. Study Design and Locality

This cross-sectional comparative study was conducted on working and non-working women who were either affiliated with or attend outpatient clinics, Mansoura campus, and Primary health care centers in Mansoura city included Talkha and Met Khamis primary health care centers, from February to April 2025.

2.2. Study Population and Sampling Method

The study population included both working and non-working in the above-mentioned settings. The inclusion criteria were being at least 18 years old, attending the outpatient clinic, or being present on the university campus as an employee, patient's companion, or visitor. In Egypt, the legal working age is 18 years; therefore, only adult women within the official working-age category were eligible for inclusion. The exclusion criteria were being under 18 years of age, pregnant, postpartum, and having chronic physical or mental diseases. A convenient sampling technique was used to recruit women.

A total of 402 women completed the questionnaire, including 210 non-working and 192 working women. While age distribution and residence did not differ significantly between working and non-working women, marital status, education, and income did. Working women were more likely to be single, have higher educational attainment, and have sufficient income (Table 1).

2.3. Sample Size Calculation

Sample size was calculated using Open Epi (https://www.openepi.com/SampleSize/). A pilot study of 35 working and 35 non-working women found that the percentages of life satisfaction were 54.3% and 40.0%, respectively. With alpha error 5%, study power of 80%, then the sample size is 186 per group. However, 402 women (192 working and 210 non-working) completed the questionnaire.

2.4. Study Tools

All study participants completed a pre-designed, interviewer-administered semi-structured Arabic questionnaire to collect the following data:

- Sociodemographic characteristics, including age, educational level, marital status, monthly income, and residence.
- Occupational history for working women, including job title, duration of employment, working hours/day, type of work shift, and working time (full-time or part-time work). Occupations were categorized according to the International Standard Classification of Occupations (ISCO-08) based on broad skill levels. Thus, high-skill occupations (skill levels 3 and 4) included managers, professionals, and associate professionals. Medium-skill occupations (skill level 2) included clerical support, technicians, service, and sales workers.

Table 1. Socio-demographic characteristics of the study groups.

Socio-demographic characteristics	Working Women N (%)	Non-Working Women N (%)	Test and P value
Age			
Mean ±SD	32.91±5.23	32.06±6.95	t=1.37, P= 0.2
≤ 35	128 (66.7)	139 (66.2)	x^2 =0.01, P=0.9
> 35	64 (33.3)	71 (33.8)	
Marital Status			
Single	49 (25.5)	12 (5.7)	x^2 =34.06, P≤ 0.001
Married	115(59.9)	174 (82.9)	
Widow or divorced	28 (14.6)	24 (11.4)	
Residence			
Urban	78 (40.6)	102 (48.6)	x^2 =2.56, P=0.1
Rural	114 (59.4)	108 (51.4)	
Education			
Secondary or Lower	63 (32.8)	160 (76.2)	x^2 =76.41, P≤ 0.001
University or Higher	129 (67.2)	50 (23.8)	
Income			
Not Enough	61 (31.8)	114 (54.3)	x^2 =20.68
Enough	131 (68.2)	96 (45.7)	P=0.004

Finally, low-skilled occupations (skill level 1) included elementary occupations and manual workers [10].

- The Life Satisfaction Scale(LSS) measures the cognitive dimension of life satisfaction and consists of 5 items. The LSS uses seven response options: Strongly Agree, Agree, Neutral, Disagree, and Strongly Disagree. The Arabic version was used [11]. Scoring was performed by summing the responses to all five items, yielding a total score ranging from 5 to 35. Higher scores indicate greater life satisfaction. Specifically, scores between 30 and 35 denote high satisfaction, 25–29 indicate satisfaction, 20–24 reflect slight satisfaction, 15–19 suggest slight dissatisfaction, 10–14 represent dissatisfaction, and 5–9 correspond to extreme dissatisfaction. The cut-off point for life satisfaction was defined as a score of 20 or higher, indicating overall life satisfaction.
- The Personal Well-being Index-Adult (PWI-A) consists of seven domains, each corresponding to a quality-of-life domain comprising standards of living, health, achievements in life, relationships, security, community-connectedness, and future security. It is measured on an 11-point end-defined Likert scale, with numerical ratings ranging from 0 (extremely dissatisfied) to 10 (extremely satisfied). An additional item, administered first, probes participants' satisfaction with their life as a whole. The question about satisfaction with religion or spirituality is optional and not included in the total score of the scale [12, 13]. The Arabic version was used [14]. Scoring was done by summing the responses to the seven core items, each rated on a 0-10 scale (0 = no satisfaction, 10 = very satisfied), and then converting the total raw score to a standard score between 0 and 100. The raw score can range from 0 to 70, which is then converted to a standard score between 0 and 100 by dividing the raw score by 7 (the number of items) and then multiplying by 10. Individual scores on the PWI can be interpreted using the following guidelines [15]: 70+ = 'normal' levels of Subjective Wellbeing, 50–69 = 'compromised' levels of Subjective Wellbeing, and 49 or less = 'challenged' levels of Subjective Wellbeing.

2.5. Data Collection and Recruitment Process

Data were collected through direct, face-to-face interviews over a period of approximately three months (February to April 2025). The interviews were conducted by two of the study authors, both with a background in public health, nursing and community medicine. Prior to data collection, they standardized the interviewing approach to ensure consistency and minimize interviewer-related bias.

Eligible women were approached in waiting areas of outpatient clinics, primary healthcare centers, and public areas within the university campus. The purpose of the study was clearly explained, and confidentiality and anonymity were assured before obtaining informed written consent.

Not all approached women agreed to participate. Some eligible women declined due to personal reasons or lack of time; however, the exact number of refusals was not recorded. A total of 402 women consented to participate and completed the questionnaire, yielding a high response rate. A pilot study was conducted with 35 working and 35 non-working women to assess the clarity and completeness of the tools and to estimate the time required to complete the questionnaire. The results showed that no refinements and modifications were needed. The pilot study was excluded from the sample.

2.6. Statistical Analysis

Data were collected, coded, computed, and analyzed using SPSS software (IBM SPSS Statistics for Windows, Version 25.0. IBM Corp., Armonk, N.Y., USA). Qualitative variables were presented as numbers and percentages. Quantitative variables were tested for normality and described as means and standard deviations (SDs). Appropriate statistical tests of significance were applied. Regression analysis was conducted to identify the independent predictors of SWL and PWB.

Predictors were organized into two main categories to identify factors associated with women's Subjective well-being and Satisfaction with life outcomes: (i) Sociodemographic predictors (e.g., age, marital status, residence, education, and income), and (ii) Work-related predictors (e.g., occupation, working hours, duration of employment, and shift work). In bivariate analyses (for each category), the Chi-square test was used to compare categorical variables, and risk ratios, along with their 95% confidence intervals, were calculated for all predictors.

Significant associations in the bivariate analyses were entered into binary stepwise logistic regression analyses (for each category) to point out the independent predictors of Subjective well-being (measured by PWI-A) and Satisfaction with life (measured by LSS). Adjusted odds ratios (AOR) and their 95% confidence intervals were calculated.

In both bivariate and logistic regression analyses, the primary outcomes—subjective well-being and life satisfaction—were treated as binary (dichotomous) variables, classifying participants into two groups: those with normal or higher well-being/satisfaction and those with compromised or lower well-being /satisfaction. This approach enabled clear interpretation of factors independently associated with a higher likelihood of subjective well-being and/or life satisfaction.

Subjective well-being (PWI-A) and satisfaction with life (LSS) were dichotomized based on validated cut-off points to facilitate clinically meaningful interpretation; therefore, binary logistic regression was applied instead of linear regression, as it provides appropriate estimation of associations (adjusted odds ratios) for categorical outcome variables.

3. Results

Table 2 illustrates that working women had significantly higher mean scores for both subjective well-being PWI-A: 72.06 ± 13.03 vs. 67.42 ± 10.14) and life satisfaction (LSS: 21.52 ± 5.76 vs. 17.55 ± 5.91) compared to non-working women. A normal level of well-being was significantly more prevalent in working women (62.0%) compared to non-working women (43.3%). Similarly, a significantly higher proportion of working women were observed in the 'satisfied' (19.3%) and 'extremely satisfied' (12.0%) categories, compared to 4.3% and 0.5% among non-working women, respectively. (Table 2)

Table 3 shows that education and income were strong independent predictors of subjective well-being. Women with university or higher education (AOR = 2.81; 95% CI: 1.83–4.31) and those reporting enough income (AOR = 2.94; 95% CI: 1.92–4.50) had better well-being outcomes.

Logistic regression analysis revealed that subjective well-being (SWB) was a significant independent

Table 2. Comparison of Satisfaction with Life (LSS) and Personal Well-being (PWI-A) scores among study groups.

Measure	Level	Working Women (n=192) N (%)	Non-Working Women (n=210) N (%)	Test and P value
Subjective Wellbeing Index (PWI-A)	Mean ±SD	72.06±13.03	67.42±10.14	t=4.03, P≤ 0.001
	Challenged Level	7 (3.6)	4 (1.9)	x^2 =17.05, P≤ 0.001
	Compromised Level	66 (34.4)	115 (54.8)	
	Normal Level	119 (62)	91 (43.3)	
Life Satisfaction Scale (LSS)	Mean ±SD	21.52±5.76	17.55±5.91	t=6.81, P≤ 0.001
	Extremely Dissatisfied	1 (0.5)	24 (11.4)	x^2 =68.55, P≤ 0.001
	Dissatisfied	24 (12.5)	32 (15.2)	
	Slightly Dissatisfied	55 (28.6)	55 (26.2)	
	Slightly Satisfied	52 (27.1)	89 (42.4)	
	Satisfied	37 (19.3)	9 (4.3)	
	Extremely Satisfied	23 (12.0)	1 (0.5)	

Table 3. Bivariate and multivariate logistic regression analysis of predictors of Subjective well-being among the study groups (n=402).

	Total	Subjective Wellbeing	Bivariate Analysis		Logistic Regression	
Predictors	**n**	**n (%) [a]**	**Risk Ratio (95%CI)**	**p**	**AOR* (95%CI)**	**P**
Overall	402	210 (52.2)	-	-	-	-
Group					-	-
Working	192	119 (62.0)	1.43 (1.18-1.73)	≤0.001		
Un-working	210	91 (43.3)	r			
Age					-	-
≤ 35	267	136 (50.9)	0.92 (0.77-1.13)	0.46		
> 35	135	74 (54.8)	r			
Residence					-	-
Rural	222	122 (55)	1.12 (0.93-1.36)	0.23		
Urban	180	88 (48.9)	r			
Education					**r**	
Secondary or Lower	223	88 (39.5)	r			
University or higher	179	122 (68.2)	1.73 (1.43-2.09)	≤ 0.001	**2.81 (1.83-4.31)**	**≤ 0.001**
Income						
Not Enough	175	62 (35.4)	r		**r**	
Enough	227	148 (65.2)	1.84 (1.48-2.30)	≤ 0.001	**2.94 (1.92-4.50)**	**≤ 0.001**
Marital Status						
Unmarried	113	57 (50.4)	1.05 (0.84-1.30)	0.65	-	-
Married	289	153 (52.9)	r			

[a] *Row percentage, AOR=Adjusted Odds Ratio, n= Number, r= Reference *Adjusted for age, residence, and marital status.*

predictor of life satisfaction. Women with higher levels of SWB were more likely to be satisfied their lives compared to those with lower SWB (AOR = 21.67; 95% CI: 12.97-36.19) (Table 4)-

Some occupational factors increased the likelihood of higher subjective well-being and satisfaction with life among working women including type of occupation, working hours, and shift work. Women employed in administrative/ technical occupations (AOR = 7.40; 95% CI: 2.69–20.35) or those in professional roles (AOR = 4.76; 95% CI: 1.81–12.55) had significantly higher levels of subjective well-being compared to unskilled and/or skilled manual workers. Conversely, shift work was a significant occupational predictor of lower well-being (AOR = 6.53; 95% CI: 2.83–15.06). Furthermore, working hours per week strongly predicted life satisfaction. Women working 40 hours or less per week were significantly more likely to report higher life satisfaction (AOR = 14.34; 95% CI: 7.09–29.02) compared to those working longer hours. (Table 5)

4. Discussion

The present study highlights the significant differences in subjective well-being and life satisfaction between working and non-working women. It contributes to the literature by exploring how socio-demographic and occupational factors are associated with the well-being of Egyptian women. Our findings indicate that it is not work alone, but also the socio-economic conditions associated with it, that play the most important role in shaping well-being outcomes.

Furthermore, we highlight the influence of occupational conditions—particularly job type, working

Table 4. Bivariate and multivariate logistic regression analysis of predictors of Satisfaction with life among the study groups (n=402).

	Total	Satisfaction with Life	Bivariate Analysis		Logistic Regression	
Predictors	**n**	**n (%) a**	**Risk Ratio (95%CI)**	**P**	**AOR* (95%CI)**	**P**
Overall	402	211 (52.5)	-	-	-	-
Group					-	
Working	192	112 (58.3)	1.24 (1.03-1.49)	0.02		-
Un-working	210	99 (47.1)	r			
Age						
≤ 35	267	143 (53.6)	1.06 (0.87-1.3)	0.55	-	-
> 35	135	68 (50.4)	r			
Residence						
Rural	222	123 (55.4)	1.13 (0.94-1.37)	0.19	-	-
Urban	180	88 (48.9)	r			
Education						
Secondary or Lower	223	104 (46.6)	r		-	-
University or Higher	179	107 (59.8)	1.28 (1.07-1.54)	0.009		
Income						
Not Enough	175	73 (41.7)	r	≤ 0.001	-	-
Enough	227	138 (60.8)	1.46 (1.19-1.79)			
Marital Status						
Unmarried	113	62 (54.9)	1.06 (0.87-1.30)	0.55	-	-
Married	289	149 (51.6)	r			
Subjective Wellbeing						
Normal	210	175 (83.3)	4.44 (3.29- 6.00)	≤ 0.001	**21.67 (12.97-36.19)**	**≤ 0.001**
Abnormal	192	36 (18.8)	r		**r**	

*a Row percentage, AOR=Adjusted Odds Ratio, n= Number, r= Reference; *Adjusted for age, residence, and marital status.*

hours, and shift work—on the well-being of working women. These results highlight the importance of not only enhancing women's access to work but also improving the quality of work environments to support sustainable well-being and life satisfaction.

In the current study, working women reported higher levels of subjective well-being and life satisfaction compared to non-working women (Table 2), which may be explained by the more favorable socio demographic characteristics of working women, including higher education levels and better income status (Table 1) which was similarly reported by other research [9, 16]. These findings suggest that employment provides not only financial stability but also social and psychological benefits that enhance overall wellbeing.

Several studies support the view that employment has positive effects on the mental health of women, primarily through the benefits of job engagement and economic independence. The sense of self-worth, achievement, financial stability, and social recognition associated with employment enables working women to perceive their jobs as rewarding and psychologically fulfilling [17–19]. Moreover, being confined to home has been identified as an important factor contributing to poor mental health among non-working women [20].

On the contrary, some research have reported comparable levels of life satisfaction between working and non-working women, though the underlying determinants differ. Evidence from India indicates that despite the financial independence,

Table 5. Bivariate and multivariate logistic regression analysis of occupational predictors of Subjective well-being and Satisfaction with life among working women (n=192).

	Total	Subjective Wellbeing	Bivariate Analysis		Logistic Regression	
Occupational Predictors	**n**	**n (%) [a]**	**Risk Ratio (95%CI)**	**P**	**AOR (95%CI)**	**P**
Overall	192	119 (62)	-	-	-	-
Occupation [b]						
Unskilled/skilled manual	29	8 (27.6)	r		**r**	**≤ 0.001**
Administrative/technician	75	55 (73.3)	2.66 (1.45-4.87)	≤ 0.001	**7.40 (2.69-20.35)**	**0.002**
Professional	88	56 (63.6)	2.31 (1.25-4.25)	≤ 0.001	**4.76 (1.81-12.55)**	
Working Hours/Week						
≤ 40	106	73 (68.9)	1.29 (1.02-1.63)	0.03	-	-
>40	86	46 (53.5)	r			
Duration of Employment						
≤ 10	135	79 (58.5)	r	0.12	-	-
>10	57	40 (70.2)	1.20 (0.96-1.50)			
Shift Work						
Yes	37	10 (27)	r		**r**	
No	155	109 (70.3)	2.60 (1.52-4.46)	≤ 0.001	**6.53 (2.83-15.06)**	**≤ 0.001**
	Total	**Satisfaction with Life**	**Bivariate Analysis**		**Logistic Regression**	
Occupational Predictors	**n**	**n (%) [a]**	**Risk Ratio (95%CI)**	**P**	**AOR (95%CI)**	**P**
Overall	192	112 (58.3)	-	-	-	-
Occupation [b]			r			
Unskilled/skilled manual	29	11 (37.9)				
Administrative/technician	75	55 (73.3)	1.93 (1.19-3.14)	≤ 0.001	-	-
Professional	88	46 (52.3)	1.38 (0.83-2.29)	0.18	-	-
Working Hours/Week						
≤ 40	106	89 (84)	3.14 (2.19-4.50)	≤ 0.001	**14.34 (7.09-29.02)**	**≤ 0.001**
>40	86	23 (26.7)	r		**r**	
Duration of Employment						
≤ 10	135	78 (57.8)	r			
>10	57	34 (59.6)	1.032 (0.80-1.34)	0.8	-	-
Shift Work						
Yes	37	21 (56.8)	r			
No	155	91 (58.7)	1.03 (0.76-1.41)	0.8	-	-

[a] Row percentage, [b] occupations classified according to the International Standard Classification of Occupations (ISCO–08), AOR=Adjusted Odds Ratio, n= Number, r= Reference

autonomy, and sense of achievement associated with employment, working women often experience greater stress due to the demands of balancing multiple roles [21]. Conversely, non-working women often find fulfillment through family roles, greater involvement in childcare, and opportunities for socialization and personal development [22]. Consequently, despite differing sources of satisfaction, both groups tend to report similar overall levels of life satisfaction.

The current study revealed that subjective well-being (SWB) was a strong independent predictor of life satisfaction (Table 4), with women with higher SWB substantially more likely to report being satisfied with their lives than those with lower SWB (AOR = 21.67). This finding aligns with previous research demonstrating that higher SWB is consistently associated with improved health outcomes, enhanced productivity, stronger interpersonal relationships, and greater overall life satisfaction [23, 24].

A range of socio-demographic and occupational factors were significantly associated with women's well-being and life satisfaction, among which employment status, education, and income play particularly important roles. In the current study, education and income consistently emerged as strong independent predictors of subjective well-being (Table 3). Women with university or higher education were nearly three times as likely to report good well-being as those with secondary or lower education (AOR = 2.81). Similarly, women who reported having enough income had almost threefold higher odds of good well-being (AOR = 2.94). These results emphasize the critical role of socio-economic empowerment in improving women's quality of life and psychological wellbeing. Similar findings have been consistently reported in earlier research linking higher education and adequate income to enhanced well-being and life satisfaction [16, 25-27].

However, previous research has shown some ambiguity regarding the relationship between education and wellbeing. Some studies have argued that educational attainment does not have a significant impact on well-being [28, 29], while others report a positive association between education and individual well-being [30]. Several researchers argue that education by itself does not directly influence happiness but that higher educational attainment enhances income, which in turn improves happiness through access to desired or scarce resources [31]. Despite these conflicting perspectives, a growing body of evidence indicates that individuals with higher educational achievements tend to report greater levels of subjective well-being [26, 32].

In this study, occupational characteristics such as job type, working hours, and shift work were significantly associated with well-being among employed women (Table 5). The current results demonstrate that occupation type is a strong predictor of subjective well-being among working women, even after adjusting for socio-demographic factors. Women in administrative or technical occupations were more than seven times as likely to report good well-being compared to unskilled and/or skilled manual workers (AOR = 7.40), while those in professional roles had nearly fivefold higher odds of well-being (AOR = 4.76). The higher well-being observed among professional and administrative/technical workers may be attributed to the psychosocial and structural advantages inherent in higher-status occupations—such as greater job autonomy, cognitive engagement, decision-making authority, and opportunities for skill development. Moreover, the professional and technical roles typically provide more stable income, access to benefits, and social recognition, all of which are important determinants. These findings emphasize that occupational class and job quality are key determinants of mental health and wellbeing, especially among women balancing multiple roles [25, 33].

In contrast, women in manual or lower-skilled occupations frequently encounter job insecurity, low wages, limited advancement opportunities, and exposure to unsafe or physically demanding environments, contributing to chronic stress and diminished satisfaction [34]. Additionally, gender-based occupational segregation may exacerbate these disparities, as women in manual sectors frequently encounter limited control, lower social support, and fewer opportunities for career progression [35].

On the contrary, shift work was associated with lower well-being; women who did not work shifts were more than 6 times as likely to experience good well-being (AOR = 6.53). This aligns with evidence linking shift work to sleep deprivation, circadian rhythm disruption, and poorer physical and psychological health [36]. Sleep deprivation is known to impair cognitive and emotional functioning, thereby diminishing overall well-being [37].

Furthermore, weekly working hours were strongly associated with higher life satisfaction. Women working ≤40 hours/week were significantly more likely to report higher life satisfaction, with odds approximately 14 times greater than those working longer hours (AOR = 14.34). This finding supports previous research suggesting that shorter working hours promote life satisfaction by allowing better work–life balance [38]. International literature similarly highlights that satisfaction is driven by factors beyond income, including working hours, work–life balance, and social support [39]. Cross-cultural evidence indicates that women's preferences for work hours are influenced by sociocultural norms and role expectations. In many settings, women prefer shorter workweeks due to dual roles in employment and caregiving, whereas, in high-income contexts such as the United States, longer hours are often perceived as a means of achieving economic security and professional advancement, which may be associated with greater subjective well-being and job satisfaction [40].

The interpretation of these findings should be considered within the socio-cultural context of Egypt, where women's roles are shaped by both traditional expectations and evolving economic demands. In settings such as Mansoura, employment may be associated not only with financial benefits but also with greater social recognition and autonomy, which may, in turn, be linked to higher perceived well-being. At the same time, many working women continue to bear primary responsibility for household and caregiving roles, which may contribute to role strain. Conversely, non-working women may derive satisfaction from family roles and social cohesion, but may also experience financial dependency or fewer opportunities for personal development. These contextual factors may partly explain the observed differences between working and non-working women and underscore the importance of interpreting the findings within the local cultural framework.

5. Conclusion and Recommendation

This study demonstrates that employment is positively associated with women's subjective well-being and life satisfaction, primarily through higher education, adequate income, and improved socio-economic status. Beyond financial benefits, employment enhances women's sense of self-worth, autonomy, and social recognition. However, the nature of work significantly shapes these outcomes: women in professional or technical occupations reported greater well-being than those in manual or shift-based jobs, reflecting the importance of job autonomy, security, and supportive work environments. Shorter working hours were also linked to higher life satisfaction, emphasizing the need for balance between professional and family roles. Policymakers and employers should prioritize interventions that enhance job quality, ensure reasonable working hours, and provide job security. Future research should adopt longitudinal designs to confirm causal relationships and explore the long-term effects of employment characteristics and changing work conditions on women's wellbeing.

5.1. Strengths and Limitations

This study possesses several notable strengths. It addresses a research gap by comparing the subjective well-being and life satisfaction of working and non-working women in Egypt—particularly in the understudied context of Mansoura. The use of validated Arabic versions of internationally recognized instruments, namely the Life Satisfaction Scale(LSS) and the Personal Well-being Index–Adult (PWI-A), ensures high reliability and comparability of results. Furthermore, the analysis incorporated a wide range of socio-demographic and occupational variables, offering a comprehensive understanding of the multifactorial determinants of women's wellbeing. The relatively large and statistically well-powered sample further enhances the credibility and generalizability of the findings. Importantly, the study provides practical insights for policymakers, public health professionals, and workplace administrators seeking to improve women's quality of life and mental health through evidence-based interventions.

Despite its strengths, the authors acknowledge that this study has limitations. The cross-sectional design restricts the ability to establish causality between employment-related factors and well-being outcomes. Additionally, the use of a convenience sampling technique may limit the generalizability of the findings to all Egyptian women, as participants were recruited from specific community and institutional settings within Mansoura. Furthermore, significant baseline differences between working and non-working women—particularly in education, income, and marital status—may have influenced the observed associations, as these factors are closely linked to employment status. Although multivariate regression was used to adjust for these potential confounders, residual confounding cannot be excluded. In addition, some adjusted odds ratios were associated with relatively wide confidence intervals, which may reflect small subgroup sizes after dichotomization of outcomes. Therefore, these findings should be interpreted with caution.. The reliance on self-reported data may introduce reporting or recall bias, as responses reflect participants' subjective perceptions rather than objective measures. The assessment of chronic physical and mental illnesses relied on self-report, which may have led to underreporting, particularly for mental health conditions due to stigma or lack of formal diagnosis. Additionally, some participants were recruited as patients' companions or visitors, and their responses may have been influenced by temporary stress related to their companion's health condition, which could have affected subjective well-being and life satisfaction measures. Moreover, the study was geographically confined to Mansoura city and may not capture variations in employment conditions or cultural contexts across other regions of Egypt. Finally, certain potential confounders—such as personality traits, family support, and broader cultural influences—were not assessed. Additionally, some factors that may influence women's well-being—such as the quality of marital relationships, the number of children, the availability of childcare support, and household responsibilities—were not assessed, which may have affected the observed associations.

INSTITUTIONAL REVIEW BOARD STATEMENT: The study protocol was approved by the Institutional Review Board (IRB), Faculty of Medicine, Mansoura University, (code number: R.24.12.2926).

INFORMED CONSENT STATEMENT: Following the assurance of confidentiality and anonymity of data, which are never to be used for any other purpose than scientific research, informed written consent was obtained from participants who agreed to share in the study.

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DECLARATION OF INTEREST: The authors declare no conflict of interest

AUTHOR CONTRIBUTION STATEMENT: HM, AA, AE, SA and EE were responsible for the initial conception and design of the study. HM and EE collected data. AA ran statistical analysis and participated in the interpretation of the data. AE revised the data analysis and interpretation. AA and SA drafted the initial version of the manuscript and all authors were involved in critically revising the manuscript. All authors have reviewed and approved the final version of the manuscript.

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The Impact of Robotization on Employees' Mental and Physical Health: A Qualitative Study in the Czech and Slovak Republics

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Keywords: Occupational Health; Mental and Physical Well-Being; Central Europe; Industry 4.0; Automation; Manufacturing

Abstract

Background: *This study aimed to explore the impact of introducing robotic technologies on employees' mental and physical health across three Czech and one Slovak manufacturing company. It sought to understand employees' subjective experiences and perceptions of organizational support during the robotization process.* **Methods**: *A qualitative study was conducted between 2023 and 2025 in four companies. 27 semi-structured interviews with employees (15 women, 12 men) and 19 unstructured interviews with managers (3 women, 16 men) were conducted on-site during working hours. Data were analyzed using thematic analysis with open and axial coding.* **Results**: *Nine thematic areas were identified: general perceptions, change, fear, problems, support, social relations and communication, employee characteristics, physical health, and mental health. Employees generally perceived robotization positively, noting reduced physical workload and less monotonous or hazardous work. Initial adaptation was challenging, especially due to technical demands and workflow changes. Technical staff experienced higher stress than operators, related to responsibility, troubleshooting, and learning new skills. Technical malfunctions, elevated noise and vibration, and cognitive demands were reported as stressors. Personifying robots and informal knowledge sharing facilitated adaptation. No significant sleep disturbances or substance use linked to robotization were reported. Managers faced a high mental workload due to coordination and supervision responsibilities.* **Conclusions**: *When the implementation process is well designed, robotization does not necessarily negatively affect employees' mental health. The findings highlight the importance of actively addressing organizational and human factors that influence the success of technological change in the workplace. Attention should focus on monitoring cognitive and psychological demands, particularly for technical and managerial staff.*

1. Introduction

The implementation of robotic technologies in the workplace, including both industrial robots and collaborative robots (cobots), is a key characteristic of the so-called Fourth Industrial Revolution (Industry 4.0), which fundamentally reshapes the organization of production processes, management

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systems, and employee requirements [1]. Robotization contributes to increased production efficiency, reduced exposure to risk factors, and fewer workplace injuries [2, 3]. In addition to its effects on physical health, researchers are increasingly focusing on the implications of these changes for employees' psychological well-being [2-4]. The rapid pace of technological change, combined with the need for continuous adaptation, can create uncertainty and stress among workers.

For organizations, robotization represents a profound transformation—often involving not only technological upgrades but also team restructuring, changes in job descriptions, and a shift from routine activities to more technically and mentally demanding tasks [5]. Research shows that such transformations may lead to various forms of psychological stress. Some authors have pointed to elevated stress levels, particularly in connection with robot malfunctions [6], software failures [7], time pressure, and fear of failure [1, 8]. Another important factor is the fear of job loss or demotion, which is especially prevalent among less-skilled workers and may result in frustration or disruption of work identity [9, 10].

On the other hand, some studies [11, 12] highlight potential positive aspects of robotization for mental health, such as reduced workload and monotony, and opportunities for employees to engage in more creative tasks [3], which may enhance job satisfaction and employee engagement. Evidence also suggests [3, 13] that when the introduction of robotic technologies is accompanied by adequate training, accessible technical support, and active communication from management, the negative psychological impacts can be significantly mitigated or even prevented. Furthermore, some research [14, 15] emphasizes the role of organizational culture and individual resilience in shaping employees' perceptions of technological change, suggesting that psychological outcomes are not solely determined by the technology itself but also by the broader social and managerial context.

Global trends indicate that the adoption of industrial robots is accelerating, particularly in Asia and Australia, with continued growth in Europe [16]. These developments highlight the importance of understanding how robotization affects employees and of proactively preparing organizations and their workforce for ongoing technological change [13, 17].

Given the contradictory findings in the literature and the lack of in-depth qualitative analyses in the Central European context, there was a need to capture employees' subjective experiences. A qualitative approach enables a deeper understanding of how workers in different positions perceive robotization, what factors shape their experiences, what helps them cope with change, and where they identify the main risks. By exploring employees' narratives, researchers can identify both systemic issues and practical strategies that organizations can implement to support workers' well-being during technological transitions.

The objective of this study was to explore the impact of introducing robotic technologies on employees' mental and physical health across three Czech and one Slovak manufacturing company. The study draws on qualitative data obtained from interviews, complemented by workplace observations and the authors' professional expertise in occupational safety and health and workplace mental health.

The secondary objective is to identify risk and protective factors that may inform the development of preventive measures to minimize the psychological burden associated with the process of robotization.

2. Methods

This qualitative study was conducted between 2023 and 2025 in four manufacturing companies where various types of robots (collaborative and industrial) had been implemented in the production process. The participating companies represented different manufacturing sectors, including automotive production, industrial assembly, and mechanical engineering, and varied in company size, degree of automation, and type of robotic technologies used. Additional characteristics of the participating companies are presented in Table 1.

Companies across the Czech Republic and Slovakia were invited to participate voluntarily. The inclusion of organizations representing different levels and forms of automation enabled the exploration of a broad range of employee experiences related to robotization. Within participating companies, employees from production, supervisory, and technical positions were invited to participate. The participating organizations coordinated recruitment; however, participation was entirely voluntary, and employees could decline involvement without any consequences. Eligibility criteria required participants to have at least three months of experience working in a robotized work environment.

The final number of participating companies, interviews, and employees reflected not only the purposive sampling strategy but also several practical considerations inherent to workplace-based qualitative research. These included the substantial time required to conduct in-depth interviews, the research team's time and personnel capacity, and the need to minimize disruption to production processes, as interviews were conducted during working hours. Consequently, participant recruitment and interview scheduling were coordinated closely with company management to ensure employees could participate without adversely affecting operational activities.

In total, 27 semi-structured interviews were conducted with employees (15 women and 12 men/14 production operators, 6 shift coordinators (supervisors), and 7 technical staff members). In addition, 19 unstructured interviews were conducted with company managers (3 women and 16 men). The age of participants ranged from 20 to 61 years.

Before data collection, all respondents signed informed consent. The study was conducted in accordance with ethical standards for research involving human subjects. To ensure anonymity, participants' names and other identifying information were removed from the transcripts, and all data were stored securely and accessible only to the research team. Given the non-interventional nature of the study, voluntary participation, and the anonymous collection of data, formal approval by an institutional ethics committee was not required.

Interviews were conducted directly at workplaces during regular working hours to ensure participants' convenience and contextual relevance. Semi-structured interviews with employees followed an interview guide covering topics such as perceived changes in work tasks, physical and mental health, stress, adaptation to robotization, cognitive demands, social relations, training experiences, and organizational support. The interview guide is provided in the Supplementary Materials. Unstructured interviews with managers focused on implementation processes, organizational challenges, employee adaptation, and management experiences related to robotization.

For organizational and confidentiality reasons, participating did companies was not permit audio recording to due of concerns about protecting internal processes and employee privacy.

Two trained researchers conducted each interview. One researcher facilitated the interview by asking the predefined questions, while the second researcher was solely responsible for contemporaneous note-taking. Detailed written notes were recorded throughout the interviews to capture participants' responses as accurately and comprehensively as possible, thereby enhancing the completeness and reliability of the qualitative data. Clarifying questions

Table 1. Characteristics of participating companies.

	Country	Manufacturing Sector	Company Size	Type of Robotization	Employee Interviews	Manager Interviews
1	Czech Republic	Automotive Industry	Large (~600)	Collaborative Robots	8	3
2	Czech Republic	Industrial Assembly	Medium (~160)	Industrial Robots	6	4
3	Slovak Republic	Mechanical Engineering	Large (~900)	Industrial and Collaborative Robots	10	6
4	Czech Republic	Automotive Industry	Large (~2000)	Industrial Robots and Automated Production Lines	3	6

were used during interviews when necessary to verify understanding, and interview notes were reviewed and supplemented immediately after data collection to improve completeness and accuracy.

Each interview lasted between 30 and 50 minutes. Variation in interview length reflected differences in participants' job roles and individual communication styles. Unstructured interviews with managers were generally longer because they addressed broader organizational issues and implementation processes.

The interviews were complemented by direct observations of the work environment and employees' interactions with technologies, conducted by members of the research team during site visits.

Data were analyzed manually using thematic analysis. Written interview records were reviewed repeatedly to ensure familiarity with the data and to identify meaningful units of text. In the first stage, open coding was applied to generate preliminary codes reflecting the key concepts, perceptions, and experiences reported by participants. In the subsequent stage, axial coding was used to examine relationships among codes and to organize them into broader conceptual categories. The initial open coding produced a wide range of preliminary codes, which were then refined, merged, and structured into higher-order categories during the axial coding process.

To strengthen analytical rigor, three members of the research team were involved in the coding process. Individual interviews were initially coded by one researcher and subsequently reviewed by another team member. Coding discrepancies were discussed in research team meetings until consensus was reached. Based on these discussions, the coding categories were further refined and applied consistently across the dataset. Through this iterative process of comparison, discussion, consensus-building, and refinement, the codes were grouped into nine final thematic areas representing the key dimensions of employees' experiences with robotization. An example of the coding structure and code development process is provided in Supplementary Material S2.

The multidisciplinary research team comprised psychologists, a psychiatrist, an occupational physician, and occupational health professionals with expertise in workplace mental health and occupational safety and health. The researchers acknowledged that their professional backgrounds could influence both data collection and interpretation. To minimize individual bias, coding decisions and thematic interpretations were regularly discussed within the research team, and the findings were developed through consensus among researchers with different professional perspectives.

Formal approval by an institutional ethics committee was not required for this study, as it involved qualitative interviews with adult employees conducted as part of workplace research and did not include any medical interventions, the collection of biological samples, or the processing of sensitive personal data beyond participants' voluntary responses. Participation was entirely voluntary; all participants provided informed consent before the interviews, and anonymity and confidentiality were maintained throughout the study. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki.

3. Results

The analysis of the interviews identified nine thematic areas: general perceptions, change, fear, problems, support (training and ongoing assistance), social relations and communication, employee characteristics, physical health, and mental health (including stress, fatigue, cognitive functions, and substance use). To provide greater clarity, the results are presented thematically, with each section summarizing the key findings and supported by selected quotations from employees and managers. These are coded according to the company and participant number, for example, C1P1 refers to Company 1, Participant 1.

3.1. General Perceptions

Employees in all four companies generally perceived robotization positively, particularly because it reduced physical strain, simplified certain tasks, and limited exposure to monotonous, hazardous, or physically demanding environments. Some employees had expected that the introduction of robots would also lead to a reduction in the pace of work; however, the reality was different—performance requirements remained high, or even increased. Despite this, employees reported that robots had become an integral part of the workplace and were generally accepted without significant resistance. As one respondent noted: “From my point of view, it is positive that the work is physically less demanding, and looking ahead, robotization is inevitable as it continues to expand in other forms.” (C3P6).

3.2. Change

The process of introducing robots was described as a period of significant adjustment. As one participant put it: “Nobody likes changes, and nobody initially approaches them positively, but in my experience, these changes have always been for the better.” (C1P5). Employees highlighted the need to adapt quickly to new workflows, learn new technical procedures, and reorient themselves to an altered work environment. The initial adaptation period was generally considered the most demanding phase, both mentally and organizationally. “During the first month, I would have dismantled the robots and taken them back to the office… but later it became a relief for me and especially for my team.” (C1P2). Managers noted that implementing new technologies often required parallel organizational changes, such as redefining job roles and responsibilities.

3.3. Fear

Although most employees did not express strong fears related to working with robots, some concerns were reported. These included worries about making mistakes and the fear of being unable to keep up with technological demands. At the same time, respondents emphasized that they did not perceive robots as a direct physical threat. They trusted the well-established safety measures, adhered to protective instructions, and often described their feelings more as a form of healthy respect rather than fear. As one operator noted: “Well, we have fairly strict safety rules in place. If I follow them, I’m not afraid.” (C3P1). Both employees and managers repeatedly emphasized confidence in established safety procedures, which appeared to reduce concerns regarding physical injury.

3.4. Problems

Technical malfunctions and operational issues were among the most stressful aspects of robotization, particularly during the initial phase of implementation. Employees and managers emphasized that many of the difficulties were not caused by the robots themselves, but rather by inconsistent or unsuitable input materials. In several cases, participants attributed these operational problems to inconsistencies in the characteristics of input components, such as clips and other parts used in automotive manufacturing. Although these components generally complied with manufacturing specifications, even minor variations in their dimensions, shape, or positioning could interfere with the reliable operation of automated production processes. As one employee explained: “The robot itself is quite trouble-free. But the problem is with the bulk material… in my opinion, that is the worst part of robotization.” (C2P5). Another recurring concern was the absence of immediate technical support. One operator noted: “The worst thing is that on the second shift we don’t have a technologist. Some things just cannot be fixed by us; it has to be someone who really understands it.” (C4P2). Frequent troubleshooting increased stress, especially for technical staff who bore responsibility for ensuring continuous operation.

3.5. Support

Statements from both employees and managers indicated that high-quality training, accessible technical support, and a clearly defined *hierarchy* for assistance played a key role in the implementation and use of robots. As one participant noted: “There is technical support, we have manuals, and we have supervisors who are directly assigned to this, and we can communicate with them.” (C4P5). While training was considered important for all employees, it proved particularly demanding for technical staff, who had to master complex problem-solving skills, advanced programming, and the ability to manage unexpected malfunctions under time pressure. Supportive measures also included regular team meetings and informal sharing of experiences.

3.6. Social Relations and Communication

Robotization influenced team dynamics in several ways. In many workplaces, it fostered closer cooperation among operators, technicians, and managers, as employees needed to coordinate more frequently and share experiences to address emerging challenges. Open communication and informal knowledge exchange were described as essential for successful adaptation. As one employee explained: "I go to my colleagues who work before me in the process, and we solve problems together; now we are dependent on each other, so we communicate much more." (C3P2). Interestingly, in some teams, acceptance of the technology was further supported by personifying the robots—for example, by giving them names or even "talking" to them. This practice reduced distance, lightened the atmosphere, and helped employees cope with the stress of working alongside automated systems.

3.7. Employee Characteristics

The characteristics of employees that proved important for managing work with robots varied by position type. For operators, formal qualifications were not decisive; rather, a positive attitude toward technology, motivation to work, manual skills, and the ability to learn new tasks were key. "I still believe that a person must want to do it, must have the will to learn; in my opinion, this work is for everyone." (C4P2). For technical staff, crucial factors included technical competencies, the ability to solve problems under time pressure, and willingness to work in a three-shift system. "Certainly, they should understand programming to know what they are dealing with… And they must not act rashly, because a robot is a robot… Sometimes it is better to do it manually; there is no need to overthink it, one should act with deliberation and calm." (C4P5). In practice, managers emphasized that one of the biggest challenges was not employees' direct work with robots, but rather the recruitment and retention of qualified technical staff.

3.8. Physical Health

Employees consistently reported that robots helped reduce physical workload, limited repetitive strain, and minimized exposure to hazardous environments. Several respondents highlighted that this contributed to lower fatigue and fewer musculoskeletal complaints. In the words of one respondent: "When I started, everyone worked alone at their tables and did everything by hand. If it had always been done like that, I wouldn't still be here. My hands were raw and bleeding. Robotization was a huge relief." (C2P3). Conversely, in some workplaces, employees raised concerns about elevated noise levels and strong vibrations, which negatively affected employees' overall psychological well-being during work. One technician noted: "…the noise and vibrations are enormous. It lasts all day. At the back, where you were, it's fine, but here at the front, where the girls are, you have to shout and compete with the noise, and it's psychologically exhausting. That noise wasn't there before; it used to be much calmer." (C2P5). It should be noted that these observations were based on employees' subjective perceptions and were not verified through objective environmental measurements.

3.9. Mental Health

The greatest psychological demands were observed in the initial phases of robotization, when work processes were being reorganized, and employees had to rapidly adapt to the new system, acquire technical skills, and orient themselves in a changed work environment. As one employee described, "At first, it was a bit stressful until you understood what the robot was going to do." (C4P8). Over time, however, employees reported that the situation improved as they became more familiar with the robots and the new workflows. Most stress was then experienced only in relation to technical problems, such as issues with input materials or malfunctions, rather than routine operations. One respondent noted, "…then you start getting frustrated, or when it happens near the end of the shift (those problems with the clips), it annoys you, and it affects your mental state." (C1P5). It should also be noted that increased noise and vibration levels in some workplaces continued to affect employees' psychological well-being negatively.

The mental health effects of robotization varied by job role. In most cases, operators did not perceive working with robots as particularly mentally or physically demanding, especially when they had received adequate training and had access to technical support. As one operator stated: "A person is not tired and can function even outside of work, at home." (C4P7).

Technical staff described increased stress related to their responsibility for the proper functioning of technologies, constant problem-solving, higher demands on learning, and limited opportunities for substitution by colleagues. "Like any job, working with robots can sometimes be stressful. It creates a certain stress in the worker—in my case, when something doesn't go well, when you can't set something up, or when the robot doesn't perform 100 percent as expected. From a mental perspective, one has to carefully think about what and how to do, which makes it more demanding." (C4P3). Although managers did not explicitly mention experiencing stress, their accounts indicated substantial psychological demands associated with coordinating robotized production processes.

They were responsible for coordinating and supervising the entire robotized production process, ensuring smooth operation, troubleshooting technical issues, managing staff schedules, and maintaining safety standards. In addition, they had to mediate between operators and technical staff, make rapid decisions under pressure, and respond to unexpected disruptions, all of which contributed to a high level of mental workload. Managers also faced considerable challenges in securing qualified employees, particularly technical staff, as there is no specialized education for these roles, and highly skilled individuals are often unwilling to work in a three-shift system.

The cognitive demands of work with robots largely stem from the nature of the tasks themselves. Operators particularly emphasized that working with robots required sustained attention and constant monitoring – a requirement that existed even before robotization. With robots, however, they now have to closely observe the machines' processes, tracking both what the robot has completed and what remains to be done, which increases the level of attention required. "You have to be attentive, you need to watch whether the robot is stuck… it requires greater attention, to observe… to notice whether the robot is moving." (C1P3). Several respondents noted that the novelty of the robotized system initially increased cognitive load, as they had to remember new procedures, sequences, and safety protocols. Over time, as routines became established, cognitive demands decreased. Still, attention remained essential, especially during problem-solving tasks or when managing multiple processes simultaneously, which was particularly demanding for technical staff. The need to quickly detect and respond to errors or malfunctions placed high demands on concentration and working memory. As one employee reflected on the increased cognitive demands of working with robots and how they manifest beyond the workplace: "I think so, ever since I started working with robots, even at home when I do something, I find myself concentrating more, thinking about whether I'm doing it the right way or another way. It's like the focus I learned at work has carried over, and I find myself thinking more about everything I do." (C3P3).

Employees did not report any negative impact on sleep or the development of addictive behaviors. "For me, it's more about the three-shift system and the resulting irregular sleep. I don't think it's really about the robots, but rather about fatigue." (C1P1).

The implementation of robotic systems can mitigate certain occupational stressors, such as physical demands and exposure to hazardous environments. However, it may also introduce new risks, including technical malfunctions, elevated noise and vibration levels, and increased cognitive demands, which necessitate careful monitoring and proactive management to safeguard employee well-being.

Taken together, these nine thematic areas—ranging from general perceptions and adaptation to changes in physical and mental health—highlight the complex and multifaceted impacts of robotization on employees, encompassing both benefits and potential challenges that require ongoing attention.

4. Discussion

The present qualitative study provides novel insights into the subjective experiences of employees and managers in Czech and Slovak manufacturing companies that have implemented industrial and collaborative robots. An analysis of interviews across four companies and multiple job positions highlights both the opportunities and challenges that accompany robotization in the workplace. The findings show that employees' perceptions and impacts of robotization are complex and significantly influenced by job type, the quality of organizational support, and the phase of adaptation. Altogether, these findings highlight that the perception and impact of robotization are dynamic and context-dependent, rather than uniform across all employees.

Overall, employees perceived robotization positively, particularly for reducing physical workload and eliminating monotonous or hazardous tasks. This is consistent with international evidence demonstrating that robots can decrease occupational risks and musculoskeletal strain [2, 3]. However, concerns were raised about elevated noise and vibration levels in certain workplaces, which employees linked to psychological fatigue. While large-scale European analyses have not identified statistically significant effects of robotization on the physical work environment [18], engineering research confirms that industrial robots inherently generate mechanical vibrations [19], and empirical measurements from robotic systems in healthcare settings indicate that noise levels may approach or exceed occupational safety thresholds [20]. These findings suggest that although not all participants reported experiencing excessive noise and vibration, such stressors may arise in specific workplace settings and should therefore be considered when designing occupational health and preventive interventions. Future research should integrate employees' subjective perceptions with objective environmental measurements to provide a more comprehensive assessment of these potential exposures and their impact on workers' health.

This duality—physical relief but potential sensory or cognitive strain—illustrates the complex health impacts of robotization and confirms earlier calls for more holistic assessments of Industry 4.0 techno logies [3].

Adaptation to new workflows was perceived as the most demanding phase, especially in the early stages of implementation. This is in line with Körner et al. [7], who noted that initial periods of human–machine interaction tend to involve heightened stress due to uncertainty and steep learning requirements [7]. Employees in this study often acknowledged that although change was initially difficult, it was eventually viewed as beneficial, echoing broader organizational research showing that technological transitions can yield long-term improvements despite initial resistance [1, 9]. On the other hand, some respondents noted that performance demands remained high or even increased, which, in some cases, contradicted their expectations. Such findings align with prior research emphasizing that while robots may reduce manual strain, they do not automatically lead to a slower or less pressured work environment [7, 13].

Regarding mental health, the findings were more nuanced. While operators generally did not experience increased psychological stress and trusted safety measures, technical staff perceived working with robots as more demanding, mainly due to higher responsibility, the unpredictability of technical problems, and the pressure to learn new skills continuously. Although managers did not explicitly report experiencing significant stress, they described a range of psychologically demanding responsibilities, including managing organizational change, reorganizing work processes, retraining employees, implementing technological innovations, and maintaining responsibility for production outcomes. Studies [1, 7, 8] confirm that technological changes do not necessarily affect employees' mental health equally across different positions or roles.

The finding that technical staff and managers experienced greater psychological demands than production operators may be interpreted in light of the Job Demands–Resources (JD-R) model [21]. According to this framework, employee well-being is influenced by the balance between job demands and available job resources. In the present study, production operators generally performed more structured, predictable tasks and often had access to established procedures and technical support when problems arose. In contrast, technical staff faced elevated job demands associated with troubleshooting, responsibility for production continuity, continuous learning, and rapid problem-solving during technical failures. Managers were exposed to substantial organizational demands, including coordinating technological change, supervising employees, managing staffing challenges, and maintaining production performance. Although access to resources such as training, technical support, collaborative teams, and clear communication appeared to mitigate some of these demands, the overall balance between demands and resources seemed less favorable for technical staff and managers than for production operators. These findings suggest that the psychological impact of robotization is shaped not only by the technology itself but also by the balance between job demands and organizational resources available to different occupational groups.

Technical malfunctions and operational issues were identified as some of the most stressful aspects of robotization. Importantly, employees reported that problems often arose not from the robots themselves, but from inconsistent input materials. This mirrors findings from previous studies that highlight the interconnectedness of technological and organizational factors in shaping employees' experiences [5, 8]. Employees reported frustration when immediate support was unavailable, illustrating the psychological burden of being solely responsible for resolving malfunctions under time pressure. Similar observations have been made in studies showing that perceived lack of control over technology is a significant driver of stress and reduced well-being [7, 8].

The findings suggest that adequate training and accessible technical support are crucial protective factors in mitigating stress and ensuring the successful integration of robots. Employees praised and emphasized the importance of proper onboarding, noting that hands-on experience on the job ultimately provided the most valuable learning. This aligns with prior research showing that training combined with participatory communication can enhance confidence and reduce psychological strain, thereby mitigating the potential mental health challenges associated with robotization [3, 13]. For technical staff, however, training remained more intensive, requiring the development of programming and troubleshooting skills, highlighting the need for tailored programs that reflect the specific demands of different roles.

Furthermore, respondents emphasized the importance of clear hierarchies of assistance. These organizational practices resonate with the broader literature on the role of organizational culture in shaping employee well-being during technological change [14].

An interesting observation was that employees in several teams tended to personify the robots they worked with by assigning them human names and, in some cases, even "talking" to them. Although such practices have received limited attention in the occupational health and human–robot interaction literature, participants described them as contributing to a more familiar and friendly relationship with the technology, reducing stress, and facilitating its acceptance in the workplace. By attributing human-like characteristics to robotic systems, employees may reduce the psychological distance between themselves and the technology, increase familiarity, and perceive the work environment as more approachable and less threatening. In this context, personification may represent an adaptive coping mechanism that supports successful human–robot collaboration, facilitates acceptance of technological change, and promotes the social integration of robotic systems into everyday work practices.

One notable finding is that employees expressed very little fear of physical harm from robots. Instead, they reported strong trust in established safety protocols and emphasized adherence to protective measures. This supports previous recommendations that well-designed safety standards are critical to preventing accidents and alleviating anxiety [22]. Rather than fear, employees described a form of "healthy respect" toward robots, which may serve as a psychological coping mechanism that balances awareness of potential risks with confidence in protective systems. This contrasts with studies in other contexts, such as China, where rapid robot adoption has been linked to heightened job insecurity and health concerns [10]. The respondents in this study appeared more reassured, possibly due to structured safety procedures and closer integration of operators in the robotization process.

Cognitive demands were highlighted across several positions, with operators and technicians reporting the need for sustained attention, rapid error detection, and adherence to new procedures. This aligns with experimental evidence showing that human-robot collaboration can elevate perceived cognitive load, particularly in the early stages of adaptation [11]. An intriguing finding of the present study was that several employees reported heightened attentional focus extending beyond the workplace into their personal lives. This observation may reflect two distinct, but not mutually exclusive, mechanisms. On the one hand, it may indicate a positive cognitive adaptation, whereby working with robotic systems enhances sustained attention, vigilance, and attentional control.

On the other hand, it may represent a spillover of work-related cognitive demands into non-work domains, suggesting incomplete psychological detachment from work and a persistent cognitive load beyond working hours. Owing to the qualitative design of the present study, it was not possible to determine which of these mechanisms predominated or whether both contributed to the reported experiences. Future research should further investigate this phenomenon using longitudinal designs and objective measures of cognitive workload, attentional performance, recovery processes, and work–life spillover to clarify its underlying mechanisms and potential implications for employee well-being.

Notably, employees did not report significant sleep disturbances or substance use linked directly to robotization. Instead, fatigue was primarily associated with shift work, confirming that contextual work-organization factors may outweigh technological factors in shaping certain health outcomes [23].

Robotization also influenced team dynamics. Respondents reported closer cooperation between operators, technicians, and managers, with communication becoming more frequent and problem-solving more collaborative. This observation is supported by Berkers et al. [5], who found that robotization reshapes work design in logistics by fostering interdependence and new patterns of cooperation.

The presented study, together with previous research [13, 22], suggests several recommendations that may be highly beneficial when implementing robots within a company's structure. Support from supervisors, the availability of technical staff, and regular opportunities for feedback appear to be crucial factors in reducing psychological stress and increasing employees' willingness to embrace change. Other important protective factors include accessible technical support, thorough training, a clearly defined hierarchy for assistance, and a work environment that promotes collaboration. Some teams also employed informal strategies to reduce psychological distance, such as naming robots and "personalizing" technology, which facilitated faster acceptance of the systems.

5. Strengths and Limitations

A major strength of this study lies in its qualitative design, which enabled an in-depth exploration of employees' and managers' subjective experiences with robotization. By including participants from different occupational groups, including production operators, technical staff, supervisors, and managers, the study captured a broad range of perspectives on the organizational, psychological, and physical impacts of robotization. The inclusion of companies that differed in size, sector, and degree of automation further enhanced the diversity of experiences represented in the data.

Several limitations should be considered. The participating companies were recruited voluntarily, and recruitment within each organization was coordinated by company representatives, which may have introduced selection bias. Organizations with more positive experiences of robotization may have been more inclined to participate. Furthermore, only current employees were included in the study. Individuals who had left their jobs because of difficulties associated with robotization, such as unsuccessful adaptation, job displacement, or dissatisfaction, were not represented, creating a potential survivorship bias.

At the same time, this characteristic of the study provided an opportunity to explore the factors associated with long-term retention in robotized work environments. The findings offer insights into the individual, organizational, and workplace characteristics that may facilitate successful adaptation to robotization and influence which employees remain in these work settings over time. Although these observations should be interpreted cautiously, they represent an important complementary outcome of the study. They may inform future longitudinal research on workforce adaptation and retention in increasingly automated workplaces.

The relatively small sample size and the study's context-specific nature limit the direct transferability of the findings to other organizational settings. Nevertheless, the identified themes and underlying mechanisms provide valuable insights into employees' experiences of workplace robotization that may apply to organizations with similar technological and organizational characteristics. Thus, while the findings are not intended to support statistical generalization, they contribute to the conceptual understanding of employee adaptation to robotized work environments and may guide future research, policy, and practice.

Another limitation of this study is that the interviews were not audio-recorded due to organizational confidentiality requirements. Although the absence of verbatim transcripts may have limited the capture of certain nuances in participants' responses, this potential limitation was mitigated by conducting each interview with two trained members of the research team. While one researcher led the interview and asked the predefined questions, the second researcher was dedicated exclusively to contemporaneous note-taking. The notes were reviewed immediately after each interview to ensure completeness and accuracy, thereby enhancing the credibility of the qualitative data collected.

In addition, several participants reported experiencing excessive workplace noise and vibrations during the interviews. However, these observations were based exclusively on employees' subjective perceptions. They were not verified by objective environmental measurements, as assessing physical workplace exposures was beyond the scope and primary objectives of the present study. Consequently, these findings should be interpreted as exploratory observations rather than evidence of actual exposure levels.

Despite these limitations, the study provides valuable insights into employees' experiences of robotization and identifies organizational factors that may support successful adaptation to technological change.

6. Conclusion

This study explored the impact of introducing robotic technologies on employee health in manufacturing companies in the Czech Republic and Slovakia, and identified key risk and protective factors to inform the development of preventive measures to reduce the psychological burden associated with robotization.

The results suggest that robotization can have positive effects on employees' physical health and, under certain conditions, pose no significant risks to their psychological well-being. A key factor is the quality of the implementation process—comprehensive training, accessible technical support, a clearly defined hierarchy of assistance, and effective team communication. It appears crucial to monitor the mental workload of technical and managerial staff, to create new job profiles (e.g., a mechatronics specialist serving as an intermediary between operators and technicians), and to foster a team culture in which mutual support and knowledge sharing are standard practice.

The results underscore that the impact of robotization on health and well-being is shaped not only by the technology itself but also by organizational support, team dynamics, and individual characteristics. By addressing these factors proactively, companies can harness the benefits of robotization while minimizing its risks.

Further research is necessary to gain a deeper understanding of the complex impacts of robotization and to identify strategies to minimize its risks while fostering a healthy work environment.

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APPENDIX

Supplementary Material S1: Interview Guide

1. Demographic and Occupational Characteristics
   - Age
   - Education
   - Current job position and work tasks
   - Length of employment in the company and current position
   - Previous experience with robotic technologies
2. General Perceptions of Robotization
   - How do you perceive the introduction of robots into your work?
   - What advantages and disadvantages do you associate with robotization?
   - Has working with robots affected your health or quality of life?
3. Changes Associated with Robotization
   - How has your work changed since the introduction of robots?
   - How has robotization affected your personal life?
   - What other changes have you noticed following robot implementation?
4. Employee Characteristics and Suitability for Robotized Work
   - What personal characteristics or skills are important for working with robots?
   - Are there groups of people who may find working with robots more difficult?
5. Training and Organizational Support
   - How was your training for working with robots organized?
   - What aspects of the training were useful or insufficient?
   - What forms of support are available when problems occur?
   - What additional support would help you perform your work more effectively?
6. Technical Problems and Error Management
   - What problems most commonly occur when working with robots?
   - How do you usually respond when technical problems arise?
   - How stressful are such situations?
   - To what extent can you rely on colleagues and supervisors when difficulties occur?
7. Cognitive Demands
   - Has working with robots influenced your attention or concentration?
   - Has it influenced your memory or information processing?
   - How mentally demanding do you find robotized work?
8. Mental Health and Well-being
   - How stressful is working with robots compared with previous work methods?
   - How does robotized work affect fatigue and recovery during shifts?
   - Has robotization influenced your sleep, well-being, or psychological health?
   - Do you have any concerns about safety, accidents, or health risks associated with robots?
9. Physical Health
   - Has robotization influenced your physical workload?
   - Have you noticed any positive or negative effects on your physical health?
10. Social Relationships and Communication
    - Has robotization changed communication with colleagues?
    - Has it changed communication with supervisors?
    - How important is cooperation in robotized work environments?
11. Lifestyle and Habits
    - Has working with robots influenced any lifestyle habits or behaviors, such as alcohol consumption, smoking, social media use, gaming, or other activities?
12. Overall Evaluation
    - Looking back, how would you summarize your experience with robotization?
    - Is there anything you would like to add or revise regarding your previous answers?

Table S2. Example of coding procedure.

Raw data extract:	Initial code	Category	Final theme
"It is a relief, but at the beginning it was terrible until everything was set up properly."	Positive perception of robotization	General evaluation of robotization	General perceptions
"Nobody likes changes at first, but later it became a relief for me and my team."	Adaptation to new technology	Organizational change	Change
"If I follow the safety rules, I am not afraid."	Trust in safety measures	Safety perception	Fear
"The clips are not always the same and the robot gets stuck."	Technical malfunction	Operational difficulties	Problems
"The worst thing is when technical support is not available."	Lack of support during problems	Operational difficulties	Problems
"We have supervisors, manuals, and technical support available."	Available assistance	Organizational support	Support
"We help each other and solve problems together."	Knowledge sharing	Team cooperation	Social relations and communication
"We call the robot Lojza."	Personification of robots	Informal coping strategy	Social relations and communication
"A person must want to learn new things."	Motivation to learn	Employee readiness	Employee characteristics
"Robotization was a huge relief for my hands."	Reduced physical strain	Lower physical workload	Physical health
"The noise and vibrations are enormous."	Noise and vibration exposure	Workplace environmental stressors	Physical health
"I have to concentrate more and constantly monitor the robot."	Increased attention	Cognitive demands	Mental health
"When something does not work, it creates stress."	Stress during troubleshooting	Psychological demands	Mental health
"I do not think robotization affects my sleep."	No sleep-related problems	Sleep and recovery	Mental health

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Sleep Quality and Physiological Stress in Emergency Physicians During 24-Hour Shifts: A Wearable-Based Occupational Health Assessment

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Keywords: Circadian Rhythm; Heart Rate Variability; Sleep Architecture; Occupational Stress

Abstract

Background: *Shift-based work schedules may disrupt circadian rhythm and adversely affect sleep and stress regulation among healthcare workers. Emergency physicians are particularly exposed to high-intensity workloads and prolonged 24-hour shifts, which can increase occupational strain.* **Methods:** *This study was conducted between April and December 2024 in a tertiary emergency department. Participants wore a smartwatch continuously for 30 days to record sleep architecture and heart rate variability-based stress indicators. Physiological parameters obtained during 24-hour shifts and at home were compared. Analyses were performed across four professional-experience groups.* **Results:** *Sleep duration, deep sleep percentage, deep sleep continuity score, breathing quality score, and overall sleep score were lower during 24-hour shifts, whereas light sleep percentage, nighttime awakenings, and stress levels were higher (all $p < 0.001$). Physicians with greater professional experience had longer sleep duration and lower stress levels during 24-hour shifts compared with early-career residents. Multiple linear regression analysis indicated that professional experience was the only independent predictor of stress levels during shifts.* **Conclusions:** *Shift work in emergency medicine is associated with impaired sleep quality and increased physiological stress. Professional experience appears to mitigate stress responses but does not fully restore sleep quality. These findings support the need for targeted occupational strategies, including optimized workload distribution and structured support for early-career physicians, to improve well-being and ensure patient safety in high-intensity clinical environments.*

1. Introduction

The continuity of healthcare services in secondary and tertiary institutions relies heavily on shift-based and on-call duty systems. However, these demanding work patterns often conflict with the endogenous circadian rhythm, potentially precipitating chronic biological and psychological health challenges [1]. Circadian misalignment among healthcare professionals is a primary driver of sleep disturbances, which can subsequently impair cognitive functions, emotional regulation, and clinical decision-making performance [2, 3]. From an occupational safety perspective, these impairments not only increase the incidence of clinical errors but also elevate the risk of workplace accidents, thereby compromising the physical integrity of the medical staff [4]. Sleep serves as a fundamental restorative process for cognitive stability and immune homeostasis;

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consequently, sleep deprivation or fragmentation increases susceptibility to infections and occupational hazards, including needle-stick injuries and accidental exposure to hazardous biological materials [5, 6].

Beyond cognitive decline, irregular sleep-wake cycles disrupt metabolic and hormonal equilibrium, diminishing clinicians' physiological resilience [7]. Emergency departments (ED) represent a unique occupational environment characterized by high-intensity, 24/7 operations that necessitate rapid interventions under unpredictable conditions. Constant exposure to critical trauma, acute time pressure, and the necessity of managing incomplete clinical data significantly escalate occupational stress levels [8-10]. Within the framework of occupational medicine, this sustained exposure constitutes a major risk factor for physician burnout and suboptimal patient outcomes.

While the repercussions of shift work on clinician well-being are documented, the specific influence of professional experience on objectively measured sleep architecture and physiological stress responses remains inadequately characterized. Recent advancements in wearable sensor technology now enable continuous, non-invasive monitoring of these parameters in real-world clinical settings. Elucidating these physiological patterns is essential for refining occupational risk management strategies and ensuring the long-term sustainability of healthcare workforces.

In this context, exacerbated stress and impaired sleep quality among emergency physicians may jeopardize patient safety and increase the prevalence of occupational morbidity. Therefore, identifying moderating factors—such as professional seniority—is vital for the development of targeted preventive workplace interventions.

This study used smartwatch-derived metrics to assess sleep quality and stress levels among emergency physicians across different career stages. The primary objective was to determine whether professional experience influences physiological stress responses and sleep parameters during 24-hour shifts. The findings suggest that increased professional experience mitigates stress responses but does not fully restore sleep quality.

2. Methods

This prospective observational study was conducted between April 1 and December 1, 2024, at the Emergency Department (ED) of Manisa Celal Bayar University Hafsa Sultan Hospital. The study was designed and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines. The study site is a high-acuity academic center characterized by unpredictable shift patterns and complex cognitive workloads. The primary objective was to evaluate and compare the sleep architecture, sleep quality, and physiological stress levels of emergency physicians during 24-hour on-call shifts versus off-duty periods at home, while examining the modulating role of professional experience.

A total of 13 Huawei Watch GT3 (Huawei Technologies, Shenzhen, China) smartwatches were utilized for continuous physiological monitoring. Due to the limited number of sensors, a rotation protocol was implemented; each participant underwent a continuous 30-day monitoring period before the device was reassigned to the next physician. All devices operated under identical firmware versions and application environments to eliminate potential inter-device measurement variability. This specific model was selected for its non-invasive form factor and high battery autonomy, which ensured high user compliance without interfering with clinical procedures. Data synchronization was performed via the manufacturer's health application, with automated transfer to the study database to prevent manual entry errors.

Ethical approval was granted by the Manisa Celal Bayar University Clinical Research Ethics Committee (Date: 03.04.2024; Decision No: 20.478.486//2361). The study adhered to the principles of the Declaration of Helsinki, and written informed consent was obtained from all participants prior to enrollment.

2.1. Smartwatch Measurement Protocol

The smartwatches provided automated analysis of sleep architecture and heart rate variability (HRV)-based stress levels. Sleep was categorized into light, deep, and rapid eye movement (REM) phases, with summary scores for deep sleep continuity and respiratory quality generated on a 0-100 scale. Stress levels, also reported on a 0-100 scale, were derived from automated HRV analysis approximately once per hour. Daily averages were utilized to reflect the autonomic nervous system response to occupational stressors and mental workload.

To ensure measurement integrity, participants were instructed to charge the devices only during daytime intervals. Nights with a wear-time of less than 90% were excluded from the final dataset. The utilization of these wearable sensors allowed for continuous, objective data collection in a real-world occupational setting, reflecting the physiological state of the participants during actual clinical practice.

2.2. Professional Experience Groups

Participants were categorized into four groups based on their professional experience:

- 0–1 year;
- 1–2 years;
- 2–4 years;
- specialists/academic faculty.

Residents in the first two groups primarily worked in high-volume green and yellow zones (~150-200 patients per shift). Those with 2-4 years of experience managed moderate-to-high acuity patients in the yellow and red zones (~50-100 patients per shift). Specialists and faculty physicians oversaw critically ill patients in the red zone (~15-20 patients per shift). As professional experience increased, the number of patients per 24-hour shift decreased, while case severity and clinical complexity increased. This stratification allowed for the assessment of how clinical workload and professional seniority modulate physiological stress responses in a real-world occupational setting.

The study included residents, emergency medicine specialists, and academic staff working in a shift-based system in the Emergency Department of Manisa Celal Bayar University Hafsa Sultan Hospital.

2.3. Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics version 26.0 (Armonk, NY, USA). Descriptive data were expressed as frequencies and percentages for categorical variables, while continuous variables were presented as mean ± standard deviation (SD), along with minimum and maximum values. The normality of data distribution was evaluated using the Kolmogorov–Smirnov test.

To compare physiological metrics obtained during 24-hour shifts and off-duty periods at home, the paired samples t-test was employed. For all paired comparisons, Cohen's d was calculated to determine the magnitude of the effect size. Differences across the four professional seniority groups were assessed using one-way ANOVA, followed by Tukey's post-hoc test for multiple comparisons where appropriate. Furthermore, univariate ANOVA models were utilized to evaluate the independent effect of professional experience on sleep and stress parameters after adjusting for potential covariates.

The relationship between continuous variables was assessed using Pearson correlation analysis. To identify independent predictors of occupational stress levels, a multiple linear regression model was constructed, incorporating variables that demonstrated significant univariate associations. To prevent multicollinearity, age was excluded from the final regression model due to its high correlation with professional experience. In all analyses, a two-tailed p-value < 0.05 was considered statistically significant.

3. Results

3.1. Participant Characteristics

The mean age of the participants was 29.5 ± 5.4 years, and 78 of them (78.8%) were male. Regarding professional experience, 27 participants (27.3%) were residents with 0-1 year of service, 27 (27.3%) had 1-2 years, another 27 (27.3%) had 2-4 years, and 18 participants (18.1%) were physicians with specialist or academic status.

3.2. Comparison of Sleep and Stress Parameters: Shift vs Home

The comparison of sleep and stress measurements obtained during 24-hour shifts and at home is summarized in Table 1. Total sleep duration, deep sleep percentage, deep sleep continuity score, breathing quality score, and overall sleep score were lower during shifts, whereas the number of nighttime awakenings, light sleep percentage, and stress levels were higher during shifts (all $p < 0.001$). No difference was observed in REM sleep percentage ($p = 0.856$).

Effect size analysis supported these findings, with Cohen's d values indicating medium to large effects for most parameters. These patterns are illustrated in Figure 1, highlighting the decline in overall sleep quality and the increase in stress levels during shift work.

3.3. Sex-Based Differences

No differences were observed between male and female participants in variables measured during 24-hour shifts. In off-duty measurements, men had a higher percentage of light sleep ($p = 0.012$), while women had a higher percentage of deep sleep ($p < 0.001$). No other differences were observed between male and female participants.

3.4. Analysis According to Professional Experience

Comparison of measurements during 24-hour shifts across professional experience groups showed differences in sleep duration, overall sleep score, and stress levels (all $p < 0.001$; Table 2). Post-hoc analyses indicated that participants with 2-4 years of experience and specialists/academic physicians had longer sleep durations and higher overall sleep scores compared to residents in the 0-1 and 1-2 year groups. Stress levels were highest in residents with 0-1 year of experience and lowest in specialist/academic physicians.

Effect size estimates for these outcomes were large (sleep duration $\eta^2p = 0.820$, overall sleep score $\eta^2p = 0.550$, and stress level $\eta^2p = 0.717$). A linear trend was observed across increasing levels of professional experience (p for trend < 0.001). No differences were observed in home-based sleep or stress parameters across professional experience groups.

3.5. Correlation and Regression Analyses

The correlations between physiological variables during 24-hour shifts are presented in Table 3. Stress levels were negatively correlated with age ($r = -0.595$, $p < 0.001$), overall sleep score ($r = -0.619$, $p < 0.001$), and sleep duration ($r = -0.762$, $p < 0.001$). In home-based measurements, stress level was negatively correlated with overall sleep score ($r = -0.382$, $p = 0.028$).

The results of the multiple linear regression analysis are shown in Table 4. The model was associated with stress levels ($F(3,29) = 23.18$, $p < 0.001$) and explained 67.5% of the variance (adjusted $R^2 = 0.675$). Professional experience was the only

Table 1. Comparison of Participants' Analyzed Values During 24-Hour Shifts and at Home.

Variables	During 24-Hour Shifts (Mean ± SD)	At home (Mean ± SD)	p-value*	Cohen's d_p
Sleep Duration (hours)	4.8±1.5	7.7±0.8	<0.001	–1.81
Number of Night Awakenings	2.0±1.3	1.0±0.5	<0.001	+0.91
Deep Sleep Percentage (%)	18.4±5.6	28.9±4.4	<0.001	–1.70
Deep Sleep Continuity Score	60.7±8.6	76.1±6.4	<0.001	–1.41
Light Sleep Percentage (%)	60.3±9.4	49.6±5.4	<0.001	+1.12
REM Sleep Percentage (%)	21.3±8.6	21.6±4.7	0.856	–0.03
Breathing Quality Score	81.8±8.3	89.6±6.8	<0.001	–1.12
Overall Sleep Score	66.6±7.9	79.6±4.4	<0.001	–1.56
Stress Level Score	65.1±20.1	29.3±9.3	<0.001	+1.33

** Paired-samples t-test. Effect size reported as Cohen's d_p.*

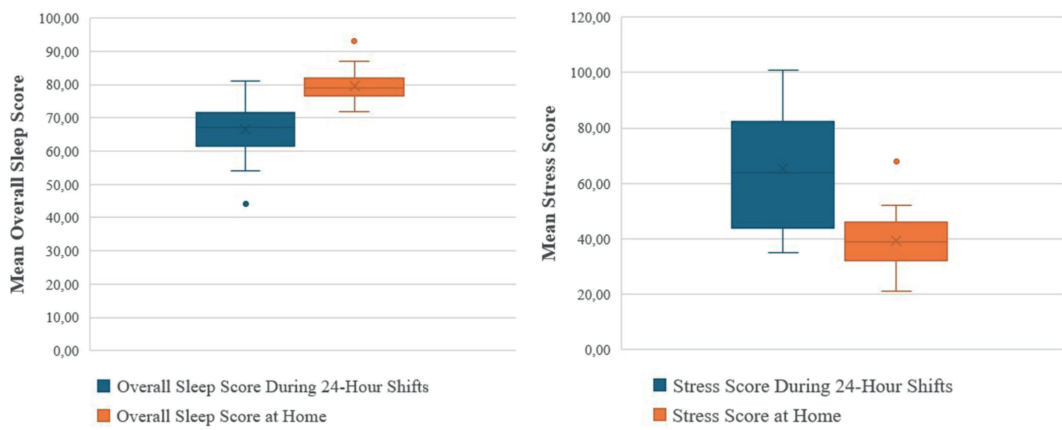

Figure 1. Comparison of participants' overall sleep and stress scores during work shifts and at home.

Table 2. Comparison of Participants' Analyzed Values During 24-Hour Shifts According to Length of Service.

	Length of Service				
Variables	**0-1 year Residents**	**1-2 years Residents**	**2-4 years Residents**	**Specialist/ Academic**	**p***
Sleep Duration (hours)	$3.3±0.3^a$	$4.0±0.6^a$	$5.9±1.0^b$	$6.8±0.6^b$	<0.001
Number of Nighttime Awakenings	2.1±1.7	2.2±1.1	1.2±0.9	2.9±0.8	0.080
Deep Sleep Percentage (%)	15.3±5.9	18.3±7.5	20.5±3.0	20.2±2.7	0.191
Deep Sleep Continuity Score	56.6±8.8	58.9±9.9	64.5±6.8	63.8±6.5	0.172
Light Sleep Percentage (%)	63.0±6.8	61.2±13.2	60.3±6.0	54.8±10.0	0.416
REM Sleep Percentage (%)	21.7±5.7	20.6±13.4	19.2±5.3	25.1±8.3	0.647
Breathing Quality Score	82.3±9.1	81.3±10.3	81.6±6.9	82.2±8.1	0.995
Overall Sleep Score	$60.4±7.0^a$	$62.4±5.5^a$	$71.7±4.0^b$	$74.3±5.2^b$	<0.001
Stress Level Score	$88.3±9.5^a$	$67.1±13.4^b$	$56.8±13.4^b$	$40.0±3.5^c$	<0.001

** One-way ANOVA test was used. Different superscript letters within the same row indicate statistically significant differences ($p < 0.05$). Effect sizes for the primary outcomes were large (sleep duration $\eta^2p = 0.820$, overall sleep score $\eta^2p = 0.550$, stress level $\eta^2p = 0.717$). All significant variables demonstrated a strong linear trend across seniority levels (p for trend < 0.001).*

independent predictor of stress level (β = –0.753, p = 0.002). Sleep duration and overall sleep score during shifts were not associated with stress levels in the final model (p = 0.743 and p = 0.946, respectively).

4. Discussion

Sleep is a fundamental physiological need with profound metabolic, cognitive, and psychological implications, playing a pivotal role in maintaining the long-term health of the workforce [11]. While current guidelines recommend that adults obtain at least seven hours of sleep per night to maintain optimal health [12], the emergency physicians in our study averaged only 4.8 hours during 24-hour shifts. By contrast, their off-duty sleep averaged 7.7 hours, exceeding the optimal threshold. This pattern likely reflects a compensatory "rebound sleep" response, in

Table 3. Correlation of Participants' Analyzed Values During 24-Hour Shifts.

Variables	Age	Sleep hours	No. of Night Awakenings	Deep Sleep Percentage (%)	Deep Sleep Continuity Score	Light Sleep Percentage (%)	REM Sleep Percentage (%)	Breathing Quality Score	Overall Sleep Score
Sleep hours	r=0.739 p<0.001	-	-	-	-	-	-	-	-
No. of Night Awakenings	r=0.093 p=0.608	r=-0.145 p=0.421	-	-	-	-	-	-	-
Deep Sleep (%)	r=0.370 p=0.034	r=0.414 p=0.017	r=-0.484 p=0.004	-	-	-	-	-	-
Deep Sleep Score	r=0.368 p=0.035	r=0.480 p=0.005	r=-0.492 p=0.004	r=0.736 p<0.001	-	-	-	-	-
Light Sleep (%)	r=-0.510 p=0.002	r=-0.390 p=0.025	r=0.068 p=0.706	r=-0.428 p=0.013	r=-0.288 p=0.104	-	-	-	-
REM Sleep (%)	r=0.313 p=0.076	r=0.155 p=0.389	r=0.240 p=0.178	r=-0.183 p=0.307	r=-0.163 p=0.363	r=-0.810 p<0.001	-	-	-
Breathing Score	r=-0.039 p=0.828	r=-0.042 p=0.816	r=-0.207 p=0.248	r=0.139 p=0.439	r=0.020 p=0.913	r=0.263 p=0.139	r=-0.371 p=0.033	-	-
Sleep Score	r=0.645 p<0.001	r=0.819 p<0.001	r=-0.384 p=0.027	r=0.550 p<0.001	r=0.679 p<0.001	r=-0.410 p=0.018	r=0.092 p=0.610	r=0.088 p=0.627	-
Stress Score	r=-0.595 p<0.001	r=-0.762 p<0.001	r=-0.047 p=0.796	r=-0.328 p=0.063	r=-0.225 p=0.209	r=0.202 p=0.260	r=-0.009 p=0.960	r=-0.009 p=0.958	r=-0.619 p<0.001

**Pearson correlation coefficient.*

Table 4. Multiple Linear Regression Analysis of Factors Associated With Stress Level During 24-Hour Shifts.

Independent Variable	B	95% CI for B	Standardized β	p-value
Professional Experience*	-13.98	-22.14 to -5.82	-0.753	0.002
Sleep Duration During Shift (Hours)	-1.16	-8.33 to 6.00	-0.087	0.743
Overall Sleep Score During Shift (0–100)	-0.03	-0.95 to 0.89	-0.012	0.946
Constant	105.85	61.36 to 150.33	—	<0.001

** Dependent variable: stress level score during 24-hour shifts. The model was statistically significant ($F(3,29) = 23.18$, $p < 0.001$) and explained 67.5% of the variance in stress level (adjusted $R^2 = 0.675$). Seniority was the only independent predictor of stress level in the final model.*

which clinicians extend their sleep at home to mitigate the significant sleep debt accumulated during high-intensity clinical duties. From an occupational health perspective, sustained chronic sleep deprivation poses a significant risk to the sustainability and mental well-being of the emergency medicine workforce.

In our study, sleep duration during shifts increased with professional experience. This disparity likely reflects the distribution of occupational workload, as junior residents typically manage high-volume "green" and "yellow" zones with continuous patient flow, whereas senior physicians and specialists primarily oversee high-acuity clinical decision-making and supervision. This suggests that the opportunity for restorative sleep is limited not only by the shift system itself but also by the clinical hierarchy and the resulting volume of patient encounters.

Interestingly, home-based sleep duration did not vary across experience groups, indicating that the physiological drive for recovery sleep is a universal response to shift-work-induced exhaustion, regardless of career stage.

Consistent with prior research indicating compromised sleep quality among shift-working hospital personnel [13], our participants' global sleep scores were lower during shifts (66.6 ± 7.9) than at home (79.6 ± 4.4). Junior residents (0–2 years of experience) had the lowest sleep quality during active duty. This finding likely reflects the cumulative burden of a higher mental workload, limited clinical autonomy, and the inherent stress of early-career training. The strong positive correlation between age and sleep quality during shifts (r = 0.645) further supports the hypothesis that professional experience facilitates better physiological adaptation to the demands of prolonged duty cycles.

Deep sleep is essential for physical recovery, immune regulation, and cognitive consolidation; its deficiency is linked to emotional dysregulation and impaired cognitive performance [14]. As observed in intensive care nurses who exhibit highly compromised overall sleep quality [15], our study found a significant reduction in deep sleep percentage and continuity during 24-hour shifts. A critical finding of our research, however, is that although more experienced physicians obtained longer total sleep, they did not show superior deep sleep quality. This suggests that the "alert-readiness" required in the emergency department environment prevents physicians from reaching deeper, more restorative sleep stages, regardless of seniority. The unpredictable nature of acute clinical emergencies seems to keep the autonomic nervous system in a state of sustained arousal. In the long term, shift-based work schedules have been associated with persistent sleep disturbances, which may increase the risk of cardiovascular disease, metabolic disorders, and depression [16]. In addition, previous studies have reported that women may have higher baseline deep sleep proportions and a greater physiological need for restorative sleep than men [17, 18].

The most striking finding from our multiple linear regression analysis was that professional experience was the sole independent predictor of occupational stress. Although sleep duration and quality were associated with stress in univariate analyses, their effects were secondary to the protective role of experience. Greater seniority likely enhances situational awareness, decision-making efficiency, and perceived control—psychological buffers that mitigate the physiological stress response. These results underscore the need for structured mentorship and supportive supervisory models to help early-career physicians build the resilience required for shift-based care.

These findings underscore the importance of integrating experience-based risk assessment into occupational health policies. Seniority appears to function as a physiological buffer, suggesting that organizational support should be preferentially directed toward early-career physicians who may lack these adaptive mechanisms.

From a preventive medicine perspective, implementing objective fatigue-monitoring systems and tailoring shift intensity to clinical seniority may help reduce burnout and occupational accidents, thereby improving physician well-being and patient safety.

This study has several limitations that should be considered. First, the use of consumer-grade wearable sensors (Huawei Watch GT3), while highly practical for continuous monitoring in real-world settings, may introduce measurement variability compared to gold-standard polysomnography [19]. However, the non-invasive form factor of these devices was essential to ensure user compliance and prevent ergonomic interference during acute clinical procedures. Second, the single-center design may limit the generalizability of the findings to different healthcare systems or hospital tiers. Third, although we adjusted for major variables, unmeasured confounders such as caffeine and nicotine consumption, chronotype variations, and individual lifestyle habits may have influenced the physiological outcomes. Finally, the observational nature of the study precludes establishing direct causality. Despite these limitations, the use of wearable technology provided high ecological validity by capturing the objective physiological strain of emergency physicians during actual clinical practice. These findings support the feasibility of utilizing objective, real-time data to refine occupational risk management and improve the health and safety of healthcare professionals in high-intensity environments. Future multicenter studies with larger populations and advanced physiological monitoring techniques are needed to confirm these findings and further explore long-term occupational health outcomes.

5. Conclusion

In conclusion, our study demonstrates that 24-hour shifts in emergency medicine are directly linked to a significant decline in subjective sleep quality and a marked increase in physiological stress markers. While professional experience appears to confer psychological resilience and attenuate acute stress responses, it is insufficient to counteract the biological toll of sleep deprivation during active duty.

These findings underscore a critical vulnerability in current emergency department staffing models. To preserve the health of the workforce, particularly among early-career physicians, who are most sensitive to stress, healthcare administrators should prioritize evidence-based interventions. Such strategies must include optimized workload distribution, protected rest periods, and shift planning that accounts for seniority.

Ultimately, addressing the physiological demands of long-duration shifts is essential to sustaining high-quality emergency care. Enhancing physician well-being is not merely an occupational health goal; it is a fundamental necessity for ensuring patient safety and the long-term operational resilience of emergency healthcare systems.

Institutional Review Board Statement: This study was conducted in accordance with the principles of the Declaration of Helsinki. Ethical approval was obtained from the Manisa Celal Bayar University Clinical Research Ethics Committee (Approval Date: 03.04.2024; Decision No: 20.478.486//2361).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

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La Medicina del Lavoro

Case report

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Interstitial Lung Disease in a Dental Technician Diagnosed Through an Integrated Multidisciplinary Approach

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Summary

Evidence from the literature indicates that workers in the dental care sector are routinely exposed to hazardous airborne agents, particularly mineral and metal dusts. This occupational risk is further amplified in settings with inadequate ventilation and insufficient use of personal protective equipment. We discuss the case of a dental technician with a significant history of occupational exposure who presented to our clinic with exertional dyspnea. High-resolution computed tomography revealed fibrosing nonspecific interstitial pneumonia associated with emphysema, and spirometry showed a severe reduction in carbon monoxide diffusion capacity. After a multidisciplinary evaluation, we performed bronchoscopy with bronchoalveolar lavage (BAL), which revealed an abundance of macrophages. Scanning electron microscopy with energy-dispersive spectrometry of the BAL detected silicon oxide, calcium, iron, and traces of metals typical of dental alloys (titanium, aluminum), as well as rare earth elements and toxic pollutants. These findings supported the diagnosis of interstitial lung disease due to long-term occupational exposure in dental laboratories, enabling a minimally invasive diagnostic approach. This strategy obviated the need for transbronchial biopsy, thereby avoiding a procedure associated with a high risk of pneumothorax in this patient.

1. Introduction

Dental care workers are exposed to a variety of physical, chemical, and biological hazards in their work environment. Dental technicians, in particular, are exposed to numerous potentially toxic substances and airborne residues, such as silica, alloys, and acrylic plastics, during their work. As a result, they are at risk of developing occupational lung diseases, including pneumoconiosis, with severity often depending on the duration of exposure [1].

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Research by Ali et al. in Saudi Arabia found limited awareness among dental technicians and dental technology students of the risks associated with exposure to dental material dust [2]. Most self-employed dental technicians work more than eight hours a day without adequate ventilation, increasing their risk of occupational lung conditions [3]. When ventilation is inadequate, dust particles, particularly those generated during grinding, have been shown to significantly exceed safety standards [4].

Pneumoconiosis is a subset of occupational ILDs characterized by the accumulation of inhaled inorganic particles in the lungs. It is one of the most prevalent occupational diseases globally, particularly in low- and middle-income countries [5, 6]. Based on the type of mineral dust responsible for the condition, pneumoconiosis can be classified into several categories: silicosis, the most common form, triggered by dust primarily containing free crystalline silica; silicate pneumoconiosis, caused by dusts containing silicates, including asbestos, talc, cement, and mica; pneumoconiosis resulting from coal dust and carbon-based dusts, such as coal worker's pneumoconiosis, graphite pneumoconiosis, and carbon black pneumoconiosis; and metal pneumoconiosis, caused by metal dusts, including aluminum pneumoconiosis, welder's pneumoconiosis, and caster's pneumoconiosis [7].

China accounts for more than half of the global burden of pneumoconiosis, representing 68.7% of incident cases, 44.3% of deaths, and 66.2% of disability-adjusted life years (DALYs). In 2019, 15,898 new cases were reported, and more than 20 million workers were exposed to occupational risk factors [8]. Recent reports from North-Eastern Italy have described an unexpectedly high incidence of silicosis among workers processing quartz conglomerates, a new artificial stone with high silica content used for kitchen and bathroom countertops, highlighting the occurrence of occupational pneumoconiosis in newer productive settings outside traditional mining and quarrying industries [9]. Currently, no curative treatment is available for pneumoconiosis, and most therapeutic approaches are aimed at slowing disease progression and alleviating symptoms [8].

Dental technicians' pneumoconiosis (DTP) is recognized as a distinct condition, often linked to exposure to cobalt-chromium-molybdenum (Co-Cr-Mo) alloys [1]. A 2014 Turkish study of 893 dental technicians admitted to the hospital in the past 5 years found that 10.1% had pathological chest X-rays. The main HRCT abnormalities were hilar and mediastinal lymphadenopathy, micronodules, reticulonodular infiltration, increased linear density, and interlobular septal thickening. In this study, the highest rate of DTP was observed among individuals working in the leveling and polishing department. Among the subgroup with a history of sandblasting, pneumoconiosis was found in 46.1% of cases [1]. In another article by Kahraman et al., 46% of participants had radiological signs compatible with the condition, with the most common finding being round consolidations. In terms of pulmonary function, forced expiratory rate and diffusing capacity of the lungs for carbon monoxide (DLco) were significantly lower in those who had worked for 16 years or more [10].

In 2022, Di Lorenzo et al. described a case of pneumoconiosis in a dental technician with 15 years of experience, diagnosed by scanning electron microscopy-energy dispersive spectrometry (SEM-EDS) of a surgical biopsy. The procedures he performed in his job, sometimes in non-closed-circuit systems and often without protective equipment, generated aluminum, crystalline silica, chromium, and titanium dust. HRCT of the chest showed bilateral micro nodulation, with a greater concentration in the upper lobes. Histological examination revealed aggregates of pigment-laden macrophages forming perivascular macules or arranged in a radial pattern around a core of sclerohyalinosis. SEM-EDS identified several mineral particles, typically characterized by the presence of crystalline silica and metal (aluminum, iron, chromium, cobalt, titanium) aggregates. The environmental concentrations of total dust and its respirable fraction were below the respective Threshold Limit Value – Time-Weighted Average (TLV-TWA) (according to the American Conference of Governmental and Industrial Hygienists – ACGIH), but still not negligible [11].

Among the few cases reported in the literature on mineralogical analyses of BAL, De Vuyst et al., in 1986, identified, by means of SEM/EDS, particles of chromium, cobalt, and molybdenum—constituents of Vitallium®, a chromium–cobalt–molybdenum alloy—in the BAL of two patients with biopsy-proven pulmonary fibrosis. The authors also demonstrated the chemical stability of this alloy in BAL fluid and lung tissue several years after cessation of exposure. Indeed, in one patient, mineralogical analysis of BAL was performed both during occupational exposure and one year after cessation of work activity, showing similar findings. In the second patient, BAL analysis demonstrated the presence of Vitallium® particles two years after occupational exposure had ended. In the BAL of one patient, bare asbestos fibers as well as typical asbestos bodies were also detected, whereas in the other patient iron, iron–chromium, and silica particles were identified [12]. More recently, Tiraboschi et al. reported the case of a ceramic dental technician with a two-year occupational history who developed pneumoconiosis. SEM/EDS analysis of the patient's BAL revealed numerous inorganic particles containing zirconium and aluminum, as well as tungsten [13].

2. Case presentation

We discuss the case of a 69-year-old male patient born in 1956, a non-smoker who was exposed to passive smoke for 5 years, from 1973 to 1978. He was a dental technician who worked in the field, exposed to ceramic and plaster dust, monomers, and polymers, particularly from ages 18 to 23. Subsequently, he was provided with personal protective equipment and extraction hoods in his workplace. However, regarding plaster, the patient specified that the material undergoes an initial processing phase during which the operator does not use personal protective equipment. His grandfather had an unspecified lung disease. As comorbidities, the patient reported only type 2 diabetes and arterial hypertension. He denied early graying. As home therapy, the patient was taking metformin and tamsulosin.

In 2020, the patient began experiencing shortness of breath during intense exertion (modified Medical Research Council – mMRC – Dyspnea Scale 0). On the general practitioner's advice, he underwent cardiopulmonary exercise testing (CPET), which was negative, and pulmonary function tests (PFTs), which showed normal lung volumes and a moderate reduction in the DLco to 45%.

The patient discontinued follow-up until 2024, when persistent respiratory difficulty prompted further evaluation. On a pulmonologist's recommendation, he underwent high-resolution computed tomography (HRCT) of the chest, which revealed a pattern of fibrosing nonspecific interstitial pneumonia (fNSIP) associated with emphysema (Figure 1). He also had new PFTs, showing a DLco of 34% and a mild restrictive defect, with a total lung capacity (TLC) of 78%. An echocardiogram showed normal pump function, with an estimated systolic pulmonary artery pressure of 28 mmHg, below the commonly proposed echocardiographic thresholds for pulmonary hypertension.

At our first visit, the patient was eupneic at rest on ambient air, with peripheral oxygen saturation of 98%. Upon auscultation of the chest, vesicular murmurs were reduced throughout the lung fields, with rare crackles at the bases. Digital clubbing was noted.

We recommended a blood sample for antibody panel testing, which was entirely normal except for a weakly positive rheumatoid factor (53.9 kU/L) in the absence of clinical features suggestive of connective tissue disease. Therefore, this finding was considered to be of no pathological significance. Moreover, alpha-1 antitrypsin levels were tested and found to be within the normal range. The case was then discussed at the multidisciplinary meeting dedicated to interstitial and rare lung diseases. The fNSIP radiological pattern was confirmed, with ground-glass areas, particularly in the lower lobes, associated with paraseptal emphysema. Radiological findings typically associated with fibrotic hypersensitivity pneumonitis (fHP), particularly those reflecting a bronchiolocentric interstitial pneumonia (BIP) pattern, such as mosaic attenuation, lobular air trapping, and the three-density sign, were not observed. Other differential diagnoses were carefully considered. No other relevant elements emerged from the history, apart from the well-documented long-standing occupational exposure commonly encountered in dental laboratories. Moreover, although not classically considered a predominant feature of these conditions, emphysematous changes have

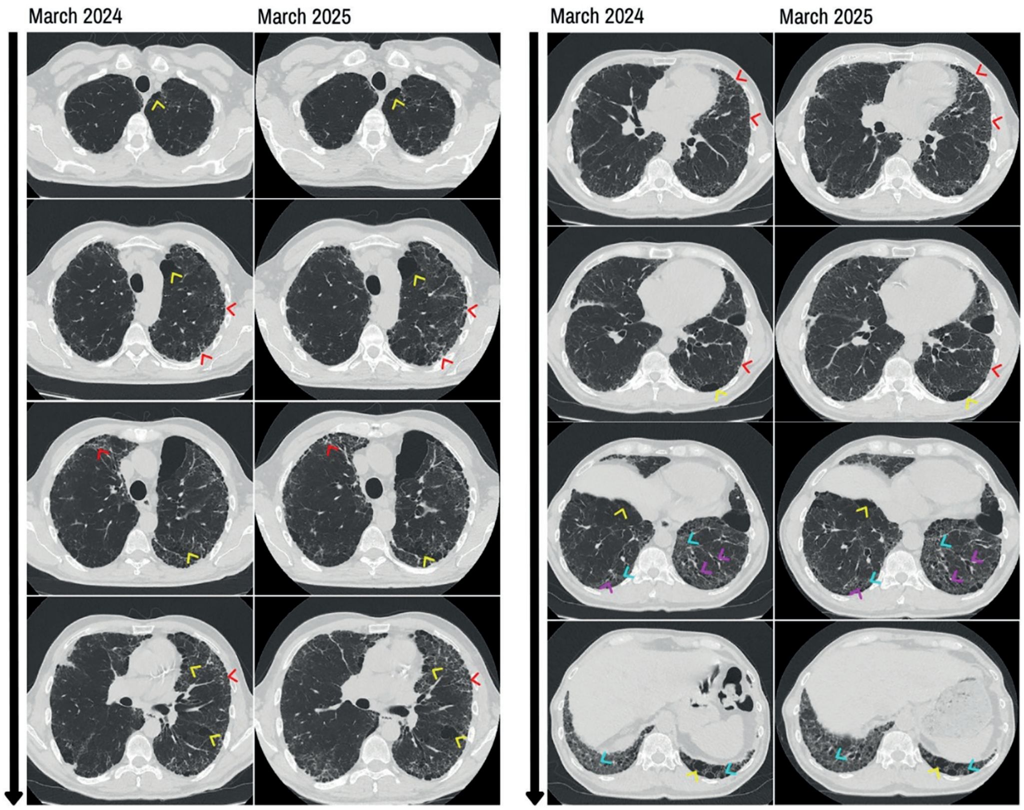

Figure 1. Chest HRCTs obtained one year apart. Findings include emphysematous changes with enlargement of the air spaces (yellow arrowheads), subpleural interstitial thickening (red arrowheads), areas of mild ground-glass opacity predominantly involving the lower lobes (light blue arrowheads), and traction bronchiectasis (purple arrowheads). A mild radiological progression is observed from March 2024 to March 2025.

been described in several forms of pneumoconiosis associated with inorganic dust exposure and were therefore considered compatible with the patient's occupational history. Considering the overall clinical, radiological, and occupational picture, the disease was considered compatible with occupational pneumopathy rather than an idiopathic process.

From the consultation with the occupational physicians, additional details emerged regarding the patient's work history. He worked as a dental technician from 1973 to 1990. During the early years as an apprentice, he mixed plaster with water and poured wax to create models for removable prostheses. Later, he worked in the production of acrylic resin, derived from a mixture of liquid monomer and powdered polymer. The acrylic prosthesis was then deflashed (removed from the plaster mold) and baked at 90°C. This was followed by milling, first polishing with pumice stone and final polishing with brushes. These procedures involved inhalation of fumes and dust from the materials being worked. Since 1990, he has been involved in orthodontics, with work activities similar to those above but characterized by reduced exposure to dust and fumes (including those from welding the tempered steel wires of prostheses with a butane torch, a very sporadic activity) due to the constant use of direct suction systems. From the technical data sheets of the products used, it was found that the materials contained gypsum (calcium sulfate dihydrate), iron oxides, titanium oxides, and methacrylate (in orthodontic resins), as well as nickel, chromium, and manganese in the stainless-steel wires and screws used in orthodontics, and therefore present in welding fumes, although the welding activity was reported as very sporadic. The use of pumice stone also exposed him to silica and aluminum. Thus, the occupational history documented multiple exposures and strengthened suspicion of a possible causal role of work in the onset of the radiological alterations previously described.

For further diagnostic purposes, the patient underwent a bronchoscopy with BAL, whose cytologic analysis showed a predominance of macrophages (80%), many of which were pigmented, with a significant proportion demonstrating activation (Figure 2). SEM-EDS on the BAL revealed the presence of silicon oxide, calcium, iron, and traces of metals typical of dental alloys and pumice stone (titanium, aluminum), as well as rare earth elements such as terbium, in addition to toxic pollutants like antimony (Figure 2). Transbronchial biopsies have been avoided due to the risks associated with the procedure, first pneumothorax, given the condition of lung parenchyma.

Combined radiological and functional findings, along with BAL analysis using SEM-EDS that identified minerals and metals consistent with those present in the products used, allowed us to confirm the occupational etiology of the pneumopathy affecting the patient. Moreover, BAL analysis did not demonstrate lymphocytosis, a finding that further argued against fHP. Based on the mineralogical findings, this condition can be classified as an occupational ILD compatible with mixed-dust exposure, presenting with a fNSIP pattern. The patient was advised to cease all work activities involving direct exposure to the dust and fumes mentioned above and has been placed under clinical, radiological, and functional follow-up.

The patient discontinued work-related activities; however, in March 2025, radiologic (shown in Fig. 1) and functional disease progression was observed, with a 7% decline in forced vital capacity compared with the previous year. In light of the observed disease course, antifibrotic therapy with nintedanib was initiated to slow disease progression. By July 2025, the patient developed reduced exercise tolerance; the 6-minute walk test demonstrated oxyhemoglobin desaturation occurring at the second minute of exercise, prompting the initiation of exertional oxygen therapy.

3. Conclusion

To the best of our knowledge, this is one of the few documented cases of ILD with an occupational etiology in a dental technician, confirmed by SEM–EDS analysis of BAL. We find the use of this technique and its diagnostic yield on endobronchial fluid particularly interesting, as it allowed us to avoid a biopsy in a patient for whom such a procedure would have been highly risky. Consequently, we achieved a definitive diagnosis with high confidence using an invasive procedure that can still be safely performed even in high-risk patients. The combination of chronic occupational exposure to multiple inorganic dusts and metals and BAL SEM-EDS findings strongly supported the occupational etiology of the ILD. In this context, we considered the case to be more appropriately classified as an occupational ILD compatible with mixed-dust exposure. The coexistence of fibrotic interstitial features and documented exposure to heterogeneous inorganic particles may also place this condition within the spectrum of mixed-dust pneumoconiosis.

The fibrogenic potential of the substances identified by BAL SEM-EDS analysis is likely heterogeneous. In our opinion, the occupational relevance of this case is primarily related to chronic inhalational exposure to inorganic dusts and metals with recognized pneumotoxic and fibrogenic potential, particularly silica-containing materials and iron particles, followed by aluminum, and, although less commonly reported, titanium particles [11], all of which were consistent with the patient's occupational history. In contrast, elements such as antimony and rare earth elements were interpreted as ancillary findings detected by the high sensitivity of SEM-EDS analysis rather than as primary drivers of the fibrotic lung disease.

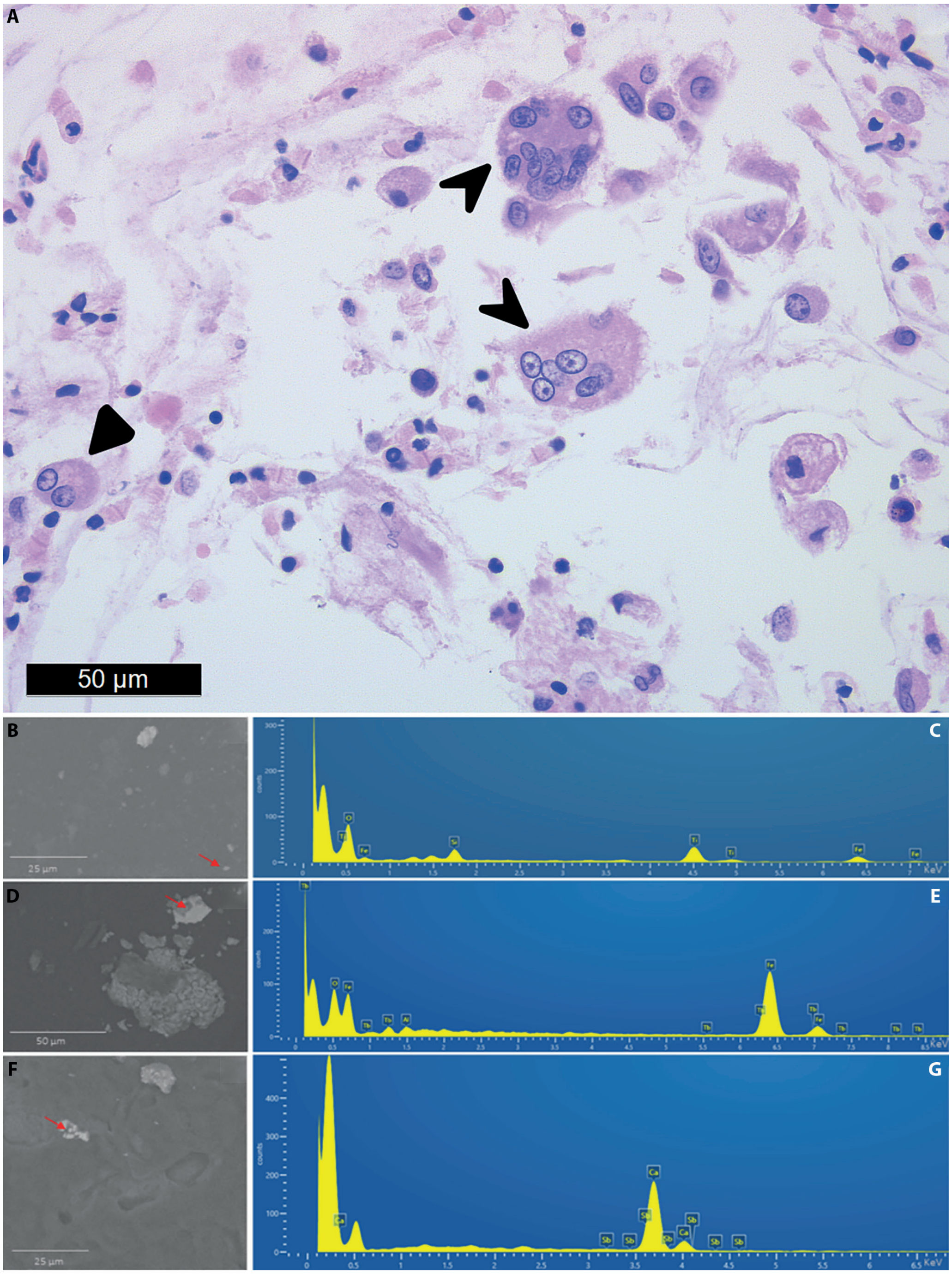

Figure 2. Bronchoalveolar lavage cytoblock H&E slides (A) demonstrated an abundant amount of macrophages. A significant proportion of these (between 15-20%) demonstrated activation (notched arrowhead); another 5% were multinucleated macrophages (full arrowhead). SEM-EDS on BAL paraffin sections (B-G): B) A particle (arrow) contains silicon, iron and titanium (C); D) in a particle (arrow) aluminum, iron, terbium are detected (E); F) A particle (arrow) contains calcium and antimony (G).

This case report aims to draw attention to this category of workers, highlighting how occupational exposures in these environments can harm the lungs. Moreover, it seeks to raise awareness of the importance of using personal protective equipment and implementing adequate suction and ventilation systems to improve workplace health and safety. Last but not least, our patient was a freelancer and did not undergo regular health surveillance examinations provided by an occupational physician; thus, the lack of periodic monitoring has reasonably delayed the diagnosis. Pneumoconiosis is a completely preventable but irreversible disease; for this reason, it is necessary to strengthen awareness among workers about occupational risk management tools and the importance of undergoing regular occupational health surveillance.

Additionally, we believe that bronchoscopy with BAL and SEM-EDS analysis may be a valuable diagnostic tool in selected cases of suspected occupational ILD. A potential surveillance strategy for at-risk individuals, such as those employed in the dental care sector, could initially rely on non-invasive approaches, including PFTs, potentially supplemented by lung ultrasound when an experienced pulmonologist with expertise in ILD is available within the screening setting. Subjects with findings suggestive of interstitial involvement could then undergo chest HRCT for further evaluation. Finally, in cases where ILD is confirmed and additional etiological characterization is clinically indicated, bronchoscopy with BAL followed by SEM-EDS analysis could be integrated into the standard multidisciplinary diagnostic work-up to support correlation of lung pathology with occupational exposure.

Informed Consent Statement: Informed consent was obtained from the subject involved in the study. Written informed consent has been obtained from the patient to publish this paper.

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Author Contribution Statement: GG was the first author and was responsible for drafting the manuscript, reviewing the patient's clinical data, and conducting the literature review; GM analyzed the bronchoalveolar lavage samples and contributed to drafting the manuscript. FC led these analyses, drawing on her extensive expertise in pathology; MD contributed by performing the endoscopic examination that led to the final diagnosis; RP reviewed and interpreted the radiological findings as an expert thoracic radiologist; SR, MDB, and CB contributed their domain expertise to the scanning electron microscopy and energy-dispersive X-ray spectroscopy analyses; PM contributed to the comprehensive occupational and medical history assessment as an expert in occupational medicine and provided valuable input in the complex clinical management of the case; PS provided expert pneumological input during manuscript preparation; EB coordinated the work and provided expert guidance in drafting and revising the manuscript. All authors have read and approved the final version of the manuscript.

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